Why Workers Stay:
Pay, Beliefs, and Attachment*

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December 2025

Abstract

Workers often remain with their employer despite higher outside pay. This may reflect information frictions or preferences. We surveyed full-time German workers and linked responses to Social Security records. Workers believe pay varies across firms and direct their search toward higher-paying employers. Expected firm-specific pay premia are correlated with administrative pay measures and workers' amenity valuations. Yet most workers would reject take-it-or-leave-it raises at named destination firms. Separation elasticities are 2-4 when pay levels and growth are known. Attachment varies across firms in ways unexplained by ex ante amenity valuations. Limited mobility is better explained by preferences than information frictions.

*We thank researchers at the IAB for assistance fielding the survey. We thank Matthew Backus, David Card, Stefano DellaVigna, Ellora Derenoncourt, Lisa Kahn, Lawrence Katz, Patrick Kline, Attila Lindner, Alexandre Mas, Conrad Miller, Amanda Pallais, Simon Quach, Jesse Rothstein, Raffaele Saggio, Isaac Sorkin, and seminar and conference participants from the Atlanta Fed Employment Conference (especially our discussant, Lisa Kahn), the Berlin Applied Micro Seminar, Boston College, the Central European University, Chicago Harris, Cornell, the Federal Reserve Board of Governors, Harvard Kennedy School, Northwestern's Kellogg School of Management, the London School of Economics, Stanford University, Stockholm University, University College London, UC Davis, University of Lausanne, University of Virginia, the 2024 NBER Summer Institute (Labor Studies), the Norwegian Workshop on Labor Economics, the Rochester Labor Economics Workshop, the 2023 Society of Economic Dynamics annual meeting, the Society of Labor Economists' Annual Meeting, the UC Berkeley labor and macro lunches, and the West Coast Junior Labor Conference for helpful comments. We thank Christopher Ackerman, Cristian Alamos, Monica Essig Aberg, James Hickman, Selina Mäurer, Chiara Motta, Kyle Neary, and Sebastian Puerta for outstanding research assistance. We thank the National Science Foundation (SES 2242439), the UC Berkeley Department of Economics, the UC Berkeley Hellman Fund, and the Joachim Herz Foundation for funding. Caldwell: scaldwell@berkeley.edu; Haegele: ingrid.haegele@econ.lmu.de; Heining: joerg.heining@iab.de. Some of the results were previously circulated under the title "Firm Pay and Worker Search".

"...The basic difficulty is that satisfactorily employed workers are almost entirely uninterested in employment conditions in other companies. This lack of interest is an even more serious obstacle than the difficulty of compiling accurate job information."
–Reynolds (1951)

1 Introduction

The willingness of workers to switch firms in response to changes in relative pay can affect the wages firms offer (Manning, 2011). A growing literature documents that, on average, a firm which lowers its wage by 10% can expect 30% of its workers to leave (see surveys in Sokolova and Sorensen, 2018; Kline, 2025).[1] This limited response gives firms monopsony power, enabling them to set wages below the competitive level.

Workers may fail to search for—or move to—jobs that offer higher pay for different reasons. In one class of models, limited mobility stems from information frictions relating to pay: workers may not switch firms if they have incomplete or incorrect knowledge of outside pay (Manning, 2003). In another, fully informed workers stay because of preferences (Card et al., 2018; Berger, Herkenhoff and Mongey, 2022). These preferences may include switching costs or idiosyncratic tastes—such as for non-wage amenities or relationships with colleagues and managers. While both mechanisms can generate upward-sloping firm-specific labor supply, they imply different sources of misallocation, different welfare conclusions, and different policy responses.

Empirically distinguishing between information and preference-based explanations is challenging. While a large literature documents that firms differ in wage premia, existing data do not reveal whether—and to what extent—workers are aware of those differences (Card, Heining and Kline, 2013; Song et al., 2019). Workers who believe there is a single market wage for their skill may be less inclined to search than those who expect pay to vary across potential employers. Those who expect pay to vary, but do not know the pay policies of specific firms, will search differently than those who have firm-specific information on pay. If preferences are the main barrier to mobility, then providing accurate pay information may not induce job-to-job mobility.

Understanding the sources of limited firm-to-firm mobility is important for both theory and policy. Information- and preference-based models yield different predictions for the welfare gains from increased labor market competition. In preference-based models, expanding access to outside opportunities is always beneficial to workers; in models with information frictions, more

[1] The idea that quit rates are less than perfectly elastic goes back nearly a century (Robinson, 1933). Empirical work has shown the limited response is true in the long run and in situations where there are many potential outside employers.

competition may not improve outcomes (Caldwell, Dube and Naidu, 2025). If information frictions regarding pay dominate, transparency policies may enhance mobility (Council of Economic Advisors, 2016); if preferences dominate, such policies may be ineffective.

We provide new evidence linking workers' beliefs about and preferences over outside firms to their search and mobility. We fielded a novel survey through the German Institute for Employment Research (IAB), and obtained responses from nearly 10,000 full-time German workers aged 25 to 50. A key innovation of this survey is that it includes questions about specific outside firms. For a set of firms named by respondents as potential destinations, and a second set of firms randomly assigned in our research protocol, we elicit (1) whether workers would apply to the firms if considering a job change, (2) what they expect the firm would pay them, and (3) how they would rank hypothetical offers from these firms relative to each other and relative to the opportunity to stay at the incumbent firm given researcher-randomized raises.

We link survey responses to administrative data on the workers and to administrative and public-use data on the provided firms including firms' pay premia, median pay, and industries. We use these linkages to perform a number of validation exercises on the survey data. The dual linkage allows us to compare workers' expectations to objective benchmarks and to compare workers' stated preferences with observed outcomes. The discrete choice experiments allow us to estimate the responsiveness of workers' stated mobility to offers with randomized changes in pay (i.e., the labor supply elasticity to the firm), holding fixed beliefs about pay (and, in some experiments, beliefs about pay trajectories). We show that workers' responses in these experiments are aligned with each other (across distinct experiments) as well as with observed measures of workers' search behavior.

Our results indicate that preferences, rather than beliefs about pay, are the primary barrier limiting worker mobility. Many workers have information on pay. They expect both pay and non-pay factors to vary across firms and perceived pay premia are positively correlated with administrative pay measures and with workers' valuations of non-wage amenities. Workers also direct their search toward higher-pay firms. Workers who are risk averse, who consider the probability an application will be successful, or who believe that high wage firms receive more qualified applicants per vacancy exhibit somewhat lower responsiveness to pay. However, the differences are economically and statistically insignificant, suggesting perceived offer probabilities are not a significant barrier to search. When asked to rank offers from outside firms—including firms named as potential destinations—workers often prefer to remain at their current firm. When asked why they believe workers are reluctant to switch firms, respondents cite a general reluctance to undergo change, location, and social connections. Pay is the fourth most cited reason; lack of opportunities is not a major factor.

We frame the empirical analysis with a partial equilibrium search model in which workers have beliefs about the pay provided by outside firms and heterogeneous preferences across these firms. The model clarifies how workers' beliefs about the pay provided by outside firms shape search behavior—both on the extensive margin (whether to search) and on the intensive margin (where to apply). Pessimistic beliefs about outside options reduce search and, as a result, mobility. But when workers expect pay to vary across firms, they are less likely to forgo profitable search. Preferences also affect workers' search and mobility. Workers who value their current employer or face high switching costs (including tangible barriers or a general aversion to change) do not move, even when presented with outside offers at higher pay. Anticipating that they will reject offers they receive, these workers are less likely to search.

The empirical analysis proceeds in three parts. First, we document new facts about workers' beliefs about pay. Guided by the model, we first assess whether workers expect their pay to vary across potential employers. Nearly one half of workers had firm-specific information on pay when they applied to their current firm. Most workers do not believe they face a uniform outside option. Only 20% say that they would earn the same pay at each of the three firms that we randomly provided to them (researcher-provided firms). We see a similar pattern when we examine workers' pay expectations at the firms they provided as potential destinations (worker-provided firms). The within-worker variation in expected pay does not only reflect expected differences in pay across industries or regions. Instead, there appears to be a strong firm component, with workers expecting pay to vary across firms within the same industry and region.

We use a two-way fixed effects model to summarize workers' expectations at the randomly assigned researcher-provided firms. From a list of 30 large German employers, we randomly selected three firms to provide to each worker. We identify the firm-specific effects from within-worker deviations in expected pay at these firms. Workers expect pay to vary significantly: the firm-specific estimates range from -0.25 (25 percentage points less than the base firm) to 0.05 (5 percentage points more). Workers' expected pay premia are positively correlated with objective measures of firm pay premia ($\rho = 0.42$) and of firm median ($\rho = 0.54$) and mean ($\rho = 0.56$) pay. We also find strong agreement across demographic groups in their expected pay rankings, suggesting a widely shared perception of the pay hierarchy among large German firms. Taken together, these findings indicate that many workers have information about firm-specific pay. Workers' responses to general questions about information and workers' pay expectations at the firms they provide confirm this finding generalizes to a larger set of firms.[2]

[2] We emphasize the researcher-provided firms in this part of the analysis as they allow us to confirm that workers' beliefs are firm-specific and because the random assignment of firms ensures our estimates of perceived wage premia

Having established that many workers possess accurate, firm-specific priors, we then examine whether they actually use this information to direct their search toward higher-paying employers. Our primary firm-level design leverages within-worker across-firm and within-firm across worker deviations in expectations, controlling for both across-worker differences in general knowledge about firm pay or willingness to move and across-firm differences in non-wage amenities. A 10% increase in expected pay is associated with a 3 percentage point increase in the probability a worker will consider applying to a firm. The average application-wage elasticity is 1. Two complementary designs—which use different portions of the questionnaire—confirm that high-pay firms (both in workers' perceptions and in the administrative data) receive more overall interest. These findings suggest that workers use prior beliefs about firm pay to guide their search and view higher-pay firms as more attractive overall.

While perceptions of pay influence workers' search behavior, perceptions of offer probabilities have a limited effect. The application-wage elasticity is somewhat lower for workers who are more risk averse, for workers who consider the amount of competition they will face when they decide where to apply, for workers who report hesitancy to apply because they perceive a low probability of receiving an offer, or for workers who believe that high wage firms receive more qualified applicants per vacancy. However, the gaps are not statistically or economically distinct. This suggests that perceptions of differences in offer probabilities are less important than perceptions of differences in pay in explaining workers' search behavior.[3]

To separate preferences from beliefs and search frictions, the final part of the paper employs firm-specific discrete choice experiments that remove uncertainty about pay and offer feasibility. These experiments mirror those commonly used to estimate the value of specific work-place amenities and to estimate market labor supply; we use the discrete choice approach because it has been shown to produce estimates of these parameters which are similar to those from revealed preference experiments (see, e.g., Wiswall and Zafar, 2018; Mas and Pallais, 2017, 2019). All of the discrete choice experiments remove uncertainty about pay and uncertainty about whether a worker will receive an offer; some further remove uncertainty about pay growth.

Many workers are unwilling to switch firms—even to firms they explicitly identified as application targets—even for sizable wage increases. Thirty-nine percent of workers would prefer to remain in their current job over a 20% raise at a firm they indicate. The implied separation elasticity is between 2 and 4. When we specify that a worker's commute would not change (removing

are unbiased. We use the worker-provided firm module, as well as a set of firm agnostic questions, to show that our core findings generalize.

[3]This is consistent with results from the LinkedIn experiment in Gee (2019); while applicants were more likely to apply to positions if they knew the number of previous applicants, increased competition was not a deterrent to either male or female jobseekers.

uncertainty about travel location and time), we obtain similar estimates, suggesting that the incumbent premium does not simply reflect the costs of physically moving locations (Kennan and Walker, 2011; Koşar, Ransom and Van der Klaauw, 2022). We also obtain similar estimates when we specify that a worker's pattern of promotions and wage growth would remain the same as at their incumbent firm, suggesting that the incumbent premium is not driven by beliefs or uncertainty regarding job stability or other dynamic considerations.

Consistent with the model, workers' own-firm preference affects their search behavior. While workers' search behavior is strongly related to their perceptions of outside pay (a market elasticity of 7), workers with a stronger incumbent preference are systematically less likely to initiate search, even when randomized outside pay is high. This pattern aligns with the model's prediction that preferences and beliefs enter the search decision as substitutes: workers who anticipate rejecting many outside offers rationally choose not to search.

We use a simple calibration to compare the relative importance of biased beliefs and preferences and find that preferences better explain why workers remain at their firms. In our baseline scenario, providing low-wage workers information about the wage increase they would see if they worked at the median wage premium firm would increase mobility in this population by only 7 percentage points (assuming that all workers can receive offers at the specified wage). By contrast, removing the own firm preference would increase mobility by 24 percentage points. Even in our most conservative scenario—which assumes that workers do not have firm-specific knowledge of pay, that workers have next to no information on outside pay, that a worker can receive an offer from an outside employer which offers the median pay premium, and that the worker considers this firm to be as attractive on non-pay dimensions as firms in their consideration set—we find that correcting beliefs and changing preferences would have similar effects on mobility. Our finding that preferences better explain limited mobility than information frictions regarding pay generalizes to the full population of workers, and is robust to a variety of alternative weighting schemes.

We conclude our analysis by using the discrete choice experiments to examine the non-wage determinants of worker attachment. We find that workers do not prefer to stay at their firm because they believe higher pay is offset by worse amenities. Amenity valuations are of similar magnitude to—and are positively correlated with—firm-specific expected pay premia. When asked to consider firms in general, a large share of workers say they expect higher-pay firms to offer better amenities (29%), and most (71%) believe that higher-pay firms offer at least as good amenities as lower-pay firms in the same market.

However, there is evidence of sorting: workers who say they would apply to a firm value it more than those who would not apply. Firm insiders (incumbent workers) differ in valuations from both outsiders who are uninterested in the firm and outsiders who are interested in the firm. That firm insiders differ from those who say they would apply to the firm is hard to rationalize with models of the labor market in which the utility a worker receives from working at a firm (other than a cost common across firms) is independent of whether a worker already works at that firm. One possibility is that ex post valuations of amenities differ because workers develop firm-specific human capital or social ties (e.g. with coworkers or managers). Unlike pay, this match-specific capital is an experience good that cannot be easily communicated to outsiders; this creates a barrier to mobility distinct from simple pay-related information frictions. Social ties are the most cited reason workers believe others are reluctant to leave their firms.

Our results contribute to four distinct literatures. First, our approach was inspired by, and modeled after, an earlier ethnographic literature that used worker surveys to shed light on the labor market (Myers and Shultz, 1951; Reynolds, 1951; Rees and Shultz, 1970). We adapted many of our survey questions—such as those eliciting the names of specific outside firms—from this literature. We leverage the modern survey infrastructure of the IAB to field our survey to workers across multiple distinct labor markets, to link workers' responses to administrative measures, and to perform a series of validations which support this approach.

Second, we contribute to a large literature documenting heterogeneity across firms in pay and amenities (e.g., Rosen, 1986; Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013; Song et al., 2019; Sorkin, 2018). Our approach is fundamentally distinct in focusing on workers' beliefs. However, this literature shaped both our design and our empirical strategy. Our finding that workers believe firms differ in pay premia—and that these beliefs are correlated with administrative wage data—helps rationalize the high degree of sorting documented in prior work (e.g., Card, Heining and Kline, 2013; Song et al., 2019).

In focusing on workers' perceptions of the labor market, this paper contributes to a growing literature on workers' information and beliefs. This literature documents mixed effects of providing workers with pay information (e.g., Caldwell and Harmon, 2019; Jäger et al., 2024; Cullen, 2023). Previous work has shown that providing information on, e.g., what their coworkers make, can improve their outcomes (Card et al., 2012). However, other papers have shown that blanket policies of information provision do not always benefit workers (Cullen and Pakzad-Hurson, 2023).[4] Relative to this literature, our contribution lies in documenting what workers know about specific outside

[4]There is mixed evidence on whether workers have biased beliefs about their outside options. Jäger et al. (2024) use a module embedded in the German-Socioeconomic Panel to document that a large share of workers anchor beliefs about the pay changes associated with transitions to their current wage. Appendix C describes how our elicitations—which isolate beliefs about pay separate from job arrival rates—differ from those in previous studies. This appendix also documents that workers at low-wage firms receive fewer job offers than comparable workers at high-wage firms and discusses how survey-based anchoring and question order effects could have influenced previous studies.

firms. To the best of our knowledge, we are the first to document that a typical worker believes in firm-specific wage premia, that there is agreement in the wage ranking of large employers, and that workers generally (for a typical firm) perceive high wage firms to offer better amenities.

Finally, this paper speaks to the theoretical literatures on imperfect competition and job search. Models of monopsony often explain upward-sloping labor supply curves using either information frictions or preference heterogeneity (see surveys in Azar and Marinescu, 2024; Caldwell, Dube and Naidu, 2025; Kline, 2025). In these models, information frictions typically center on pay. Our results suggest that models emphasizing preferences may better capture worker behavior. Our findings that workers use firm-specific information about pay to guide search and associate higher pay with higher overall utility also support central assumptions of directed search (Wright et al., 2021). Our results suggest that lack of information about outside pay is not the central friction preventing mobility, echoing previous work by Fox (2010) on Swedish engineers, as well as early work by Reynolds (1951).

The rest of the paper proceeds as follows. Section 2 presents a simple partial equilibrium model that motivates our empirical analysis. Section 3 describes the survey design and data linkages. Section 4 examines workers' firm-specific pay expectations. Section 5 analyzes how these beliefs affect search at the firm and market level. Section 6 uses a series of discrete choice experiments to estimate incumbent premia and to compare the relative importance of preferences and beliefs in explaining workers' limited mobility. Section 7 examines the mechanisms behind workers' firm-specific attachment. Section 8 concludes.

2 Empirical Framework

We present a simple partial equilibrium search model that motivates the empirical analysis in Sections 4-6.[5] Appendix D provides details and proofs.

2.1 Setup

Consider a labor market populated by a unit mass of infinitely lived workers indexed by i who can work at any of J firms. Workers derive utility from pay and non-pay amenities provided by firms. Worker i's flow utility from employment at firm k is:

[5] A natural extension would embed this setup in a general equilibrium framework. In equilibrium, firms' wages and hiring selectivity would adjust endogenously to worker behavior and allowing workers' beliefs to update dynamically in response to these adjustments. We do not present this version because our empirical focus is on how workers' existing perceptions shape their search decisions, rather than on equilibrium wage determination.

$$u_{ik} = \beta w_{ik} + a_k + \epsilon_{ik}, \tag{1}$$

where $\beta > 0$ is the marginal utility of log pay, w_{ik} is the log pay of worker i at firm k, a_k is the total value of non-pay amenities provided by firm k, and ϵ_{ik} is an idiosyncratic preference shock drawn independently and identically across workers and firms from a type 1 extreme value distribution. The non-pay amenities may encompass a variety of factors, such as a firm's corporate culture, childcare policies, or health insurance policies. We denote worker i's employer in period t by $j(i, t)$. If a worker is not employed, $j(i, t) = 0$ and they receive flow utility $u_{i0} = a_0 + \epsilon_{i0}$ where a_0 is the value of non-employment.

Pay Beliefs. Workers hold beliefs $\tilde{w}_{ik}$ about pay at potential outside employers, which may differ from actual pay, w_{ik}. We assume that beliefs are given by:

$$\tilde{w}_{ik} = w_{ik} + b_i + \eta_{ik}, \quad \mathbb{E}[\eta_{ik}|b_i, w_{ik}] = 0.$$

In this formulation, b_i captures worker i's systematic optimism (if $b_i > 0$) or pessimism (if $b_i < 0$) relative to true pay and η_{ik} allows beliefs to vary across firms arbitrarily. Workers form beliefs based on all available information, including networks, online platforms, or past applications.[6] Worker i learns the true pay at firm j after receiving an offer.

Dynamics, Search, and Mobility. In period t worker i receives their flow utility. They then decide on a set of firms $A \subseteq \mathcal{J} \backslash j(i, t)$ to apply to where $\mathcal{J}$ is the set of all firms. They then pay an application cost κ for each application, regardless of whether this application is successful: if they apply to $|A|$ firms, they pay $\kappa|A|$. The probability of receiving an offer from firm k is p_k.[7]

Worker i then simultaneously receives job offers and learns whether their current match will be exogenously destroyed (which occurs with probability λ). After learning about their offer set, worker i chooses the best offer in their option set. If their match was destroyed, this option set includes non-employment and any firms which made them an offer. Otherwise it also includes their current firm. If the worker moves to a new firm, they pay a one-time cost s_i.[8] We refer to s_i as a switching cost, noting that it can encompass a broad set of frictions that reduce the attractiveness of

[6] We remain agnostic about the belief formation process as we focus on elicited beliefs rather than learning.

[7] This is firm-specific but worker-invariant. This implies that all workers face the same probability of success when they apply to firm k. Allowing p_k to vary across workers complicates the notation and does not change the predictions. In Section 5 we find that perceptions of offer probabilities do not meaningfully affect workers' behavior.

[8] In the empirical analysis, we estimate the annuitized flow equivalent of the 'Incumbent Premium.' This represents the permanent percentage increase in wages required to offset the utility loss of switching. This includes the structural switching cost s_i as well as the gap in non-wage values between the inside and outside firms.

going to a new employer, including uncertainty about new environments, disruption to established commuting patterns, and the cost of developing social ties with managers and coworkers.

Search is forward-looking and workers discount the future at rate $\delta \in (0,1)$. Worker i applies to outside firms if the benefit of doing so exceeds the cost. We use A to denote the set of firms a worker applies to and $B(A)$ to represent the set of firms which make an offer to individual i (a subset of A). Because there is uncertainty about what firms will offer the worker employment, workers make decisions on the basis of expected utility, accounting for uncertainty in whether an application will be successful. Worker i's perceived value of employment at firm k is:

$$\begin{aligned} \tilde{V}_{ik} &= \underbrace{\beta \tilde{w}_{ik} + a_k + \epsilon_{ik}}_{\tilde{u}_{ik}} \\ &\quad + \max_{A \subseteq \mathcal{J} \setminus \{k\}} \Bigg\{ - \kappa\,|A| + \ \delta\,\mathbb{E}_{B(A)} \Big[(1-\lambda)\ \max \Big\{ \overbrace{\tilde{V}_{ik}}^{\text{stay}}, \overbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i \big)}^{\text{switch employers}} \Big\} \\ &\quad + \lambda \max \ \Big\{ \underbrace{V_{i0}}_{\text{unemployed}}, \underbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i \big)}_{\text{switch employers}} \Big\} \Big] \Bigg\}. \end{aligned} \tag{2}$$

where we use $\tilde{V}_{ik}$ to indicate that the value function depends on the vector of perceived wages $\widetilde{\mathbf{w}}_i = (\widetilde{w}_{i1}, \ldots, \widetilde{w}_{iJ})$, rather than the vector of actual wages. We define V_{ik} similarly, except the true wage at firm k is known.

For worker i, employed at firm j, the expected utility of not searching (not applying to any firms) is

$$V_{ij}^{\text{no search}} = u_{ij} + \delta \left[\lambda V_{i0} + (1-\lambda) V_{ij} \right],$$

and the expected utility of applying to a non-empty set of firms, A, is

$$V_{ij}^{\text{search}}\big(A\big) = u_{ij} - \kappa |A| + \delta \mathbb{E} \left[\tilde{V}_i | A \right],$$

$$\begin{aligned} \text{where } \mathbb{E}\big[\tilde{V}_i|A\big] &= \mathbb{E}_{B(A)}[(1-\lambda)\ \max \Big\{ \underbrace{V_{ij}}_{\text{stay}}, \underbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i \big)}_{\text{switch}} \Big\} \\ &\quad + \lambda\ \max \Big\{ \underbrace{V_{i0}}_{\text{unemployed}}, \underbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i \big)}_{\text{switch}} \Big\}]. \end{aligned}$$

Definition of Directed Search. Formally, we say that search is directed if there exists at least one pair of firms (k, l) with $\tilde{w}_{ik} \neq \tilde{w}_{il}$ such that altering these beliefs, while holding aggregate distributions constant, changes the worker's applications. This behavior requires that: (1) workers believe their own pay would vary across firms ($\tilde{w}_{ik} \neq \tilde{w}_{il}$ for some $k \neq l$) and (2) workers can link their beliefs $\tilde{w}_{ik}$ and the values of amenities a_k to the identity of the firm k.

2.2 Firm-Specific Knowledge of Pay and Search

The model yields several predictions about the relationship between workers' beliefs and switching costs and their search and mobility. Formal proofs are in Appendix D.

1. The Probability of Search is Increasing in a Worker's Perceived Outside Options and Decreasing in $\mathbf{s_i}$. Worker i at firm j searches in period t if there exists at least one firm for which the expected gain exceeds the cost:

$$\max_k \left\{ \delta p_k[(1-\lambda)\max(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda \max(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \right\} > \kappa \tag{3}$$

The left side of this expression is increasing in perceived outside options $(\tilde{V}_{ik})$ and decreasing in the switching cost s_i. Perceived outside options depend on the wages other firms offer ($\mathbf{w}_i$), on worker optimism about outside pay (where $b_i > 0$ increases search), and on firm-specific bias η_{ik}. Workers may choose not to search even when they believe other firms offer better wages and amenities if they have a large switching cost s_i.

2. Mistaken Non-Search is Less Likely With Firm-Specific Pay Beliefs. We say that there is mistaken non-search whenever a worker does not search, but would if they were to base their decision on actual wages $\mathbf{w}_i = (w_{i1}, ..., w_{iJ})$ rather than perceived wages $\widetilde{\mathbf{w}}_i = (\widetilde{w}_{i1}, \ldots, \widetilde{w}_{iJ})$.

Consider two hypothetical scenarios in which workers believe firms vary in pay. In the first scenario, the worker can map these beliefs to firm identities (directed search). They search if there is at least one firm for which the value of applying to that firm exceeds the cost (equation 3). In the second scenario, the worker cannot map their beliefs to firm identities (non-directed search). They search if the expected benefit of a random application exceeds the cost:

$$\delta \frac{1}{J} \sum_k p_k[(1-\lambda)\max(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda \max(\tilde{V}_{ik} - s_i - V_{i0}, 0)] > \kappa. \tag{4}$$

Targeting the firm with the maximum expected gain (the left side of equation 3) yields an expected value at least as high as applying randomly (the left side of equation 4) because the worker can avoid firms with low or negative expected gains. Therefore, mistaken non-search is less common if workers have firm-specific pay beliefs. As a reminder, for the incumbent $\tilde{V}_{ij} = V_{ij}$ as we assume workers know their current wage.

3. Directed Searchers Apply More to Firms with Higher Perceived Wages. Third, if workers direct their search, the probability that worker i applies to firm k is an increasing function of their perceived wage at that firm $\tilde{w}_{ik}$. By contrast, if workers do not perceive wage variation across firms—or if they cannot link a given wage to a specific firm—the probability worker i applies to firm k is uncorrelated with its wage.

For simplicity, suppose worker i considers whether to apply to firm k, taking the rest of their applications as given. Their decision depends on whether the gains exceeds the cost:

$$\text{Apply}_{ik} := 1\{p_k \cdot \tilde{V}_{ik} > \kappa_i\},$$

where κ_i is an individual-specific threshold which depends on the per-firm application cost κ, the worker's switching cost s_i, the worker's current employer and set of applications, the probability of job destruction, and the discount factor. Each is independent of $\tilde{w}_{ik}$. Treating the worker's application threshold κ_i as a random variable with differentiable distribution function F, we can examine how the probability the worker i applies to firm k responds to changes in perceived wages:

$$\frac{d\,P(\text{Apply}_{ik}=1)}{d\,\tilde{w}_{ik}} = f_{\kappa_i}(p_k\tilde{V}_{ik})\left(\underbrace{\beta p_k}_{\text{direct utility gain}} + \underbrace{p_k\frac{\partial\,a_k}{\partial\,\tilde{w}_{ik}}}_{\text{amenities}} + \underbrace{\tilde{V}_{ik}\frac{\partial\,p_k}{\partial\,\tilde{w}_{ik}}}_{\text{selectivity}} + \underbrace{p_k\frac{\partial\,U_k(A^*)}{\partial\,\tilde{w}_{ik}}}_{\text{gains from optimal applications}}\right) \tag{5}$$

There are four channels through which applications respond to $\tilde{w}_{ik}$: (1) the direct utility gain from higher pay, (2) the relationship between wages and amenities, (3) the relationship between wages and offer probabilities, and (4) the gains associated with changes to future application behavior.[9]

An immediate implication of prediction 3 is that, because only workers who direct their search respond to a particular firm's wage, the responsiveness of a firm's applications to changes in its own wage is increasing in the share of directed searchers. In many models, firms' markdowns are decreasing (wages are increasing) in the elasticity of applications to wages (Manning, 2003;

[9]After moving to a higher wage firm a worker may apply less because the gains to moving are lower. In Appendix D, we show a similar expression arises when workers can construct application portfolios. Even then, an increase in $\tilde{w}_{ik}$ raises the expected payoff of any set of applications that includes firm k. This preserves the core intuition: higher perceived wages lead to a higher probability of applying to k, all else equal.

Dal Bó, Finan and Rossi, 2013).

4. The Probability of a Job Switch is Increasing in a Worker's Perceived Outside Options and Decreasing in s_i. Finally, incumbent workers' decisions to search and move are shaped both by their beliefs about outside option pay, and by their costs of switching employers. While workers learn the actual wage of an outside firm before they accept an offer from that firm, a worker can only switch jobs if they receive an offer. Workers switch jobs when they (1) apply to another job, (2) receive an offer, and (3) decide that offer is good enough to overcome the switching cost.

A worker's probability of searching is increasing in their perceived outside options and decreasing in s_i (prediction 1). Conditional on their applications, the probability worker i receives an offer does not depend on either their beliefs or s_i. Given that worker i has an offer from outside firm k, they prefer it to the option of remaining at their incumbent firm if:

$$V_{ik} - V_{ij} > s_i \tag{6}$$

where we use V_{ik} rather than $\tilde{V}_{ik}$ as we assume that a worker learns a firm's wages upon receiving an offer. All else equal, a worker is less likely to accept the outside offer if their switching costs are higher and more likely to accept if the firm offers them a higher wage.

2.3 Identifying the Barriers to Mobility

We use these predictions to guide our analysis of workers' beliefs and the relationship between these beliefs—and switching costs—and workers' search and mobility. Section 4 examines whether workers have firm-specific beliefs about pay (a necessary condition for prediction 2). Section 5 shows that workers' market search relates to their outside options (prediction 1) and examines whether workers direct their search to high wage firms (prediction 3). This Section also examines whether beliefs about offer probabilities meaningfully affect workers' search behavior at the firm or market level.

Our model indicates that both beliefs and preferences affect observed search and mobility. Section 6 uses a series of discrete choice experiments in which we asked workers to rank take-it-or-leave-it offers from specific outside firms with randomized pay to isolate the role of preferences. These experiments isolate the role of preferences by removing the role of beliefs about pay, and the role of beliefs about offer probabilities.[10] We show that workers who anticipate staying at their firm

[10] As detailed in Section 3, our primary discrete choice experiments ask workers to choose between offers at named firms with specified, randomized wage increases. In a more general model in which wages grow over time and firms also differ in growth rates, workers' preferences over the outside firm and the incumbent depend on an additional

are less likely to engage in search (prediction 1). We then use the model to guide an examination of the relative importance of preferences and beliefs for search.

Section 7 asks whether workers' preference to remain at the incumbent employer varies across firms in ways that can be predicted ex ante. We allow for the possibility that workers' valuations of non-wage attributes differ once they have joined a firm, e.g., through the accumulation of match-specific capital or social ties with managers or coworkers, generating firm-specific attachment. This is not incorporated in standard models which assume that, even if firms vary in the utility they provide, this variation exists before a match is formed (e.g., Moen, 1997).

Throughout our empirical tests, we leverage our ability to directly measure workers' beliefs about outside options, rather than inferring them from observed behavior, and our ability to isolate preferences from beliefs about pay. Our approach allows us to distinguish between: (1) information frictions relating to pay, where workers hold inaccurate beliefs about the pay provided by outside firms; (2) preferences, including both switching costs and firm-specific attachment. Switching costs and firm-specific attachment provide mechanisms through which workers, who are informed and actively consider outside opportunities, may nevertheless remain with their current employer.

3 Data and Survey Design

We designed a survey to elicit workers' information about pay, search behavior, and preferences. We fielded the survey through the German Institute for Employment Research (IAB), a group within the German Employment Agency with a statutory mandate to study the labor market. We linked workers' responses to their Social Security records. We also developed a linkage between workers' responses about specific firms and public and administrative data on those firms.

3.1 Survey Implementation and Descriptive Statistics

We fielded the initial wave of the survey to 134,995 full-time German workers in the fall of 2022. We selected workers for inclusion in the survey from the set of full-time employed workers between the ages of 25 and 50 who had been at their establishment for fewer than eight years. We oversampled workers from the firms included in the firm survey conducted by Caldwell, Haegele and Heining (2024). This design allows us to produce estimates that are representative of the full-time German workforce (after re-weighting), but also gives us the ability to estimate firm-specific

term: the discounted difference in growth rates. In most of the paper, we follow the literature in considering firm-specific amenity values as a bundle, without decomposing them into their constituent parts. However, in some of our analysis, we leverage a set of additional discrete choice experiments we conducted in which we told workers that future promotions and wage trajectories would not differ from their incumbent firm.

valuations among insiders (those currently employed at a firm) for a large set of firms. We weight responses according to this explicit dimension of over-sampling; this is considered a best practice in the survey literature (Solon, Haider and Wooldridge, 2015).

We designed the survey so that it was similar to other worker surveys fielded by the IAB. We sent invitations to the survey—which were signed by the director of the IAB—by mail; respondents completed the survey online. We received 13,680 total responses, for an effective response rate of 11.4%. Our response rate compares favorably to other IAB surveys targeting first-time respondents (Haas et al., 2021) and to other recent surveys that invited participants electronically (Caplin et al., 2023). The high completion rate (74%) and moderate median response time (9 minutes) suggest that survey fatigue and limited attention are not major concerns.[11] Throughout the paper, we pool data from the initial survey wave with data from a follow-up survey we conducted in the spring of 2024. The follow-up response rate (51%) is similar to those of other panel surveys (Haas et al., 2021). In both the initial and follow-up surveys, we randomized incentives to respond. The appendix to this paper and to Caldwell, Haegele and Heining (2024) provide more information on response rates and selection; we follow the approach in Caldwell, Haegele and Heining (2024) and show our results are robust to a variety of re-weighting methods.

The key innovation in our survey design is that we asked workers questions about specific firms, and linked these responses to public and administrative data on those firms. While these questions form the basis of most of our analysis, we also collected additional characteristics not observed in the Social Security records—such as collective bargaining coverage and hours worked—as well as general information on search.[12]

Table 1 describes the 9,756 employed workers who completed the initial wave of the survey and provided their consent to link their responses to the administrative records. The average individual in our sample is around 30 years of age, and 53% are male. Most are German citizens, and roughly 60% have a college degree. The majority of respondents have been at their jobs for more than two years. The average daily pay is roughly 136 euros, and half of workers are covered by a collective bargaining agreement (CBA).[13] Because we only invited full-time workers to participate,

[11] We define a response as complete if a respondent clicked through to the second to last question, which elicited consent for participating in a follow-up survey. We do not require respondents to have answered every question. Because the survey was advertised as a study of salary progression in Germany, it is unlikely that workers who felt uninformed about outside market conditions chose not to start the survey. Our procedure followed typical processes at the IAB; previous worker surveys fielded by the IAB have been used by external researchers through research data centers.

[12] The survey also contained modules on workers' bargaining behavior (Caldwell, Haegele and Heining, 2024). The questionnaires and information on field logistics can be found at https://sydneec.github.io/Website/CHH_Survey_Appendix.pdf.

[13] Pay is top-coded at the Social Security maximum. We impute the upper tail of the wage distribution. To calculate daily pay, we divide the imputed total earnings by spell duration.

our population earns more than the average German worker. However, our final sample includes thousands of workers at the lowest-paying firms. Appendix B.1 confirms our results are robust to weighting our sample to match the overall population of full-time workers in Germany. We also present a variety of heterogeneity exercises which show our results hold for both high-wage and low-wage workers.

3.2 Firm-Specific Questions

We designed two survey modules to elicit firm-specific information from workers. The *researcher-provided firm module* posed a series of questions about randomly drawn subsets of 30 large German employers. This approach ensures that the identity of the firms about which a given worker answers questions is not endogenous to the worker. Appendix Tables E3 and E4 confirm that the randomization was successful.

The *worker-provided firm module* first asked workers to provide the names of three firms they would consider applying to. It then posed a series of follow-up questions, analogous to those in the researcher-provided module, about these firms. The worker-provided firm module allows us to focus on firms that—by construction—are most relevant for a given worker. We were initially concerned that workers would provide the names of firms which they would like to work at, but which were out of reach. However, workers provided a large number (more than 2,800 unique) of firms which fall throughout the pay distribution (Appendix Figure A2) and which workers from their firm have historically moved to (Appendix Figure F1). Appendix E.1.5 describes these firms.

Much of the analysis presented in the main text relies on the responses in the researcher-provided firm module. To avoid priming workers with the names of specific firms, we asked workers to complete the worker-provided firm module first. Both modules unfolded similarly, as outlined below and depicted in Figure 1.[14]

Search. We first asked workers where they would consider applying if they wanted to switch firms. In the researcher-provided firm module, we elicited this information by providing workers with seven potential firms and asking them:

> "Suppose you planned to move to a new company in the next {one/three/six} months.
> Would you consider applying to any of these firms? Please select all that apply."

[14]We included the researcher-provided firm module in both the initial and follow-up surveys. However, in the follow-up survey, we did not present the worker-provided firm module to workers who completed this module in the initial survey. To avoid contamination from potential learning effects, we exclude worker-provided firm responses from the follow-up wave.

In the worker-provided firm module, we asked workers to name three firms they would consider applying to. We randomized the listed time frame. We asked about applications conditional on wanting to move so that our question would be applicable to workers without immediate job search plans. The measure we elicit is similar to the measures of interest used in many empirical studies of directed search but has the added advantage of not requiring firms to have open vacancies or to recruit via online job boards.

Pay. We then asked workers what they believed they would earn at each of three firms:

> "What do you think your gross annual pay would be if you worked at these companies in a position similar to your most recent position?"

In the researcher-provided firm module, the three firms were chosen at random from the set of seven firms included in the search question. In the worker-provided firm module, the three firms were those previously provided by the worker. To reduce the burden of the survey, we only posed this and the following question on preferences to a random 50% of workers in the researcher-provided firm module. We posed these questions to all workers in the worker-provided firm module.[15]

The goal of this question was to elicit information on what workers thought they would make at outside firms, not what it would take to get them to move to these firms. We focused on annual pay because our sample consists of full-time workers. We told workers to assume they had the same position at each firm to ensure within-worker comparability of pay across firms. We process the raw expectations data by winsorizing workers' pay expectations at the 90% level. This processing may, if anything, lead us to slightly understate the variation in workers' expectations (and bias us against our finding that workers have firm-specific information). Appendix Section E.1.6 provides more details on the data processing and describes robustness checks which confirm our results are not driven by this choice.

Preferences. Finally, we asked workers to rank hypothetical job offers from the same three firms under researcher-randomized raises. We told workers:

> "Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take."

[15] Given that the median respondent completed the survey—which included several background questions, a module on bargaining, and both worker-provided and researcher-provided firm modules—in just 9 minutes, it is unlikely that respondents had enough time to look up the listed firms. Further, we do not see bunching in the provided salary expectations for each firm; bunching would be consistent with workers looking up this information.

We provided three hypothetical job offers with raises that ranged from 5% to 20% in the initial survey, and -5% to 15% in the follow-up survey. We chose 5% increments to simplify comparisons. We asked workers to rank these options relative to each other and to the option of remaining at their current firm at a specified raise. We specified the incumbent firm's pay to avoid the concern that some workers may anticipate the incumbent firm would renegotiate.

We use these discrete choice experiments (DCEs) to isolate workers' preferences. We chose to elicit workers' preferences using DCEs—rather than eliciting the pay changes that would lead a worker to leave their current firm or to join specific outside firms—because take-it-or-leave-it experiments are a more reliable way to measure preferences than contingent valuation approaches (Hausman, 2012; Mas, 2025). For instance, recent work by Rodemeier (Forthcoming) found stated willingness to pay estimates that were 1388% larger than experimental estimates from the same population. DCEs are commonly used to estimate the value of workplace amenities (Wiswall and Zafar, 2018; Mas, 2025) because they yield estimates consistent with actual job market behavior (Mas and Pallais, 2017, 2019).

3.3 Researcher-Provided Firms

A goal of our study was to shed light on workers' perceptions of outside firms and the linkage between these perceptions and their search and mobility. Workers provided firms for the worker-provided firm module; we then asked them questions about those firms for the rest of the module. A key challenge in designing the survey was selecting the set of researcher-provided firms, which needed to meet five distinct criteria.[16]

First, for respondents' answers to be meaningful, respondents needed to be at least somewhat familiar with the provided firms. Second, for it to be reasonable to ask workers for their expected pay at each of the outside firms (conditional on holding the same position they currently hold), the included firms needed to employ workers in a wide variety of occupations. Further, for workers to find the firms reasonable, they could not be overly associated with a single occupation. Third, because we estimate firm-specific pay effects and valuations, we needed to focus on a relatively small number of firms to guarantee precision. Fourth, to ensure our findings were broadly relevant, we needed to select firms that were important in the German labor market. Lastly, to ensure relevance we also needed to include firms that represented realistic employment options for workers, rather than aspirational "dream" firms that were clearly desirable but out of reach to most respondents.

Based on these criteria, we selected 30 researcher-provided firms from a pool of publicly traded

[16] For privacy concerns, we cannot list the names of these firms.

and family-owned firms in Germany.[17] We included 20 firms in the initial wave of the survey and all 30 firms in the follow-up wave. We focused on private sector firms, rather than including public sector employers, since our sampling frame consists of workers in Social Security-covered employment, which excludes much of the public sector in Germany. Selecting from the sets of publicly traded and family-owned firms allows us to analyze a range of well-known firms that employ a large share of the German work force, but which differ in a range of characteristics, including pay. Appendix E.1 provides more details on the selection criteria.

The final set of 30 researcher-provided firms are important employers in Germany and are relevant for our respondents. Together, they employ over 1.8 million German workers and have received more than 39.1 million page views on Germany's leading employer rating platform, Kununu. They capture six major industries in Germany and vary in characteristics such as firm age and location (Table 2; Appendix Table E5). Nearly one quarter of respondents have worked at least at one of the 30 firms in the past ten years and 47% of the researcher-provided firms were among the top 100 firms named by respondents. The fact that the firms have similar median pay and pay premia as the firms respondents currently work at (Panel A of Appendix Figure A2) weighs against the concern that these firms may be out of reach.

3.4 Validity of Survey Approach

While there is a long history in labor economics of using surveys to study workers' information and mobility (see e.g., Reynolds, 1951), a natural concern with this methodology is that respondents may provide inaccurate information or may provide responses which do not align with how they would behave in real-world scenarios. We validate our data using four complementary strategies: comparison with external surveys, comparison with administrative records, assessment of internal consistency, and adherence to experimental design best practices.

Accuracy of Responses. Respondents may not accurately describe their beliefs if they lie, are inattentive, or rely on external resources. We find strong evidence against these concerns. First, in previous work on bargaining, we report that workers' responses to a set of questions on bargaining behavior (including potentially sensitive questions about the use of outside offers to negotiate) align with the responses of HR executives at their firm (elicited in a separate survey fielded a

[17] An alternative approach would have been to select firms on the basis of firm-to-firm flows or networks, but this was infeasible. Although we could link survey responses to the administrative data by linking firm names we provided to establishment identifiers, we were not permitted to retrieve firm names from the establishment identifiers.

year apart by a different research partner) (Caldwell, Haegele and Heining, 2024).[18] After we fielded the survey, the IAB received e-mails from respondents who were interested in receiving a summary of our findings, which suggests they expected others to provide truthful and accurate responses. Appendix F presents a number of validity checks which confirm there is cross-question consistency in responses.

To accurately characterize beliefs, we avoided incentivizing "correct" responses, removing the motive for respondents to look up information. Ex post, the mean survey response time of 9 minutes and the absence of bunching in reported salaries indicate that respondents provided their immediate priors and did not simply report results from external research. Additional robustness checks discussed in Section 5 further support the reliability of these responses.

Relevance of Consideration Sets. We also validate that workers provided realistic, rather than aspirational, outside options. Appendix F shows that the firms workers listed in the worker-provided firm module are empirically the same firms that coworkers from the respondent's current firm have historically moved to. This confirms that respondents are describing their actual consideration sets. We also included a variety of firm-agnostic questions (e.g. search probabilities in response to randomized outside market pay) in the survey, with which we validate many of our core results.

Revealed vs. Stated Preferences. Both our data on search and on mobility come from workers' stated preferences, rather than revealed choices. For the analysis in Section 5—which examines how workers search across firms–this is a necessary limitation. For workers who are not actively searching for outside employment or who are not interested in switching firms, it is not clear how to otherwise obtain data on their search behavior. In Section 6 we present specifications which relate overall search probabilities to randomized changes in outside pay; this is consistent with what has been done elsewhere in the literature.[19]

Our analysis of mobility relies on stated preferences elicited via discrete choice experiments. This approach offers a distinct identification advantage over revealed preference data: in observational data, immobility may reflect either a preference for the current firm or the failure to receive an outside offer (feasibility). By contrast, our experimental design presents workers with hypothetical "in-hand" offers. This allows us to structurally isolate preferences from search frictions

[18]This is a joint test that respondents in both surveys are attentive and truthful; the analysis in this paper only requires respondents in the worker survey to be attentive and truthful.

[19]For instance, the experimental evidence presented in the main text of Jäger et al. (2024) comes from examining how the stated search intention of workers (recruited via online platforms) responds to gaps between provided information on outside pay and workers' just-elicited beliefs about outside pay.

(such as the probability of receiving an offer or the fear of rejection). While revealed preference data conflate the desire to move with the ability to find a job, our design identifies the separation elasticity conditional on feasibility.

We used discrete choice experiments rather than open-ended contingent valuation (e.g., asking workers for firm-specific reservation raises) because recent literature demonstrates that discrete choice experiments yield non-wage valuations and labor supply elasticities comparable to revealed preference approaches. This suggests the approach is well-suited for our setting. By contrast, stated valuation approaches can yield valuations that diverge from experimental estimates by orders of magnitude (Rodemeier, Forthcoming; Mas, 2025). We further validate our approach in Appendix F.[20]

3.5 Linkages to Firm-Level Datasets

We linked workers' firm-specific responses to firm-level information. Most of this information comes from administrative data collected by the IAB. Following IAB linkage procedures, we linked the researcher-provided and the worker-provided firms to the establishment panel (BHP) and to the employment records of workers at these firms (IEB). We constructed employment-weighted averages to create firm-level characteristics from the establishment-level data. Some specifications include controls for the distance between a worker's current workplace and the indicated firm. Our baseline measure is the driving distance between a worker's current workplace and the 7-digit municipality of a firm's headquarters. In robustness checks (See Appendix Tables A3 and A6), we use an alternative measure of distance based on the distance between a worker's current workplace and the closest establishment of the outside firm.

We supplement IAB data with Orbis and hand-collected data on firm size, age, and worker ratings. Orbis provides us with firm age and financial characteristics. Most of our hand-collected data come from an employer review platform, Kununu, and from a variety of firm rankings, which we detail in Appendix Section E. We also hand-collected the number of employees in Germany.

4 Workers' Firm-Specific Pay Expectations

We begin our empirical analysis by presenting several novel facts about workers' firm-specific pay expectations, guided by the model in Section 2. We show that a large share of workers say they have firm-specific knowledge of pay and that their firm-specific knowledge is positively correlated

[20] Because we found that most workers are inframarginal to their firm, it is not surprising that—in unreported results—we saw no impact of the survey on future mobility.

with objective predictions. We find similar results based on general questions about pay, questions about researcher-provided firms (which allow for fixed comparisons across workers) and questions about firms workers indicate are their outside options.

4.1 Workers Believe Firms are Heterogeneous in Pay

In the model presented in Section 2, workers can direct their search only if they expect firms to vary in pay and if they possess firm-specific information on pay. We first examine whether workers have firm-specific information on pay; we then examine within-worker heterogeneity in these beliefs.

The first bar of Figure 2 shows that about half of workers had firm-specific information about pay when they applied to their current firm. Eighteen percent of workers say they knew the exact pay they would earn when they applied; a further 27% had a rough idea of what that specific firm would offer them. The remaining workers said they had no idea, or only knew pay in their region or industry.

As in the United States, it is rare for job ads in Germany to include pay information (Batra, Michaud and Mongey, 2023; Caldwell, Haegele and Heining, 2024). However, workers may obtain pay information from other sources, such as both public pay aggregators or social networks.[21] Consistent with the idea that workers obtain information through informal networks, nearly 60% of workers who knew a current or former employee when they applied to their firm report that they had firm-specific information on pay. Less than 40% of unconnected workers had such information. Workers covered by collective bargaining agreements, which specify pay ranges for covered employees, are more likely to report that they had firm-specific information.[22]

Workers do not believe firms would offer them the same pay. Rather, most workers believe firms would offer them different pay for the same position. The first column of Table 3 shows the fraction of workers who report that they would earn the same pay at each of the three researcher-provided firms (first two rows) or at the firms they provided (remaining rows). The first two rows indicate that less than a third of workers expect the same pay for the three researcher-provided firms in each of the initial and follow-up surveys. Because we told workers to assume that they hold the same position at each of these firms, the within-worker variation in expected earnings does not reflect the fact that workers believe they will be engaged in different tasks at each employer. Because we focus on within-worker comparisons, the variation in expected earnings is also not driven by unobserved differences in worker ability.[23] The remaining rows of the table confirm

[21] Pay data from Kununu, a widely used online aggregator, are positively correlated with administrative records.

[22] Because bargaining groups are not always indicated in job ads, we would not expect all covered workers to have specific information on pay when they apply (Caldwell, Haegele and Heining, 2024).

[23] Standard concerns with survey elicitations—such as anchoring bias—would lead us to understate the true amount

that—for the firms workers themselves provided (their perceived outside option firms)—workers also perceive significant heterogeneity in pay.

The within-worker variation in expected pay could reflect a variety of firm-specific factors, including differences in amenities and hours worked. However, it does not simply reflect the fact that workers expect pay to vary across geographic regions or sectors. Rather, workers believe that pay varies across firms within a detailed labor market. Rows 3 to 6 of Table 3 show the fraction of workers who provide the same expected pay for each of the three firms in the worker-provided firm module. For a subset of these workers, we collected additional information on the specific location they had in mind when listing the worker-provided firms. We find similar results when we focus on workers who indicated that the three firms they provided were in the same state, district, or municipality (Rows 4 to 6). Further, Appendix Table A1 shows that firm fixed effects have explanatory power beyond sector fixed effects, state fixed effects, or sector-by-state fixed effects.[24]

4.2 Workers' Pay Expectations are Firm-Specific

We summarize workers' firm-specific pay expectations using a two-way fixed effect model,

$$\log \tilde{w}_{ij} = \tilde{\alpha}_i + \tilde{\psi}_j + \epsilon_{ij}, \tag{7}$$

where $\tilde{w}_{ij}$ is worker i's expected pay at firm j. This specification provides us with estimates of workers' perceived pay premia, $\tilde{\psi}_j$, up to a normalization; we normalize $\tilde{\psi}_1 = 0$. For instance, we interpret $\tilde{\psi}_2$ as the average log difference in pay workers expect to receive at firm 2, relative to (the arbitrarily chosen) firm 1. If $\tilde{\psi}_2 = 0.2$, workers expect firm 2 to pay 20% more than firm 1.

This specification is similar to the two-way fixed effects models often used in the wage-setting literature (Abowd, Kramarz and Margolis, 1999). However, we identify the firm-specific pay premia using within-worker variation in expectations, rather than job-to-job mobility. Our baseline specification includes data from the researcher-provided firm module in both the initial and follow-up surveys. Because we randomly assigned workers to firms, there is no mechanical correlation between workers' characteristics and their firm-specific pay expectations (Appendix E.1.2).

To understand the stability and range of estimates, we split the sample into two random samples and plot the estimates for each sample in Figure A3. This figure shows that the estimates range from -0.2 to 0.05: the lowest-pay firm (in workers' view) in our sample pays about 25% less than

of variation in workers' expectations. As we describe in Appendix E.1.6, we winsorize workers' firm-specific expectations to reduce the influence of outliers. If anything, this procedure biases against detecting firm-level variation.

[24]We selected the researcher-provided firms to ensure sufficient overlap to estimate these effects. Many workers provided firms in the same sector and state.

the highest-pay firm in our sample. This figure also shows that the estimates do not simply reflect noise: the correlation between the estimates from two random samples is 0.88. Workers simply believe some firms pay more than others.

Appendix Table A2 presents a variance decomposition of workers' pay expectations. Column 1 shows that most of the variation in workers' expectations can be explained by variation in the person effects α_i. However, both informed and uninformed workers see a role for firms in wage-setting. Columns 2 and 3 split the sample according to whether a worker knew pay when they applied to their current firm and Columns 4 and 5 split the sample according to whether the worker recently engaged in job search. Even for workers who are less informed according to these definitions, we see that there is a role for firm effects and we can reject the hypothesis that the firm effects are jointly zero. Further, the between-group gap in the standard deviation of firm effects is small. Column 6 describes the objective predictions for these firms. A comparison of Columns 1 and 6 suggests that, relative to the objective predictions, workers' perceived firm effects are about a third as variable as the firm effects observed in the data.[25]

4.3 Workers' Pay Expectations are Correlated with Objective Predictions

Workers' firm-specific pay expectations are positively correlated with objective predictions. For each worker-firm pair in the researcher-provided firm module, we construct an objective prediction (ω_{ik}) by taking the logarithm of an individual's current total annual pay, subtracting the pay premium of her current place of work, and adding the pay premium of the specified outside firm:

$$\omega_{ik} := \underbrace{\log(w_i)}_{\text{observed earnings}} - \underbrace{\psi_{\mathbf{j}(i)}}_{\text{current firm}} + \underbrace{\psi_k}_{\text{provided firm}} .$$

The firm premia come from Bellmann et al. (2020), who fit two-way fixed effect models to 2010–2017 population data. By exchanging the firm effects, we are able to account for the fact that pay may vary across workers for observed (e.g., education, labor market experience) and unobserved (e.g., bargaining ability, skill) reasons.[26] We asked workers to assume they hold the same position in each of the outside firms so that these unobserved factors would not vary within worker.

Figure A4 shows that there is a strong positive relationship between the objective predictions ($y-$axis) and workers' pay expectations ($x-$axis). A simple regression shows that workers' firm-

[25]One explanation for the reduced variability is the observed firm effects—based on firm-to-firm movers—are noisily estimated. Our estimates of perceived wage premia are relatively precise as we focused on a relatively small set of firms. Another is that question-order effects lead us to understate the amount of variation in workers' beliefs.

[26]We use the firm pay premia from Bellmann et al. (2020) rather than estimating firm pay premia ourselves with observed pay data, because we do not have access to full population records. One limitation is that we are not able to account for measurement error in the estimation of these premia.

specific expectations explain about 20% of the variation in the objective predictions. Further, Figure 3 shows that workers' expected pay premia ($\tilde{\psi}_j$) are correlated with firms' objective pay policies: there is a correlation of 0.42 with the objective pay premia and of 0.54 (0.56) with firms' median (mean) pay.[27] Workers' expected pay premia are more positively correlated with observed firm characteristics such as age and size than the objective premia are.

Heterogeneity in Perceived Premia. A natural question is whether the perceived pay premia vary across demographic groups, and whether some groups have beliefs which are more correlated with objective firm pay policies. To answer this question, we re-estimate equation 7 for different subsamples of workers. We report the correlation between the estimates for each sample, as well as p-values from formal tests of equality in Columns 1-2 of Table 4. To assess the stability of the perceived pay ranking of firms, we use a rank-ordered logit model to fit workers' firm-specific pay expectations and report analogous results from this model in Columns 3 and 4 of Table 4.

Even when groups disagree on the pay premia of these firms (when a formal test of equality suggests the estimates are distinct), the estimates are similar in magnitude and highly correlated. For instance, we can reject the hypothesis that men and women have the same perceived pay premia at each of the 30 researcher-provided firms. However, there is a correlation of 0.73 between the pay premia men and women associate with the researcher-provided firms. We find a similar relationship when we analyze the correlation between the estimates of workers with and without a CBA or workers with and without a college degree. Column 4 of Table 4 shows that we cannot reject equality of estimates when we use the rank-based model.[28] However, we see the most disagreement between workers with and without a college degree; this is consistent with the empirical fact that these workers differ in firm-specific earnings trajectories (Friedrich et al., 2024).

4.4 Discussion and Robustness

Our results indicate that many workers have firm-specific information on the pay provided by outside firms.

[27] Because there is substantial noise in our measurement of firms' objective pay premia, not all deviations can be interpreted as reflecting bias in workers' beliefs. Rather, we interpret the positive correlation with objective premia as a sign that workers have information on the pay policies offered by different firms. We refrain from taking a stand on whether workers who deviate from our prediction are overly optimistic (in the case of over-prediction) or overly pessimistic (in the case of under-prediction).

[28] In unreported results we find that men, younger tenured workers, and workers covered by a collective bargaining agreement have perceived premia which are somewhat more correlated with objective measures. We do not find that workers at low-wage firms have perceived premia which are less correlated with objective measures.

Interpreting Survey Elicitations. A natural concern is that respondents might have looked up pay information during the survey. Section 3.4 provides information on the overall validity of the survey responses, including specific tests which indicate this is unlikely. Several facts weigh against the concern that, by asking workers to provide pay expectations for multiple firms, we primed them to believe outside firms are heterogeneous. If respondents were merely primed to invent differences, their responses would be white noise. Instead, we find they are structurally correlated with objective administrative data (AKM effects) and exhibit strong within-worker consistency across split samples (Appendix Figure A3). Further, as we show in Section 5.2, workers' stated search behavior over firms not mentioned to them in the pay module due to randomization) is strongly predicted by the wage perceptions of other workers. This confirms that the elicited beliefs reflect genuine, pre-existing knowledge rather than survey artifacts.

Given the substantial noise in estimates of the objective wage premia (from Bellmann et al. (2020)) it is difficult to compare the variance in workers' pay expectations to those of objective measures. One possibility is that, even if workers' beliefs are directionally correct and not anchored to their current salary, they are more compressed than the true wage distribution. Another possibility is that they simply appear compressed because the variance in objective predictions is inflated due to measurement error.[29] We do not take a strong stance on this point. When we compare the relative importance of beliefs and preferences in Section 6, we consider a range of assumptions about workers' beliefs. Our baseline calibration assumes workers' perceived variance is 1/4 that of the true variance; this is conservative, given the results in Appendix Table A2. We also present independent evidence on the importance of the preference channel using discrete choice experiments wherein we presented workers with take-it-or-leave-it offers.

Representativeness of Firms. Another natural concern is that some of our analysis of workers' pay information relies on their beliefs about pay at a relatively small set researcher-provided firms. While these firms allow for precise identification of firm-specific priors, they may not represent the typical worker's outside option. To confirm that our findings generalize, we leverage the worker-provided firm module, where respondents listed their own relevant outside options.[30] Table 3 shows that the share of workers perceiving pay heterogeneity is similar across both firm sets. Appendix Figure A4 demonstrates that the positive correlation between subjective beliefs and objective pay measures also holds for the worker-provided firms. This confirms that our core finding—that many

[29] Survey effects such as anchoring may also lead us to understate the variance in workers' beliefs.

[30] A priori it is not clear that focusing on researcher-provided firms would bias us towards a finding that workers have information about wages. That they are larger may make it more likely workers know the pay of these firms; that they are not necessarily in a worker's consideration set makes it less likely workers know the pay of these firms.

workers possess directionally accurate firm-specific priors—extends to the broader labor market. Appendix F suggests that workers whose firm-specific predictions exceed (or fall below) objective predictions in the researcher-provided firm module also typically have firm-specific predictions which exceed (or fall below) objective predictions in the worker-provided firm module.

Relationship With Other Work. The descriptive facts contrast with work in Germany which suggests workers anchor beliefs to their current wages—low-wage workers underestimate and high-wage workers overestimate outside pay (Jäger et al., 2024). Appendix C reconciles differences between studies. We highlight both differences in questionnaire design and that we elicited beliefs about pay, rather than beliefs about expected pay changes following voluntary or involuntary transitions. The latter depend on beliefs about pay, as well as beliefs about offer probabilities and preferences. Beliefs about preferences may explain differences in the studies. If anything, low wage workers are more optimistic about their outside options than workers at high wage firms (Appendix Figure C4), despite the fact they receive offers at lower rates, both with and without active job search (Appendix Table C1). Our results align with other recent work suggesting workers have information about their employment and earnings risk (Guo, 2025; Caplin et al., 2023). In Section 6 we show that many workers are reluctant to move even when told they could have higher pay elsewhere.

5 Firm Pay and Worker Search

We next show that workers use their firm-specific information to direct their search to firms with higher perceived pay. Our estimates imply an average application-wage elasticity of 1. The fact that workers direct their search to firms with higher perceived pay reflects the fact that workers value pay, and the fact that workers believe higher pay firms have better non-wage amenities. Differences in risk tolerance or perceived offer probabilities have a statistically insignificant and economically small impact on the application elasticity or on the market search elasticity.

5.1 Relationship Between Firm-Specific Pay and Consideration

The model in Section 2 suggests that a worker's application decisions depend on perceived wages, firm-specific amenity values, and offer probabilities. Having established in the previous section that many workers possess firm-specific information on wages, we next examine whether these beliefs guide their search behavior and whether perceptions of offer rates mediate this relationship.

Our primary design which links workers' beliefs to their search behavior leverages the fact that, in the researcher-provided firm module, we observe both whether a worker would consider applying to the specified firm and what the worker believes they would earn at that firm. We regress an indicator for whether worker i says they would consider applying to the randomly provided firm j on the worker's expected pay at that firm $(\log \tilde{w}_{ij})$, on an indicator for whether the data come from the follow-up survey ($\zeta_{t(i)}$), and on worker (λ_i) and firm (γ_j) fixed effects.[31] Some specifications include additional worker-firm covariates X_{ij}. We estimate the following linear probability model:

$$\text{Consider}_{ij} = \eta^s \log \tilde{w}_{ij} + X_{ij} + \gamma_j + \lambda_i + \zeta_{t(i)} + \epsilon_{ij}, \tag{8}$$

and cluster the standard errors at the worker level. Given the large number of fixed effects, we estimate this equation by OLS instead of using a nonlinear model.

The coefficient of interest, η^s, identifies the link between workers' pay expectations and consideration by relying only worker-firm specific deviations in expectations. The worker fixed effects account for differences in workers' overall level of pay expectations (including systematic optimism) and for heterogeneity in switching costs (s_i). The firm fixed effects account for the fact that firms vary along multiple dimensions, including amenities (a_j) and offer probabilities (p_j).[32]

Main Estimates. The estimate in Column 1 of Table 5 suggests that a 10% increase in workers' firm-specific pay beliefs increases a worker's probability of consideration by 3 percentage points. Columns 2 and 3 show that this estimate is stable when we add a control for the logarithm of the driving distance between the provided firm and the worker's current place of work (Column 2), and when we additionally control for whether the provided firm is in the same sector as the worker's current firm (Column 3). The positive semi-elasticity confirms prediction 3: workers with firm-specific information on pay are more likely to apply to firms they believe pay higher wages. We obtain the application-wage elasticity by dividing η^s by the baseline consideration rate. As reported at the bottom of Table 5 our estimates imply an application-wage elasticity of 1.

Role of Offer Probabilities. These fixed-effect models do not account for variation in worker-firm specific offer probabilities. If workers believe higher pay is associated with lower offer probabilities, their application behavior may be less responsive to pay. Columns 4 to 7 of Table 5 split

[31] This is a slight abuse of notation: in our model $\tilde{w}_{ij}$ refers to beliefs about log pay. We write $\log \tilde{w}_{ij}$ in the specifications provided here and in the rest of the paper to clarify that we take the log of workers' firm-specific expectations.

[32] The firm fixed effects also account for the possibility that workers may (as a group) have misperceptions about the pay of specific firms.

the sample according to workers' risk tolerance (Columns 4 and 5) or whether the worker says they would be reluctant to apply to a position if the probability of an offer were low (Columns 6 and 7). While we see smaller estimates and elasticities among workers who are less tolerant of risk (Column 4 relative to Column 5) or who are more sensitive to competition (Column 7 relative to Column 6), we cannot reject equality of estimates or elasticities across groups.[33]

This heterogeneity does not reflect the fact that workers do not believe there is a positive correlation between wages and the amount of competition for a vacancy. Appendix Figure A5 shows that the majority of workers believe that higher wage firms receive more qualified applicants per vacancy. Appendix Table A4 presents additional specifications which split by whether workers consider the amount of competition associated with a position (Columns 1-2) or believe higher wage firms receive more qualified applicants (Columns 3-4). Across all definitions, we find little evidence that competition meaningfully affects workers' behavior.

5.2 Robustness and Alternative Designs

Appendix Table A3 shows that our results are robust to a variety of alternative specifications and weighting schemes. We obtain similar estimates when we use alternative specifications or measures of distance (Columns 1 to 3). Columns 4 and 5 show stability across survey waves, mitigating concerns that respondents acquired new pay information after the initial wave. Columns 6 and 7 show robustness to alternative weighting schemes.

Because our primary design includes firm fixed effects it does not identify whether high-pay firms receive more overall consideration. We use two alternative designs that allow us to examine whether, within a market, high-pay firms receive more overall consideration. These designs do not hold fixed the non-wage characteristics of firms and use non-overlapping subsets of the data that differ from the data used in our main within-worker, within-firm design.[34] These designs also confirm that differences in risk tolerance or perceived offer probabilities do not meaningfully affect workers' search behavior.

Alternative Design 1: Consideration without Pay Expectations. The first design shows that there is a positive correlation between whether a worker considers applying to a firm and the pay premia other workers associate with that firm (as well as with the firm's observed pay premium). This design uses data on consideration for the four researcher-provided firms per worker-wave for

[33]In unreported results, we analyzed demographic heterogeneity. While we found somewhat larger estimates for male workers, workers who are unmarried, and workers without children, none of the differences were significant.

[34]Together with our main design, the data subsets used in each of the alternative designs span the data collected in the researcher-provided and worker-provided firm modules.

which we elicited consideration, but not pay expectations.[35] We regress an indicator for whether worker i said they would consider applying to firm j if they wanted to switch firms on a measure of the firm's pay policy (Pay_j), on worker fixed effects, on the (log) driving distance between the worker and firm j, and on the additional covariates and fixed effects listed in Appendix Table 6,

$$\text{Consider}_{ij} = \beta^{RP}\text{Pay}_j + \lambda_i + X_{ij} + \epsilon_{ij}. \tag{9}$$

Our baseline specification in Column 1 of Table 6 mimics the specification in Column 4 of Table 5, but exchanges the firm fixed effects—which would be collinear with Pay_j—with dummies for sector, along with a control for the log number of employees at the firm. The coefficient β identifies whether, within a sector, firms that have higher pay—or firms that workers believe have higher pay—receive more interest. We cluster the standard errors at the worker level.

Column 1 of Table 6 shows that there is a strong positive relationship between both the subjective pay premia (Panel A) and objective pay (Panels B) of these firms, and the probability a worker says they would consider applying to that firm. The estimate in Column 1 suggests that, relative to firms in the same sector and with the same size, workers are 9.5 percentage points more likely to consider applying to a firm with 10% higher perceived pay premium. We obtain similar estimates after controlling for a firm's brand recognition (Column 2) and for CBA coverage (Column 3).

Alternative Design 2: Free-Text Provision of Firm Names. The second design illustrates that the positive link between consideration and pay also emerges when we consider workers' free-text responses. As we document in Appendix F, workers provide firms which workers from their firm have historically gone to. For each worker, we construct a dataset that has one observation for each of the researcher-provided firms we did not provide to them in the initial survey's researcher-provided firm module. For each worker-firm pair, we construct an indicator for whether the worker independently provided the name of that firm.[36] For workers who did not provide any firm names, this indicator is zero for each firm. We regress this indicator on the same measures of firm pay policies and worker-firm covariates as in our first alternative design, again clustering the standard

[35] To reduce the survey burden for respondents, we elicited consideration for seven firms, but only asked for their pay expectations for three (randomly chosen) of the seven firms.

[36] Workers completed the worker-provided firm module before the researcher-provided firm module. We only use data from the initial survey because we asked all workers to provide the names of firms in this survey. In the follow-up survey, we only asked workers to provide the names of firms if they failed to in the initial survey. Our main dataset includes one observation for each of the 23 firms we did not provide to the worker. Because one of the researcher-provided firms could not be linked to AKM wage premia, the sample includes 22 firms in Panel B.

errors at the worker level,

$$\text{List Firm}_{ij} = \beta^{WP}\text{Pay}_j + \lambda_i + X_{ij} + \epsilon_{ij}. \tag{10}$$

Columns 4 to 6 of Table 6 shows there is a strong positive relationship between the objective and subjective pay premia of these firms and the probability a worker lists the firm. The effect sizes are an order of magnitude smaller in this design, reflecting the fact that workers could name any firm they wanted in this module: the mean of the dependent variable is only 1.7%. However, the ratio of the effect size to the mean of the dependent variable is similar across the two designs.

Robustness. The alternative designs confirm that high-wage firms receive more overall consideration and serve as robustness checks for our main design. Because neither alternative design controls for firm fixed effects (unlike our main design), the estimates are not directly comparable to those from that design. Here the positive relationship with search documented may reflect workers directing their search towards these firms because of pay or because of non-pay characteristics positively correlated with pay. [37] Appendix Table A5 shows that, using these designs, we also obtain similar results when focusing on workers who are risk averse or who are sensitive to the probability they will receive an offer.[38] Appendix Table B4 shows these results are robust to re-weighting and to using a firm's log mean pay, rather than pay premium.

5.3 Market Search Responses

Our central estimate of an application-wage elasticity of 1 is similar to estimates of the application-wage elasticity from studies in which posted pay is randomized within a position. This includes estimates from observational and experimental studies of jobseekers applying on online job boards (Marinescu and Wolthoff 2020 and Belot, Kircher and Muller 2022 obtain elasticities of 0.8) and from the public sector experiment by Dal Bó, Finan and Rossi (2013).

This similarity is surprising. Our sample consists of full-time employed workers, many of whom are not actively searching. Further, our variation comes from workers' own beliefs about firm pay, not from posted wage differences. Many firms do not provide pay in job ads; when

[37]Because they leverage differences in perceived pay across firms, the estimates in Panel A produce elasticities which are most comparable to our main design. Columns 1-3 suggest an implied overall application elasticity with respect to perceived wage premia on the order of 3; Columns 5-6 suggest an elasticity closer to 5. That these exceed the estimates of our preferred design suggests that workers believe high-wage firms also offer better non-wage amenities; Figure A5 confirms this.

[38]In unreported results we also found significant (albeit smaller) results when we focus on workers who provided uniform pay expectations; these workers do not provide beliefs which identify $\tilde{\psi}_j$. This is consistent with the idea that at least part of the uniformity reflects survey anchoring.

required to do so by law, many choose to provide large ranges (Batra, Michaud and Mongey, 2023). Our estimates contribute to a literature examining the responsiveness of applications to wages (see, e.g., Belot, Kircher and Muller, 2022).[39] While our estimates rely on stated search behavior (rather than observed click rates from job boards) this was necessary for us to obtain data from a representative population including both searchers and non-searchers.

Market-Level Estimates. This application-wage elasticity suggests that, at the firm level, workers are responsive to differences in pay. However, this does not account for the extensive margin of search (i.e. whether workers apply to any firms). In the follow-up survey, we asked workers how likely they would be to search for outside employment if they learned pay was 5%, 10%, or 20% higher (randomized across workers) at other firms, following a literature that uses survey experiments to estimate the responsiveness of search to information about outside pay.[40] This allows us to estimate the market search elasticity on the extensive margin. Panel A of Figure 4 plots the average response for each level of the randomized outside pay. There is a clear positive relationship between the stated outside pay and the probability a worker would begin searching. However, the average probability of search is far from 100%, even when the specified outside pay is significantly higher than what the worker currently makes.

Column 1 of Table 7 shows that search is highly responsive to beliefs about outside pay. A 10 percentage point increase in the stated outside wage increases the probability workers will search by 18.8 percentage points. This suggests that the costs of search are low: relatively small changes in beliefs about outside pay lead to large changes in search.

The implied market search-wage elasticity is 7.14 (bottom of Column 1). As with our firm-specific estimates in Section 5.1, we find no evidence that perceived offer probabilities have a large impact on workers' search behavior (remaining columns). This suggests that low search initiation rates are not driven by the perception it is difficult to obtain outside offers.

[39] While we provide evidence in support of the core assumptions of directed search models—that workers have firm-specific information on pay to direct their search—we do not take a stand on whether the data are most consistent with all of the standard predictions of random or directed search models. While most directed search models feature wage posting, in previous work we documented that workers reject a substantial share of the offers they receive and that bargaining is common (Caldwell, Haegele and Heining, 2024).

[40] See, for instance Jäger et al. (2024), who randomize whether respondents see the average wages of similar workers in their respective labor markets. Instead of randomizing whether to provide any information to workers (and leveraging variation in respondents' pre-treatment error), we randomized the magnitude of hypothetical raises.

6 Beliefs, Preferences, and Mobility

Workers use firm-specific pay beliefs to direct their search to higher-paying firms. Yet many do not search, even when they believe they could earn more. We use discrete choice experiments in which we ask workers to rank potential offers from firms they indicated are application targets to isolate the role of worker preferences. In these DCEs we provide the wage; we find that most workers are unwilling to switch employers for even sizable raises. Our estimates imply separation elasticities of 2-4. Workers who are reluctant to leave their firm are less likely to search, even when market wages are high. A quantitative exercise indicates preferences better explain limited mobility than biased beliefs.

6.1 Most Workers Would Not Switch Firms for Known Pay Raises

We isolate the role of preferences from information and search frictions using the discrete choice experiments described in Section 3.2. These experiments show that, even when offered a significant pay raise, many workers say they would prefer to stay at their incumbent firm (with no raise). By presenting the offer as 'in-hand', we shut down concerns about queuing, competition for vacancies, or rejection probability. Most of this analysis relies on workers' rankings of the firms they named as potential destinations, rather than on workers' rankings of the researcher-provided firms.

Panel B of Figure 4 shows a positive relationship between the randomized outside pay and the share of workers saying they would prefer the outside job offer at one of the firms they listed in the worker-provided firm module. However, the share who prefer the outside offer is far from 100% even for relatively large raises (20%). We see similar results when we focus on workers' preferences over the same firms under the assumption that their commute would not change or that their future career path would not change.[41]

The monopsony literature often emphasizes separation elasticities, which quantify how responsive workers are to wage changes at their current employer or outside firms. Table 8 presents results from ordinary least squares regressions of stated mobility on the randomized outside pay, and the separation elasticities implied by these models. Columns 2-4—which use data from the module in which workers rank the incumbent relative to potential destination firms—suggest separation elasticities on the order of 2-4. These estimates are strikingly similar to the typical estimate in the monopsony literature (between 3 and 5).[42]

[41] In the initial survey, we asked a random 50% of workers to re-rank the hypothetical offers from worker-provided firms under two scenarios: (1) if their location and route to work would not change and (2) if their career growth (future raises and promotions) would be identical across options.

[42] In unreported subgroup analysis we find somewhat larger separation elasticities for men (relative to women), for

This design approximates an ideal—but infeasible—experiment in which researchers randomize real job offers to employed workers. Because we provide pay—and indicate that this is an in-hand offer—elasticities estimated from these experiments are not shaped by workers' beliefs about outside pay. Because we specify that the worker has an offer, they are also not shaped by search frictions, such as the fear of rejection or workers' beliefs about the likelihood firms would make them an offer. Rather, our estimated separation elasticities reflect a preference on the part of workers to remain at their incumbent firm, even when alternatives of interest offer more pay. The fact that we obtain estimates similar to those in the literature—when we remove these factors—suggests that preferences alone can explain a significant portion of immobility.

6.2 The Incumbent Preference Is Meaningful

The model in Section 2 indicates that whether worker i prefers an offer from firm k to the option of remaining at the incumbent firm j depends on (1) the gap in log wages, (2) the gap in firm-provided amenities, and (3) the switching cost (or match-specific preferences). Under the standard assumption that workers' idiosyncratic preferences are identically and independently distributed and follow a type 1 extreme value distribution we can estimate the parameters of workers' utility functions using a rank-ordered ("exploded") logit (Luce and Suppes, 1965).

Our discrete choice experiments generate exogenous variation in the offered log wage at the outside firm, which is, by construction, uncorrelated with worker and firm characteristics. Our initial specification includes two parameters: (1) the randomized raise associated with firm k and (2) an indicator for whether firm k is the incumbent.[43] Because we do not include firm-specific dummies, we interpret the incumbent dummy as capturing both switching costs (s_i) and the preference for the incumbent employer relative to a random employer (researcher-provided firm module) or the preference for the incumbent employer relative to another in the worker's consideration set (worker-provided firm module). We therefore refer to this as the "incumbent premium", rather than a switching cost. Because we have workers rank multiple outside firms (including their incumbent firm), our rank-ordered logit specification naturally accounts for firm-invariant worker-specific heterogeneity.

Table 9 shows that the incumbent premium is large. The table reports the coefficients, as well as estimates of the incumbent premium expressed in monetary terms (i.e. scaled by β). The average implied cost to get a worker to move to one of the researcher-provided firms is 18% of their annual

workers at high-wage (relative to low-wage) firms, and for workers without a college degree.

[43]We do not multiply the randomized raise by the worker's current wage as this does not change the comparison of utilities within a worker. Because $\log w_{ij} = \log(w_{i,t} \times (1 + r_{ij})) = \log w_{i,t} + \log(1 + r_{ij})$, the first term drops out in the conditional logit specification.

pay. By contrast, it costs 7% of their annual pay to get workers to switch to one of the firms they listed in the worker-provided module. The lower estimates for worker-provided firms likely reflect both the fact that these firms are more appealing to workers for non-wage factors (amenities) and that these firms are typically physically closer to the worker's current workplace.

The incumbent premium does not simply reflect a preference to avoid moving physical locations. We can see this by either adding distance controls (Column 2 adds the log travel time between a worker's current place of work and the indicated firm) or by using following-up discrete choice experiments in which we told workers to assume that their location of and route to work would not change. Column 2 shows that adding controls for distance shrinks the incumbent premium (relative to a random firm) to 11%. Column 4 of Table 9 shows that the incumbent premium (relative to a firm in a worker's consideration set) remains relatively stable when we consider workers' preferences over fixed commutes. This is likely because workers often provide firms which are close to their current location.

Robustness. Appendix Table A6 confirms we obtain similar estimates under alternative weighting schemes (Columns 7-8), under alternative specifications of distance (Columns 2-4), and in alternative samples (Columns 5-6). Appendix Section B.3 shows that the workers who did not provide firm names are somewhat less responsive to outside pay, weighing against concerns that these workers would be most responsive to information; if anything, we may understate workers' preference to remain at the incumbent firm. In unreported results we find that costs are somewhat larger for long-tenure workers, and for workers covered by a collective bargaining agreement. Consistent with the findings of Topel and Ward (1992), the incumbent premium is lower for workers at the beginning of their career (with less than four years of experience) than for older workers.

6.3 Preferences Influence Search

Our model posits that firm-specific attachment acts as a wedge that reduces the incentive to search: even workers who over-estimate the attractiveness of outside employed may not search if they are, to quote Reynolds, "satisfactorily employed". To validate this mechanism, we can examine the link between the two distinct elicitations in our survey: the decision to search (from the extensive margin experiment in Section 5.3) and the decision to stay (from the discrete choice experiments).

We construct the "residual probability of searching" by regressing the stated probability of searching on the randomized market wage and calculating the residual for each worker. We construct the "residual probability of choosing the incumbent" by fitting an OLS model where the outcome is an indicator for whether the worker prefers the incumbent firm and the regressors include the difference between inside and outside pay (r_{ij}) and a full set of worker fixed effects:

$$\text{Stay}_{ij} = \beta r_{ij} + \zeta_i + \epsilon_{ij}.$$

Figure 5 shows there is a strong negative relationship between these residuals: workers who exhibit a high residual preference for their current employer in the choice experiments (high $\hat{\zeta}_i$) are significantly less likely to initiate search in the search experiments. This confirms that the incumbent premium is a structural friction that actively suppresses labor market search, consistent with Prediction 1 of our model.

6.4 Preferences Better Explain Workers' Limited Mobility

We conclude this section by examining the relative importance of beliefs about pay and preferences in explaining workers' limited mobility. We simulate data from a version of the model in Section 2 that incorporates heterogeneity in workers' outside options and recognizes that offer acceptance is probabilistic.

Setup. We assume that beliefs about outside pay are compressed: if an outside firm's true wage exceeds the current firm's wage by Δw, the worker believes it exceeds it by only $\lambda \Delta w$. In our baseline we assume $\lambda = 0.5$, implying workers perceive only 25% of the actual between-firm variation in pay. This is conservative; estimates in Appendix Table A2 suggest the ratio of variances is closer to 1/3. We present a "high compression" scenario ($\lambda = 0.25$), representing extremely limited information, and a "low compression" scenario ($\lambda = 0.75$), which accounts for potential measurement error in our administrative estimates (Kline, Saggio and Sølvsten, 2020).[44]

We use our estimates of the relationship between outside pay and worker search from Section 5.3 to predict how search probabilities respond to changes in perceived net gains. The predicted search probability is:

$$\widehat{\text{Search}}_i \quad = \quad 1.88(\lambda \Delta w) + 0.263,$$

and we set $\lambda = 1$ when workers have full information. As shown in Section 2, expected wage gains and the incumbent-firm preference enter the search decision as substitutes. Correcting beliefs increases the perceived gain ($\lambda \to 1$), while eliminating preferences improves the net gain by the amount of the incumbent premium.

[44] We abstract from the insurance value of information (variance reduction). Risk aversion does not significantly dampen the search elasticity in our reduced-form results (Table 5).

To model job acceptance, we use the preference parameters from our rank-ordered logit. The probability that a worker accepts an offer depends on the true wage gain Δw and the incumbent premium:

$$P(\text{Accept}) = \frac{\exp(\beta \Delta w)}{\exp(\beta \Delta w) + \exp(\alpha)}.$$

Consistent with our framework, workers accept offers based on observed wage gains, even if their search was guided by compressed beliefs. We assume all workers who search get offers so the fraction who move is the fraction who search multiplied by the fraction who would accept an offer.[45]

Results. Table 10 presents the results of our simulation. We first focus on low wage workers (Panel A) and assume they can receive an offer from the median firm premium firm. In our baseline scenario (Columns 1-2), providing full information about median market wages raises mobility by only 7 percentage points. This relatively modest gain occurs because, while transparency increases search (by 12 percentage points), the high incumbent premium leads many searchers to reject the offers they find. By contrast, removing the own-firm preference increases mobility by 24 percentage points—more than tripling the impact of full transparency.

This dominance of the preference channel holds even in our most conservative benchmark, which uses the lower incumbent premium estimated for workers' own consideration sets. Even assuming the outside firm is as attractive non-pecuniarily as those in the consideration set, correcting beliefs raises mobility by only 9 percentage points, while eliminating the incumbent premium raises it by 12 percentage points. Only in the limiting case where we stack assumptions even further against preferences—assuming effectively zero information (high compression row, $\lambda = 0.25$) and that outside firms are as attractive as those in a workers' consideration set (low attachment column)—does transparency roughly equal the effect of eliminating attachment (rows 2 and 4 of Column 4).

We generalize these results to the full population in Panel B. Here, we relax the assumption of a fixed median target and instead simulate random search, allowing workers to draw offers from the true distribution of firm effects. This allows even high-wage workers to have biased beliefs and to want to move. Even in this unrestricted setting, the pattern remains: under baseline assumptions, transparency increases mobility by only 6 percentage points, while eliminating attachment increases it by 15 percentage points. Appendix Table B5 shows that these conclusions are robust to alternative weighting schemes.

[45] We also assume that the incumbent firm does not respond to the outside offer such that the wage gap between the two offers remains Δw.

Discussion. Our calibration likely overstates the role of information. First, we assume perfect feasibility: in the transparency counterfactuals, we assume workers can successfully generate an offer from the target firm. While the transparency treatments require this high degree of feasibility to generate value, eliminating attachment will increase search (and mobility) even when outside firms offer similar pay. Second, in Section 2 highlights that if workers have firm-specific information—even if compressed—they can effectively direct search. By treating the friction as a uniform reduction in variance rather than acknowledging the efficiency of existing directed search, we likely inflate the marginal value of transparency.

Taken together, the results imply that preferences are at least as important as, and often significantly more important than, bias in pay beliefs in shaping mobility. This conclusion is reinforced by our earlier finding that the cost of search is small—a 10% increase in beliefs about market pay leads to an 18 percentage point increase in probability of searching; by contrast, the marginal worker needs an outside firm to offer 18% more to switch firms. Because our data come from Germany—where switching employers does not alter health insurance—the own-firm preference likely does not reflect lock-in due to benefits. Instead, switching costs and home bias grant firms market power over incumbent workers, much as they do over consumers in product markets.

7 Firm-Specific Attachment

In the last part of our analysis we examine whether workers' preferences for the incumbent can be explained by a general reluctance to switch firms (common across firms). We cannot predict firm-specific attachment based on ex ante firm characteristics. Instead, attachment is firm-specific: the gap in valuations between current employees and outsiders is heterogeneous across firms. Our findings are consistent with models in which firms are experience goods. Consistent with this, the most cited reason workers say others are reluctant to leave their firm is "social connections".

7.1 There is Systematic Variation in Non-Wage Values

We can allow amenities to influence workers' preferences by augmenting our rank-ordered logit model to include a full set of firm dummies. One important non-wage factor is likely the location of a firm relative to a worker's current place of work; assuming that this is proportional to the travel time between a worker's current workplace and the listed firm's headquarters, we model workers' preferences as

$$u_{ij} = \beta \log w_{ij} + a_j + m \log d_{ij} + \epsilon_{ij}$$

where w_{ij} is what worker i would earn at firm j, a_j is the non-wage value provided by firm j, and d_{ij} is the travel time between worker i's current place of work and firm j.

We use data on workers' preferences over the 30 researcher-provided firms, excluding their rank of the incumbent firm. For consistency with our analysis in the next section—which investigates whether insiders and outsiders differ in their valuations of a firm—our main analysis sample consists of data from the researcher-provided firm module for workers who work at one of the researcher-provided firms. This is the set of firms for which we can plausibly estimate valuations for incumbent workers ("insiders") and for firm outsiders.

Estimating this model using only workers' choices across outside firms confirms that workers expect firms to vary in the overall value of amenities they provide ($a_j \neq 0$). Columns 1 and 2 of Table 11 show that the p-value from a test that all of the firm dummies are zero is well below 0.01. We refer to these as "ex ante" valuations as they are valuations perceived by those who do not currently work at a firm.

Figure 6 shows that workers expect higher-pay firms to offer better bundles of amenities. This figure plots the estimates of a_j/β (scaled so that we can interpret estimates as monetary equivalents) against the estimates of ψ_j from Section 3. The perceived variation in firm-specific amenity values is significant: the point estimates suggest a range of 25 percentage points, which is comparable to the range in our estimates of firm pay premia. This finding suggests that amenities are just as important as pay in explaining workers' choices across firms. A 10% increase in the perceived firm pay premia is associated with an 11% increase in the amenity value.[46]

Robustness and External Validity. Our results indicate that workers believe high wage firms offer better non-wage amenities. This positive correlation is consistent with models of rent-sharing or high-road management, where high-productivity firms share rents through both wages and working conditions, contrasting with a simple compensating differentials framework (Bender et al., 2018; Sorkin, 2018). Appendix B.5 shows that we obtain similar results when we fit a random coefficient rank-ordered model, which allows workers to have heterogeneous valuations of wages and distance.

While our estimates of non-wage amenities are specific to the set of 30 large German firms included in our researcher-provided firm module, workers generally believe that higher wage firms are not worse on non-wage dimensions. In the follow-up survey, we asked workers whether they

[46]Of course, these valuations capture all non-wage characteristics which vary across firms; they do not imply that high-wage firms are better on every dimension. They also do not allow us to identify the valuation of specific amenities, such as the value of childcare facilities or remote work. Instead they are comparable to estimates of the overall value of a firm provided in the macro-labor literature (Hall and Mueller, 2018; Sorkin, 2018; Volpe, 2024).

thought a firm paying 30% above market offered amenities that were better, the same, or worse than those offered by a firm paying 10% above market. The top two bars of Appendix Figure A5 show that the majority of workers (71%) believe that the higher-paying firm offers at least as good amenities; a large share (29%) believe that it offers strictly better amenities. Because workers who believe that the higher-pay firm offers lower amenities may not believe that these fully offset the value of pay, this is an upper bound on the percentage of workers who believe that amenities offset pay differentials.

7.2 Insiders and Outsiders Differ in their Valuations of a Firm

One potential explanation for the incumbency advantage is that workers value their firm simply because they dislike change. This would imply that the value firm insiders ascribe to a firm differs from the value firm outsiders ascribe to that firm by a level shift. Another explanation is that the gap in valuations of insiders and outsiders differs across firms. One way this could emerge is if some firms are more successful than others in generating the personal connections workers highlight as a key reason not to switch firms (Figure 7).

We fit an augmented version of the model which allows workers' utility to take the form:

$$u_{ij} = \beta \log w_{ij} + a_j + \iota_j \cdot 1\left\{\mathbf{j}\left(i\right) = j\right\} + \zeta_j \cdot 1\{\text{Consider}_i(j) = 1\} + m \log d_{ij} + \epsilon_{ij}$$

where $1\left\{\mathbf{j}\left(i\right) = j\right\}$ is an indicator for whether worker i is an incumbent at firm j and $1\{\text{Consider}_i(j) = 1\}$ is an indicator for whether worker i said they would consider applying to firm j. In this specification ι_j is the firm-specific premium. The ζ_j denote the difference in valuations for outsiders who would consider a firm and those who would not.[47] As before, we allow worker's utility to depend on the travel time between their current place of work and firm j ($\log d_{ij}$). We fit a rank-ordered logit model to account for between-worker heterogeneity.

We use this model to test three hypotheses. First we test whether workers who would consider applying to the firm differ in their valuations from those who would not. This is a test of whether workers sort into firms (or direct their search towards) firms on the basis of non-wage amenities. For this exercise we include only data on workers' preferences over outside firms. Column 2 shows that the data strongly reject the idea that valuations are the same for those who would and would not consider the firm. Appendix Figure A6 shows that valuations are larger for workers who would consider the firm, suggesting that workers also direct their search on non-wage dimensions.

[47]The overall valuation of amenities provided by firm j is a_j for firm outsiders who would not consider applying to the firm, $a_j + \zeta_j$ for outsiders who would consider applying to the firm, and $a_j + \iota_j$ for firm insiders.

Second, we examine the differences in valuations between insiders (incumbent workers) and outsiders (workers at other firms). We first test whether these groups of workers differ in their valuations (i.e. whether $\iota_j = 0 \forall j$). The p-values in Column 3 of Table 11 reject the idea that outsiders and insiders have the same valuations. We then test whether this difference in valuations is constant across firms (i.e. whether $\iota_j = \iota_k \forall j, k$). This would be indicative of a fixed switching cost, common across firms. The p-values in Column 3 also reject this.

One possibility is that insiders and outsiders differ in their valuations in predictable ways. This could occur if, for instance, firms vary in observed characteristics (e.g. availability of remote work, location). In this case, outsiders and insiders may differ in valuations simply because some outsiders do not value the specific amenities offered by the firm. To test whether the gap in valuations is predictable based on ex ante factors, we add an additional set of interactions for whether a worker says they would consider applying to a given firm. We then compare the valuations of incumbent workers to these workers. Column 4 confirms that the valuation of these outsiders differ from firm insiders (i.e. the p-values from a test of $\zeta_j = \iota_j \forall j$ are well below .05). Column 5 reports results from a specification which includes only worker-firm observations where the worker is either an incumbent at the firm or would consider applying to the firm. The p-values in this column show that the gap in valuations between insiders and outsiders who would consider moving to the firm is heterogeneous. In the Appendix, we show that these results are robust to alternative weighting schemes (Table B6).

The results in Table 11 suggest that firm-specific attachment varies across firms in ways that are hard to predict ex ante. One explanation is that firms are experience goods and that workers' valuations change dynamically after they have joined, potentially due to the development of firm-specific skills or relationships. While it is outside the scope of our analysis to unpack the specific factors which change once a worker has joined a firm, one potential explanation is that of social connections. When we asked workers why they believed others were reluctant to leave their employer, the most cited reason was personal connections (Figure 7). This is a firm-worker specific effect whose value is likely unknown ex ante. More generally, non-wage amenities which are hard to observe by firm outsiders, could explain the patterns in our data.

8 Discussion and Conclusion

Why do many workers stay with their employer despite higher pay elsewhere? Using novel survey data from nearly 10,000 full-time German workers linked to administrative records, we examine the roles of beliefs and preferences in shaping mobility. We show that workers expect firms to vary in pay and that perceived pay premia are correlated with observed wage premia. Workers direct their search to firms they believe would pay them more (with an average application-wage elasticity of 1), and expect higher-wage firms to offer better non-wage values. For a set of large German employers, there is agreement in the perceived pay rankings for different demographic groups. Among these firms, a firm with a 10% perceived wage premium has, on average, 11% higher non-wage amenities (in money-metric terms).

Yet even when offered a raise at a firm they identified as desirable, most workers prefer to stay. Implied separation elasticities are between 2 and 4. Quantitative exercises suggest preferences are at least as important—and, under reasonable assumptions—significantly more important than lack of pay information in explaining why workers stay at their firm. When we examine the own-firm preference, we find that attachment is both firm-specific and hard to predict based on the valuations of firm outsiders. One potential explanation is that firms are experience goods. Consistent with this, when asked why other workers might be reluctant to switch firms, respondents cite factors such as a general reluctance to undergo change, amenities, and social ties. Pay ranked fourth and lack of opportunities was not a major factor.

Our analysis contributes to the literature on competition in the labor market by examining the sources of firm monopsony power. Our results suggest that worker preferences, rather than beliefs about pay, better explain limited mobility. However, our sample is limited to full-time private sector workers and does not capture beliefs about public sector jobs or the behavior of labor market entrants. Our results also suggest that preferences change after a worker joins a firm; one explanation, suggested by workers in our survey, is that this occurs because workers form social ties. Exploring the types of ties that matter, and whether uncertainty about specific non-pay factors influences workers' behavior are important directions for future work.

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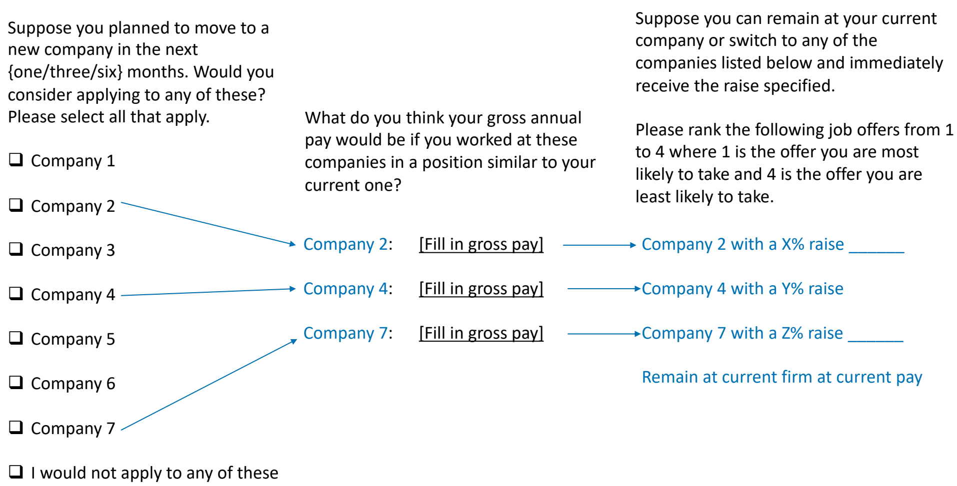
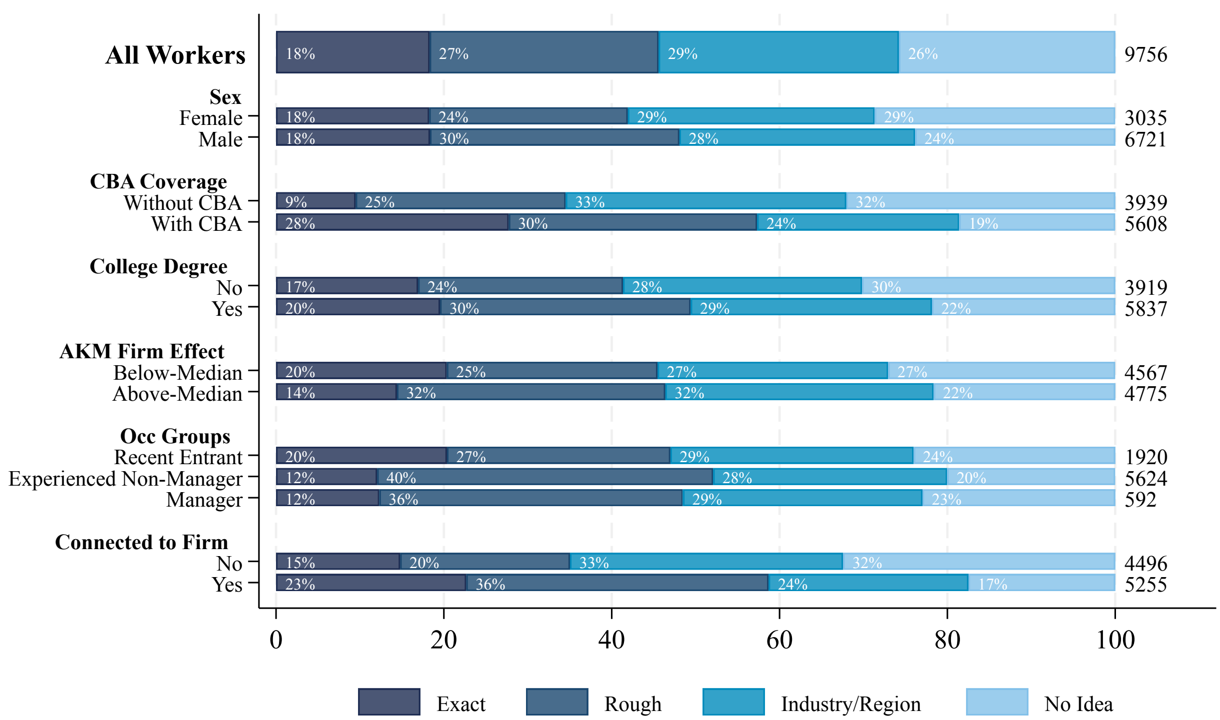
9 Figures and Tables

Figure 1: Researcher-Provided Firm Module



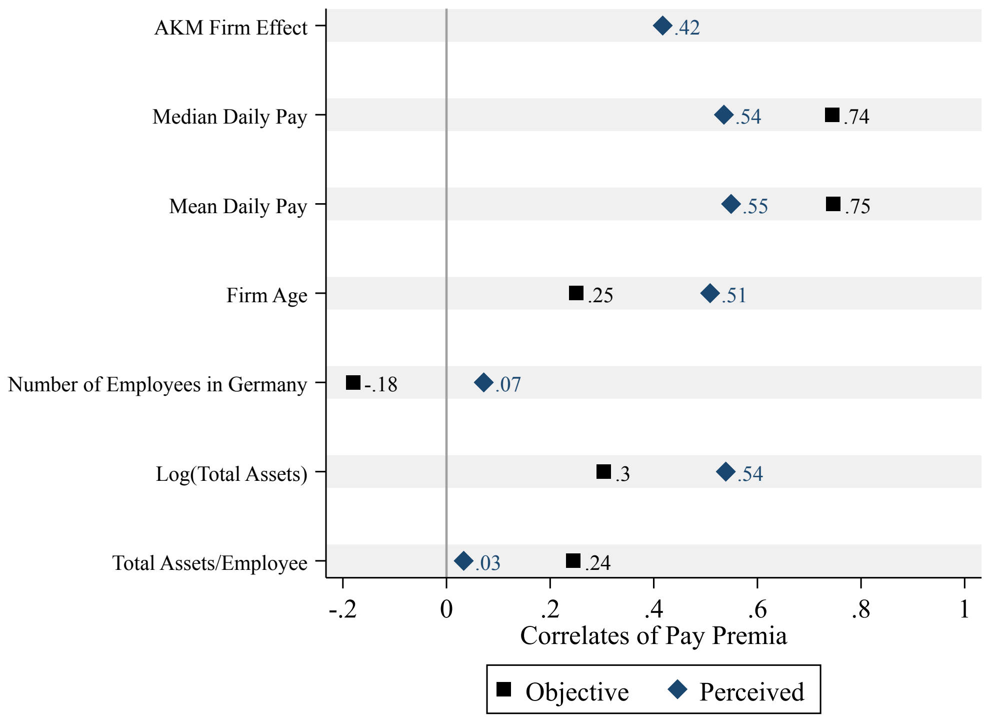
Note: This figure shows how survey respondents transitioned through the researcher-provided firm module. See Appendix Section E.1 for additional information on this module.

Figure 2: Stated Knowledge at Time Worker Applied to Current Firm



Note: Respondents could choose one of four responses: exact knowledge of how much the position pays, a rough idea of how much the position pays, a rough idea of pay in the industry/region, and no or very little idea. Results are weighted using sampling weights. Appendix Table B2 presents additional specifications. The number of observations is listed at the end of each row.

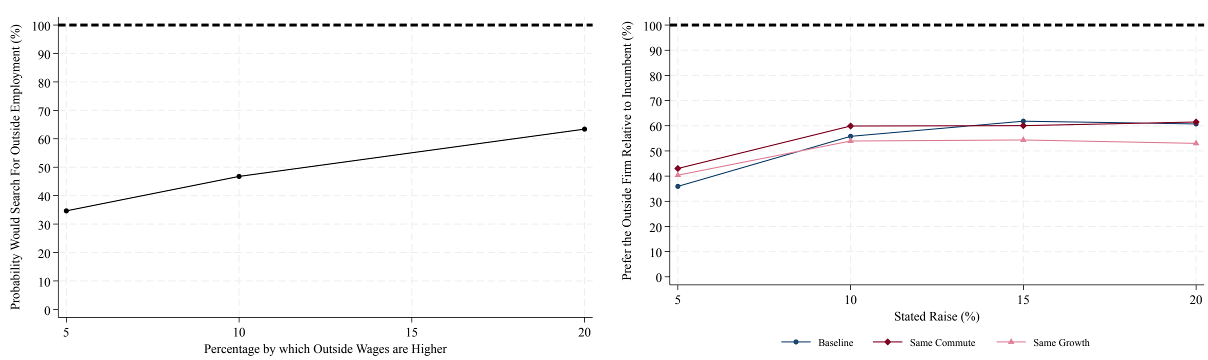
Figure 3: Correlates of Workers' Expected Pay Premia



Note: This figure presents correlations between our baseline estimates of $\tilde{\psi}_j$ (navy) and the firm-specific characteristics indicated on the y-axis. The black dots present analogous correlations between objective measures of pay premia from Bellmann et al. (2020) and the same characteristics. More information on the data is in Appendix E.

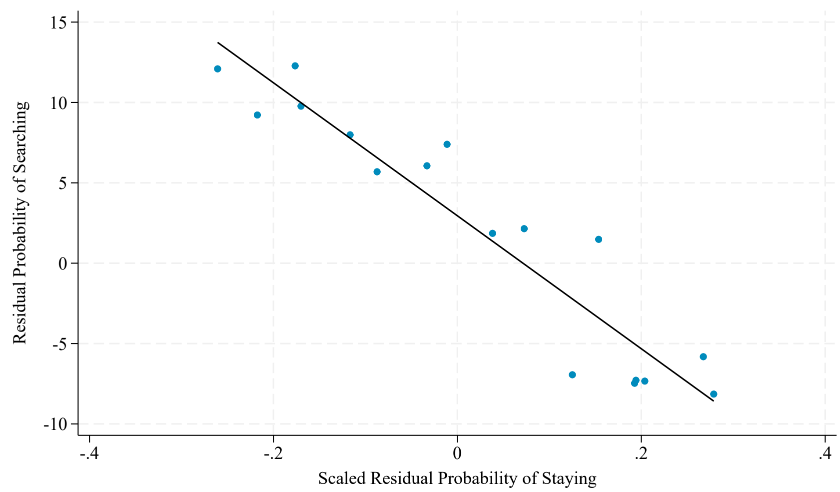
Figure 4: Stated Search and Mobility and Outside Pay

A. Search and Outside Pay B. Mobility and Pay at Provided Firms



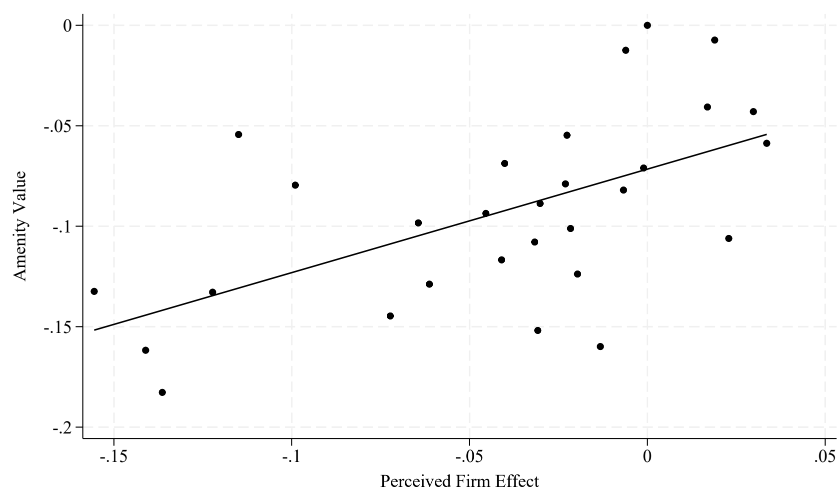
Note: This figure examines workers' propensity to search or move as a function of outside pay. Panel A shows the average probability a worker says that they would start searching for outside employment if they discovered that pay was 5%, 10%, or 20% higher in their market. Panel B shows the share of workers selecting the outside firm over the inside firm in the worker-provided module as a function of the randomized outside pay. Results are weighted using sampling weights. Appendix F shows that the provided destinations are firms where workers from a worker's firm have historically moved to.

Figure 5: Relationship Between the Residual Probabilities of Search and Staying



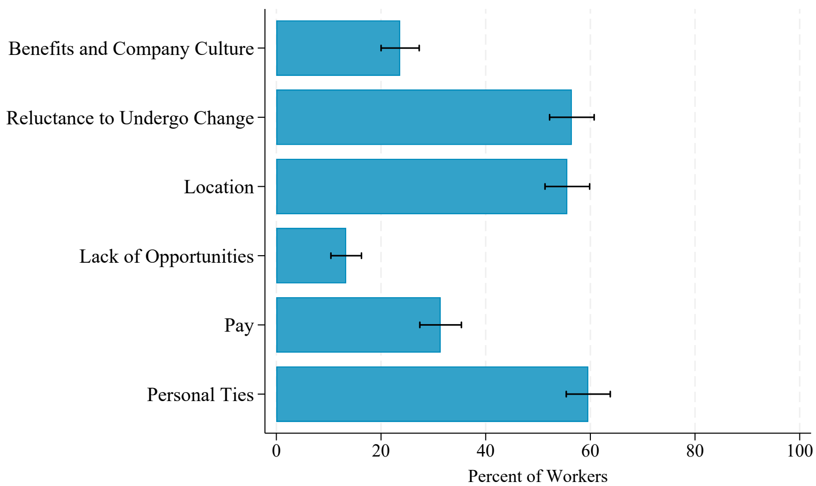
Note: We estimate the residual probability of searching by regressing search probabilities on randomized market pay and computing the residual for each worker. We calculate the scaled residual probability of staying by fitting OLS regressions (using data from the worker-provided firm module) where the outcome variable is an indicator for whether the worker prefers their incumbent firm and the independent variables include the randomized raise associated with the outside firm and a full set of worker dummies.

Figure 6: Relationship Between Perceived Firm Pay Premia and Amenity Valuations



Note: This figure compares estimates of the perceived value of firm-specific amenities and the perceived pay premia for each of the researcher-provided firms. The perceived pay premia ($\tilde{\psi}_j$) come from estimating equation 7 and controlling for whether a firm was provided in the follow-up survey. The amenity valuations come from estimating a rank-ordered logit model using workers' preferences across the researcher-provided firms. We plot estimates of a_j/β rather than a_j so that the results are directly comparable. Both models are fit using sampling weights. We plot a simple line of best fit.

Figure 7: Why Respondents Believe Workers are Reluctant to Switch Firms



Note: This figure shows the reasons that workers report for why people may be reluctant to switch firms. We asked respondents to select the top two reasons. This figure presents the share of respondents who selected each option. Whiskers present 95% confidence intervals. Results are weighted using sampling weights.

Table 1: Characteristics of Surveyed Workers

	Unweighted		Initial Wave		Initial and Follow-Up	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
	(1)	(2)	(3)	(4)	(5)	(6)
Demographics						
Female	0.31	(0.46)	0.40	(0.49)	0.37	(0.48)
Age	33.35	(6.23)	31.13	(5.18)	31.31	(5.19)
German Citizen	0.92	(0.27)	0.89	(0.32)	0.92	(0.28)
College Degree	0.60	(0.49)	0.53	(0.50)	0.61	(0.49)
Apprenticeship	0.33	(0.47)	0.37	(0.48)	0.31	(0.46)
Employment						
Daily Pay (Imputed)	180.05	(60.61)	143.42	(50.40)	150.77	(50.24)
Censored Pay	0.22	(0.41)	0.06	(0.24)	0.07	(0.25)
Hours (Survey)	40.27	(5.99)	40.36	(6.47)	40.43	(5.90)
CBA Covered (Survey)	0.59	(0.49)	0.48	(0.50)	0.45	(0.50)
Manufacturing Sector	0.47	(0.50)	0.22	(0.41)	0.23	(0.42)
Retail Sector	0.09	(0.28)	0.09	(0.29)	0.09	(0.29)
Professional Sector	0.13	(0.34)	0.15	(0.36)	0.17	(0.37)
Observations	9756		9756		3575	

Note: Columns 1-4 include workers who completed the initial survey; Column 5 and 6 includes the subset who also completed the follow-up survey. Statistics in Columns 3-6 are calculated using sampling weights.

Table 2: Characteristics of Researcher-Provided and Worker-Provided Firms

				Worker-Provided Firms					
	Researcher-Provided Firms			Unweighted			Weighted		
	Mean	Std. Dev	N	Mean	Std. Dev	N	Mean	Std. Dev	N
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Number of Employees									
1-10	0.00	(0.00)	30	0.04	(0.19)	479	0.05	(0.22)	479
11-50	0.03	(0.18)	30	0.07	(0.25)	479	0.10	(0.29)	479
51-200	0.00	(0.00)	30	0.10	(0.29)	479	0.04	(0.19)	479
201-1000	0.00	(0.00)	30	0.22	(0.41)	479	0.11	(0.31)	479
1001-10000	0.03	(0.18)	30	0.34	(0.47)	479	0.17	(0.38)	479
10001+	0.93	(0.25)	30	0.24	(0.43)	479	0.53	(0.50)	479
Sector									
Manufacturing	0.57	(0.50)	30	0.31	(0.46)	565	0.40	(0.49)	565
Retail	0.07	(0.25)	30	0.12	(0.32)	565	0.10	(0.30)	565
Professional Services	0.10	(0.31)	30	0.13	(0.33)	565	0.15	(0.35)	565
Information Services	0.10	(0.31)	30	0.07	(0.26)	565	0.08	(0.27)	565
Finance	0.10	(0.31)	30	0.07	(0.25)	565	0.05	(0.22)	565
Other Firm Characteristics									
HQ in Eastern Germany	0.07	(0.25)	30	0.07	(0.25)	565	0.07	(0.25)	565
Year of Incorporation	1936	(49)	30	2008	(835)	476	1957	(328)	476

Note: Columns 1 to 3 (4 to 8) describe the researcher-provided (worker-provided) firms. Columns 7 and 8 weight by the number of times each firm was listed in the initial survey wave. The number of employees, sector, headquarters location, and year of incorporation come from Orbis. We include all worker-provided firms mentioned at least twice, for which we hand-collected characteristics (Appendix E.1).

Table 3: Variation in Expected Pay at Other Firms

	Fraction Identical	Std. Deviation	Max/Min	N
	(1)	(2)	(3)	(4)
A. Researcher-Provided Firms				
Initial Survey	0.26	5673	1.18	3715
Follow-Up Survey	0.30	5294	1.15	3163
B. Worker-Provided Firms				
All Workers	0.25	5863	1.19	4433
All in Same State	0.22	4869	1.18	509
All in Same District	0.26	4701	1.19	173
All in Same Municipality	0.22	5084	1.21	159

Note: Column 1 provides the share of workers that provided the same pay at each of the provided firms. Columns 2 provides the average standard deviation in Euros. We compute the ratio of the max and min within worker (across observations) and average across workers. Column 4 provides the number of workers. Panel A presents data from the researcher-provided firm module. As described in Appendix E.1.2, we asked a random 50% of surveyed workers to provide pay information in the initial wave; in the follow-up survey we asked all workers to provide pay information. Panel B presents analogous results for workers who provided the names of specific outside firms they would consider in the initial survey wave. We use workers' provided location information for sub-group analysis in Panel B; we asked a random 25% of workers to provide this information. Results are weighted using sampling weights.

Table 4: Between-Sample Agreement in Estimates of ψ_j

	Baseline Model		Rank-Ordered Logit	
	Correlation	Test of Equality (p-value)	Correlation	Test of Equality (p-value)
	(1)	(2)	(3)	(4)
Split-Sample	0.88	0.85	0.94	0.25
Sex	0.75	0.02	0.87	0.81
CBA	0.84	0.19	0.91	0.84
College Education	0.65	0.12	0.90	0.52
Current Firm AKM Effect (Split at Median)	0.76	0.01	0.90	0.33
Searched in Past 6 Mo.	0.73	0.02	0.86	0.73
Knew Wages at Application	0.73	0.20	0.92	0.56
Easy to Get a Better Job	0.82	0.38	0.93	0.93
Tenure (Split at 2 Years)	0.86	0.47	0.94	0.45

Note: This table examines the agreement in the estimates of ψ_j for different subsamples of workers. The first row uses a random, non-overlapping split of workers; the remaining splits are based on the listed characteristic. Columns 1 and 2 are based on estimating equation 7 and controlling for whether a firm was provided in the follow-up survey. Columns 3 and 4 are based on estimates from fitting a rank-ordered logit model to workers' firm-specific pay expectations (ranking firms by workers' expected pay). All regressions use sampling weights.

Table 5: Relationship Between Firm-Specific Pay Expectations and Consideration

	All Workers			Risk Tolerance		Would be Reluctant to Apply if P(Success) Were Low	
				Low	High	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Own Pay Expectation	0.341***	0.313***	0.309***	0.256***	0.441***	0.297***	0.240***
	(0.050)	(0.050)	(0.050)	(0.056)	(0.103)	(0.084)	(0.087)
Distance Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Same-Sector Control	No	No	Yes	Yes	Yes	Yes	Yes
Observations	21272	21272	21272	14967	6305	5392	6507
Number of Workers	6440	6440	6440	4519	1921	1476	1781
Test of equality (p-value)	---	---	---	.116		.64	
Mean of Outcome	0.311	0.311	0.311	0.310	0.313	0.274	0.305
Implied Elasticity	1.097***	1.007***	0.993***	0.827***	1.406***	1.082***	0.788***
	(0.161)	(0.161)	(0.161)	(0.182)	(0.330)	(0.307)	(0.286)

Note: This table presents estimates of equation 8. Each regression controls for worker and firm fixed effects, survey wave, and the listed controls. The distance controls include the logarithm of the driving distance between the districts where a worker currently works and where the researcher-provided firm has its headquarters, as well as an indicator for whether the distance is zero. Regressions with the same-sector control also include an indicator for whether the control is missing. Following Dohmen et al. (2011), we elicited risk tolerance on a ten-point scale in the initial survey and define someone as having high risk tolerance if they selected seven or above. In the follow-up survey, we asked workers whether they would "be reluctant to apply for a job if the probability I would get an offer is low." We define workers as being reluctant to apply if they selected four or above on the seven-point scale. Appendix Table A3 presents alternative specifications. Regressions use sampling weights. Standard errors are clustered by worker. Elasticities are evaluated at the mean of consideration, are given at the bottom of the table. Additional specifications are in Appendix Table A4. Levels of significance: * 10%, ** 5%, and *** 1%.

Table 6: Alternative Designs Linking Pay Expectations and Consideration

	Stated Consideration			Free-Text Responses		
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of Dependent Variable		0.254			0.017	
	A. Perceived Firm Effect (Split-Sample)					
Firm Premium (Split-Sample)	0.953***	1.055***	1.059***	0.089***	0.091***	0.093***
	(0.110)	(0.127)	(0.127)	(0.011)	(0.013)	(0.013)
Observations	89742	89742	89742	224388	224388	224388
Number of Workers	9756	9756	9756	9756	9756	9756
	B. Observed Firm Effect					
Firm Premium (Observed)	0.174***	0.165***	0.173***	0.014***	0.014***	0.014***
	(0.033)	(0.033)	(0.035)	(0.005)	(0.005)	(0.005)
Observations	89258	89258	89258	214632	214632	214632
Number of Workers	9756	9756	9756	9756	9756	9756
Firm Characteristics	Size	Size, Brand Recognition	Size, Brand Recognition, CBA	Size	Size, Brand Recognition	Size, Brand Recognition, CBA
Fixed Effects	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector	Worker, Sector

Note: Each regression controls for logarithm of driving distance, survey wave, whether the firm is in the same sector as the worker, the listed firm characteristics, and indicators for whether the listed controls are missing. Regressions use sampling weights; other specifications are in Appendix Table B4. Brand recognition is from Kantar (2023). We use data from both the initial and follow-up surveys. Standard errors are clustered by worker. Levels of significance: * 10%, ** 5%, and *** 1%.

Table 7: Market Search Elasticities

		Risk Tolerance		Would be Reluctant to Apply if P(Success) Were Low		Consider the Amount of Competition		Believe High Wage Firms Get More Applicants	
	All	Low	High	No	Yes	No	Yes	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Outside Pay	1.880***	1.847***	1.896***	2.207***	1.550***	1.702***	2.062***	1.932***	1.768***
	(0.146)	(0.172)	(0.277)	(0.216)	(0.196)	(0.216)	(0.198)	(0.180)	(0.246)
Constant	0.263***	0.256***	0.289***	0.245***	0.286***	0.275***	0.252***	0.271***	0.251***
	(0.020)	(0.023)	(0.040)	(0.031)	(0.026)	(0.028)	(0.028)	(0.024)	(0.033)
Implied Elasticity	7.139***	7.207***	6.559***	9.024***	5.423***	6.180***	8.188***	7.133***	7.048***
	(1.058)	(1.272)	(1.818)	(1.963)	(1.143)	(1.375)	(1.640)	(1.269)	(1.866)
Observations	3557	2551	1006	1587	1954	1857	1685	2223	1334

Note: This table presents OLS estimates of the relationship between stated search probabilities and (randomized) outside pay. Column 1 includes the full sample; the remainder split the sample by workers' risk tolerance (Columns 2–3, elicited in the initial survey), by reluctance to apply given a low probability of success (Columns 4–5, elicited in the follow-up survey), consideration of number of other applicants in applying (Columns 6–7, elicited in the follow-up survey), and whether they believe higher pay is associated with more competition (Columns 8–9, elicited in the follow-up survey). We present the slope, intercept, and implied elasticity (slope/intercept), evaluated at a raise of zero. Standard errors are clustered at the worker level. Regressions use sampling weights. Levels of significance: * 10%, ** 5%, and *** 1%.

Table 8: Firm-Specific Labor Supply Elasticities

	Move to a Researcher-Provided Firm	Move to a Worker-Provided Firm		
		Baseline	Same Commute	Same Growth
	(1)	(2)	(3)	(4)
Outside Pay	0.391***	1.566***	1.288***	0.945**
	(0.133)	(0.236)	(0.364)	(0.368)
Constant	0.321***	0.359***	0.425***	0.408***
	(0.018)	(0.033)	(0.042)	(0.043)
Implied Elasticity	1.219**	4.364***	3.031***	2.320**
	(0.477)	(1.037)	(1.139)	(1.130)
Observations	21872	12217	6264	6272

Note: This table presents OLS estimates of the relationship between stated mobility and (randomized) outside pay. We collapse the rank-ordered data so that we have one observation for each outside firm; the dependent variable is an indicator for whether the worker ranks the outside firm higher than the incumbent and the independent variable is the randomized raise associated with that outside firm. Column 1 uses the baseline discrete choice experiment in the researcher-provided firm module, Columns 2 to 4 use the experiments in the worker-provided firm module. We present the slope, intercept, and implied elasticity (slope/intercept). Standard errors are clustered at the worker level. Regressions use sampling weights. Levels of significance: * 10%, ** 5%, and *** 1%.

Table 9: Incumbent Premia

	Move to a Researcher-Provided Firm		Move to a Worker-Provided Firm		
	Baseline	Distance Controls	Baseline	Same Commute	Same Growth
	(1)	(2)	(3)	(4)	(5)
Log Raise	6.172***	6.251***	8.112***	12.323***	10.207***
	(0.492)	(0.495)	(0.824)	(1.283)	(1.207)
Incumbent	1.132***	0.703***	0.596***	0.771***	0.774***
	(0.074)	(0.129)	(0.106)	(0.134)	(0.135)
Distance Controls	No	Yes	No	No	No
Observations	29961	29961	17539	8821	8782
Number of Workers	7735	7735	4796	2400	2385
Implied Incumbent Premium	0.183***	0.112***	0.074***	0.063***	0.076***
	(0.009)	(0.019)	(0.008)	(0.007)	(0.008)

Note: This table presents estimates of a rank ordered logit model, fit to workers' preferences. Columns 1 and 2 use data from the researcher-provided firm module. The model in Column 1 includes only the randomized raise, an indicator for whether the firm is the worker's current place of work, and a control for the survey wave. Column 2 adds a control for the log driving distance between the firm and the worker's current place of work. Columns 3 to 5 use data from the worker-provided firm module. Regressions use sampling weights. The model drops observations with ties. Appendix Table A6 presents additional specifications. Levels of significance: * 10%, ** 5%, and *** 1%.

Table 10: Relative Importance of Preferences and Beliefs

	High Attachment ("Provided Firms")		Low Attachment ("Consideration Set Firms")	
	Search	Mobility	Search	Mobility
	(1)	(2)	(3)	(4)
A. Below Median-Wage Workers				
Full Transparency				
Baseline $\lambda = 0.5$	12%	7%	12%	9%
High Compression $\lambda = 0.25$	19%	10%	19%	13%
Low Compression $\lambda = 0.75$	6%	3%	6%	4%
Eliminate Attachment				
Baseline $\lambda = 0.5$	34%	24%	14%	12%
B. All Workers				
Full Transparency				
Baseline $\lambda = 0.5$	8%	6%	7%	7%
High Compression $\lambda = 0.25$	12%	9%	10%	10%
Low Compression $\lambda = 0.75$	4%	3%	3%	3%
Eliminate Attachment				
Baseline $\lambda = 0.5$	19%	15%	10%	8%

Note: This table describes results from the simulation described in Section 6.4. We assume that workers' beliefs are compressed relative to the truth such that $\Delta\tilde{w} = \lambda\Delta w$. In our baseline scenario $\lambda = 0.5$; the high compression scenario sets $\lambda = 0.25$ and the low compression scenario sets $\lambda = 0.75$. In the transparency scenario, we tell workers they could earn the wage associated with the median wage premium firm; we assume that Δw is the gap between this firm and their current firm. We use the estimates described in Table 7 to examine the impact on worker search and the estimates in Table 9 to examine the impact on worker mobility. The sample is restricted to workers currently employed at firms with below-median wage premia. More details on estimation are in Section 6.4. This table weights results by sampling weights; Appendix Table B5 presents results for alternative weighting schemes.

Table 11: Firm-Specific Attachment

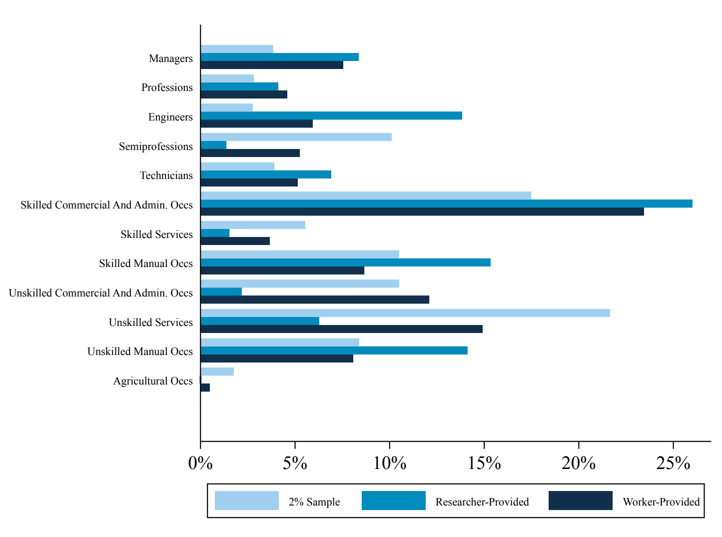
	Preferences Over Outside Firms		Preferences Over All Firms		Preferences Over Considered or Incumbent Firm
	(1)	(2)	(3)	(4)	(5)
Log Raise (β)	8.560***	14.136***	6.162***	8.712***	15.647***
	(3.090)	(3.358)	(2.322)	(2.294)	(4.409)
Observations	4217	4217	5671	5671	3001
Number of Workers (Clusters)	1177	1177	1200	1200	1192
Distance Controls	Yes	Yes	Yes	Yes	Yes
Data Include Incumbent Ranking	No	No	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Firm-Consider Interactions	No	Yes	No	Yes	No
Firm-Incumbent Interactions	No	No	Yes	Yes	Yes
Amenities are Non-Zero $a_j = 0 \forall j$					
p-value	<.001	<.001	<.001	<.001	<.001
Degrees of Freedom	29	29	29	29	29
Consideration Doesn't Affect Valuations $\zeta_j = 0 \forall j$					
p-value	---	<.001	---	<.001	---
Degrees of Freedom	---	30	---	30	---
There is an Incumbent Effect $\iota_j = 0 \forall j$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	18	18	17
Insiders and Outsiders' Valuations Differ $\iota_j = \iota_k \forall j, k$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	17	17	16
Insiders and Outsiders Who Want to Move to the Firm Have Different Valuations $\iota_j = \zeta_j \forall j$					
p-value	---	---	---	<.001	---
Degrees of Freedom	---	---	---	18	---

Note: This table presents logit coefficients (first row) and p-values (remaining rows) from fitting a rank-ordered logit model to workers' stated preferences over researcher-provided firms. Each regression controls for the randomized raise, and includes the firm fixed effects indicated in the relevant column. The baseline sample includes workers who currently work at any of the 30 researcher-provided firms. The data include workers' choices over outside firms (Columns 1 and 2) or over all firms (Columns 3 and 4). Column 5 restricts the sample to worker-firm observations in which the worker currently works at the firm or in which the worker has said he/she would consider applying to the firm if he/she wanted to switch firms. Because we do not observe workers at all 30 firms, tests for the incumbent premia are conducted for the set of identified parameters. Regression use sampling weights; alternative specifications are in Appendix Table B6. Standard errors are clustered at the worker level.

Online Appendix for "Why Workers Stay: Pay, Beliefs, and Attachment"
(Caldwell, Haegele & Heining)

A Appendix Figures and Tables

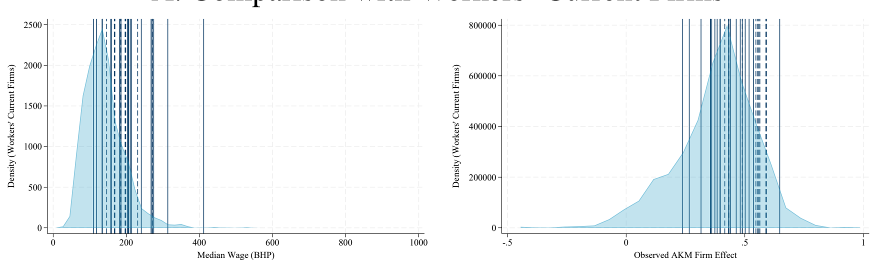
Figure A1: Occupational Distribution of Researcher-Provided Firms



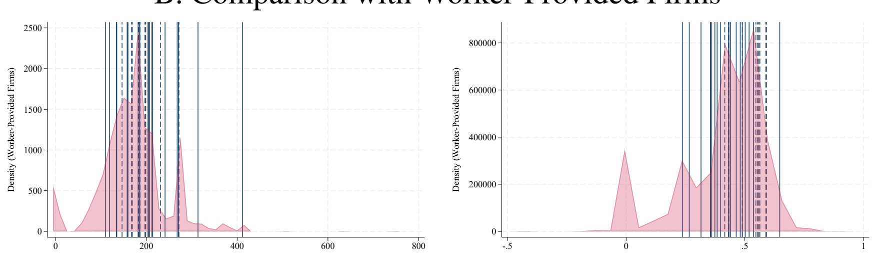
Note: This figure compares the span of major occupational groups in the overall German workforce using a 2% random sample to workers at our researcher-provided and worker-provided firms.

Figure A2: Comparison of Researcher-Provided Firms and Workers' Firms

A. Comparison with Workers' Current Firms

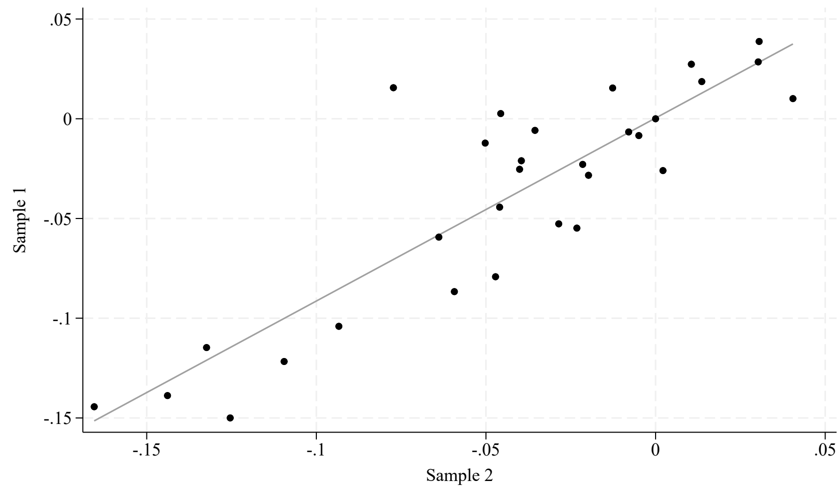


B. Comparison with Worker-Provided Firms



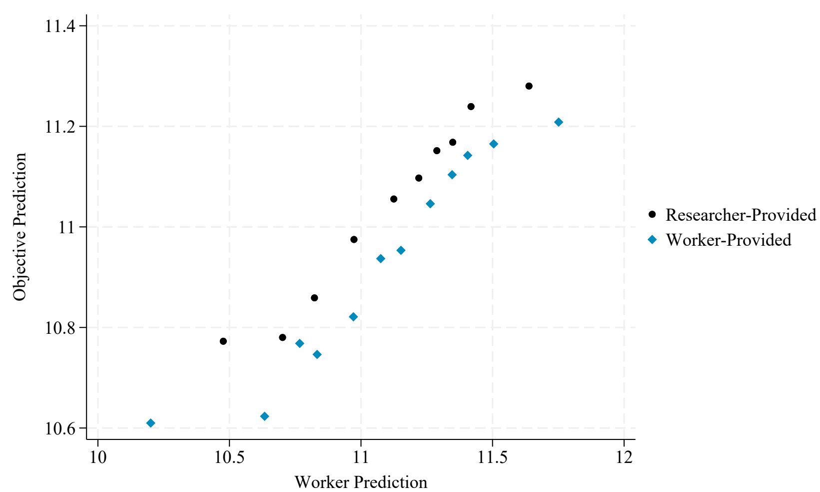
Note: This figure compares the 30 researcher-provided firms to the workers' current firms (Panel A) and to the worker-provided firms (Panel B). The left (right) figures plot the kernel density of workers' current or provided firms' median pay (firm wage premium from Bellmann et al. (2020)). The vertical lines indicate the values for researcher-provided firms. Solid lines indicate firms included in the initial survey. Dashed lines indicate firms only included in the follow-up survey. Results are weighted using sampling weights.

Figure A3: Expected Firm Pay Premia: Split-Sample Evidence



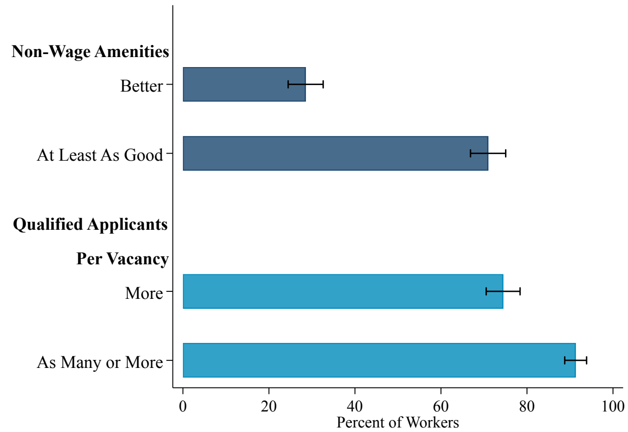
Note: This figure examines the relationship between workers' expected pay premia estimated by fitting equation 7 in two subsamples of workers in the researcher-provided firm module. We created the subsamples by randomly dividing our sample in half. Each dot represents the expected pay premium associated with one researcher-provided firm. There are 30 dots, one of which is constrained to be 0 in each sample (ψ_1). Table 4 presents tests of equality for estimates of ψ_j in different groups of workers, as well as correlations in the estimates for each sample.

Figure A4: Correlation Between Worker Expectations and Objective Pay Predictions



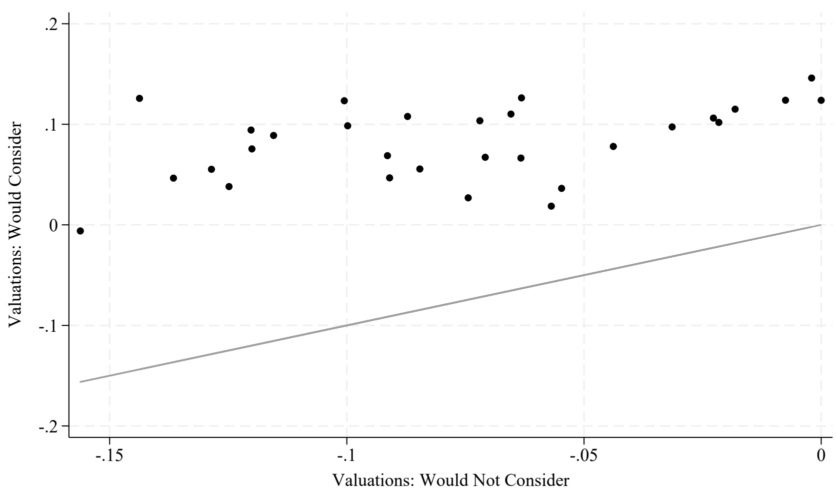
Note: This figure compares workers' expectations to the objective pay predictions. We estimate workers' expected pay premia by fitting equation 7. We construct the objective prediction by taking the logarithm of a worker's current pay, subtracting the firm pay effect of their current firm, and adding the firm pay effect of the specified firm. The black dots are based on data from the researcher-provided firm modules in both the initial and follow-up surveys. The blue diamonds are based on data from the worker-provided firm module in the initial survey. The plot is created using the package created by Cattaneo et al. (2024). Results are weighted using sampling weights.

Figure A5: Workers' Perceptions about Amenities and Application Rates



Note: In the follow-up survey, respondents compared two firms in the same labor market—one paying 10% above and one 30% above market average. They indicated which firm had better non-wage amenities and attracted more qualified applicants. The figure shows the fraction selecting the higher-paying firm as strictly better (top bar) or similar or better (bottom bar). Results use sampling weights; whiskers show 95% confidence intervals.

Figure A6: Valuations by Consideration



Note: This figure shows the valuations for each of the 30 researcher-provided firms for those who say they would (y-axis) and would not (x-axis) consider applying to the firm if they wanted to switch firms. Each dot presents the estimate from a separate firm; the estimate of a_1 for those who would not consider applying to the firm is constrained to be zero. We estimate valuations using a rank-ordered logit model applied to the discrete choice experiments in the researcher-provided firm module. The model includes a control for the (randomized) wage offer, the order in which the firm was listed, and a full set of firm dummies, interacted with worker-firm specific consideration. We convert valuations to monetary equivalents by dividing by the coefficient on the randomized wage offer.

Table A1: Firm Effects Have Explanatory Power Beyond Sector and Location

	Sector Fixed Effects		Sector and State Fixed Effects		Sector-State Fixed Effects	
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Researcher-Provided Firms					
Adjusted R-Squared	0.858	0.866	0.862	0.866	---	---
Observations	19734	19734	19734	19734	---	---
Parameters Tested	24		17		---	---
F-Statistic	7		5			
p-value	<.01		<.01		---	---
	B. Worker-Provided Firms					
Adjusted R-Squared	0.888	0.907	0.868	0.885	0.871	0.881
Observations	10733	10733	8527	8527	8138	8138
Parameters Tested	535		387		334	
F-Statistic	17370894		392000000000		18485608	
p-value	<.01		<.01		<.01	

Note: This table reports the adjusted R-squared from regressions of log expected earnings at researcher-provided (Panel A) or worker-provided (Panel B) firms. Columns 1, 3, and 5 include worker fixed effects, and the fixed effects indicated in the column. Columns 2, 4, and 6 add firm fixed effects. The F-statistic tests whether firm dummies are jointly zero. Standard errors are clustered at the worker level. Regressions use sampling weights. We exclude results with limited within-sector/state firm pairs.

Table A2: Decomposition of Workers' Firm-Specific Pay Expectations

	Worker Expectations					
		Informed at Application		Recent Search Activity		
	All Workers	Yes	No	Yes	No	Objective Predictions
	(1)	(2)	(3)	(4)	(5)	(6)
Number of Parameters						
Person Effects	5305	2643	2662	3971	1334	1103
Firm Effects	30	30	30	30	30	29
Summary of Parameter Estimates						
Std. Dev. Person Effects	0.365	0.386	0.344	0.358	0.386	0.516
Std. Dev. Firm Effects	0.051	0.055	0.049	0.051	0.053	0.091
Model RMSE	0.105	0.108	0.102	0.105	0.106	0.315
Addendum						
Std. Dev. Log(Salary)	0.378	0.401	0.357	0.371	0.401	0.254
Variance Log(Salary)	0.143	0.160	0.127	0.138	0.160	0.064
Observations	19431	9692	9739	14580	4851	3879

Note: This table describes the results of estimating equation 7 using data from the researcher-provided firm module and controlling for survey wave. Column 1 presents the baseline results. Columns 2 and 3 present results for workers who did and did not know pay at time of application to their current firm. Columns 4 and 5 present results for workers who did and did not engage in job search in the previous six months. Column 6 presents analogous statistics for our objective predictions, which use firm effects from Bellmann et al. (2020). Regressions use sampling weights.

Table A3: Relationship Between Pay Expectations and Consideration: Additional Specifications

	Alternative Specifications of Distance			Alternative Samples		Alternative Weighting Schemes	
	Quadratic in Distance	Direct Distance	Closest Establishment	Initial Survey Only	Follow-Up Only	Unweighted	Population Weights
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Own-Pay Expectation	0.312***	0.309***	0.331***	0.427***	0.249	0.225***	0.255***
	(0.050)	(0.050)	(0.050)	(0.122)	(0.174)	(0.026)	(0.085)
Observations	21272	21272	21272	15095	5980	21272	21236
Number of Workers (Clusters)	6440	6440	6440	0.396	0.309	0.330	0.335
Mean of Outcome	0.31	0.31	0.31	0.35	0.24	0.33	0.32
Implied Elasticity	1.004***	0.996***	1.067***	1.214***	1.052	0.677***	0.788***
	(0.162)	(0.161)	(0.161)	(0.347)	(0.736)	(0.079)	(0.261)

Note: Each regression uses data from the researcher-provided firm module and controls for worker and firm fixed effects, for whether the firm and worker are in the same sector, for the log driving distance between the worker and indicated firm, for the survey wave, and for indicators for whether the respective controls are missing. Columns 1–3 replace log driving distance with quadratic driving distance (1), log point-to-point distance (2), or log distance to the firm's closest establishment (3). Columns 4–5 restrict the sample to initial and follow-up survey waves, respectively. Columns 1–5 use sampling weights; Column 6 is unweighted; Column 7 is re-weighted to match the German full-time population. Standard errors are clustered at the worker level. Elasticities of consideration with respect to pay expectation (evaluated at the mean) are provided at the bottom. Levels of significance: * 10%, ** 5%, and *** 1%.

Table A4: Relationship Between Pay Expectations and Consideration: Additional Heterogeneity

	Consider the Amount of Competition		Believe High Wage Firms Get More Qualified Applicants	
	No	Yes	No	Yes
	(1)	(2)	(3)	(4)
Own-Pay Expectation	0.271***	0.366***	0.322***	0.228**
	(0.083)	(0.070)	(0.057)	(0.107)
Observations	5685	15127	16759	4513
Number of Workers (Clusters)	1544	4575	5190	1250
Test of equality (p-value)	0.830		0.439	
Mean of Outcome	0.300	0.329	0.312	0.304
Implied Elasticity	0.905***	1.113***	1.031***	0.752**
	(0.277)	(0.212)	(0.183)	(0.352)

Note: This table presents additional specifications for the analysis in Table 5. Each regression uses data from the researcher-provided firm module and controls for worker and firm fixed effects, for whether the firm and worker are in the same sector, for the log driving distance between the worker and indicated firm, for the survey wave, and for indicators for whether the respective controls are missing. Columns 1-2 divide the sample by whether a worker says they consider the amount of competition when deciding whether to apply. Columns 3-4 divide the sample by whether a worker believes high wage firms receive more qualified applicants per vacancy.

Table A5: Alternative Designs Linking Pay Expectations and Consideration: Heterogeneity

	Risk Tolerance		Would be Reluctant to Apply if P(Success) Were Low		Consider the Amount of Competition		Believe High Wage Firms Get More Applicants	
	Low	High	No	Yes	No	Yes	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. Stated Consideration							
Firm Premium (Split-Sample)	0.844***	0.929***	0.823***	0.860***	0.871***	0.825***	0.860***	0.910***
	(0.107)	(0.141)	(0.165)	(0.152)	(0.130)	(0.152)	(0.102)	(0.167)
Observations	63508	26234	20696	25480	24180	22009	72374	17368
Number of Workers	6874	2882	1592	1960	1860	1693	8420	1336
Mean of Dependent Variable	0.249	0.265	0.269	0.259	0.259	0.269	0.250	0.272
Firm Premium (Observed)	0.153***	0.223***	0.148**	0.153**	0.231***	0.076	0.175***	0.165**
	(0.040)	(0.061)	(0.071)	(0.069)	(0.071)	(0.068)	(0.036)	(0.083)
Observations	63157	26101	20478	25215	23911	21796	72064	17194
Number of Workers	6874	2882	1592	1960	1860	1693	8420	1336
Mean of Dependent Variable	0.250	0.266	0.270	0.260	0.260	0.270	0.250	0.273
	B. Free-Text Responses							
Firm Premium (Split-Sample)	0.077***	0.124***	0.098***	0.124***	0.101***	0.123***	0.089***	0.107***
	(0.012)	(0.016)	(0.023)	(0.020)	(0.020)	(0.024)	(0.012)	(0.026)
Observations	158102	66286	36616	45080	42780	38939	193660	30728
Number of Workers	6874	2882	1592	1960	1860	1693	8420	1336
Mean of Dependent Variable	0.016	0.018	0.019	0.018	0.019	0.018	0.016	0.020
Firm Premium (Observed)	0.016***	0.011	0.019	0.017	0.016	0.019	0.016***	0.004
	(0.006)	(0.010)	(0.015)	(0.012)	(0.013)	(0.014)	(0.005)	(0.016)
Observations	151228	63404	35024	43120	40920	37246	185240	29392
Number of Workers	6874	2882	1592	1960	1860	1693	8420	1336
Mean of Dependent Variable	0.017	0.019	0.020	0.019	0.019	0.019	0.017	0.021

Note: Each regression controls for logarithm of driving distance, survey wave, whether the firm is in the same sector as the worker, the listed firm characteristics, and indicators for whether the listed controls are missing. Columns 1-2 split by risk tolerance. Columns 3-4 split by whether the worker says they would be reluctant to apply to a firm if they thought the probability of success were low. Columns 5-6 split by whether a worker says they consider the amount of competition when applying. Columns 7-8 split by whether a worker believes high wage firms receive (strictly) more qualified applicants per vacancy. Regressions use sampling weights; other specifications are in Appendix Table B4. Standard errors are clustered by worker. Levels of significance: * 10%, ** 5%, and *** 1%.

Table A6: Robustness of the Incumbent Premium

	Alternative Specifications of Distance				Alternative Samples		Alternative Weighting	
	Log Distance	Quadratic in Distance	Direct Distance	Closest Establishment	Initial Survey Only	Follow-Up Only	Unweighted	Population Weights
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. Researcher-Provided Firms							
Implied Incumbent Premium	0.112***	0.123***	0.113***	0.167***	0.142***	0.083***	0.127***	0.058
	(0.019)	(0.011)	(0.019)	(0.011)	(0.025)	(0.023)	(0.007)	(0.043)
Observations	29961	29961	29961	29961	16594	13367	29961	29918
Number of Workers	7735	7735	7735	7735	4322	2351	7735	7724
	B. Worker-Provided Firms							
Implied Incumbent Premium	0.079***	0.078***	0.066***	0.072***	---	---	0.099***	0.053***
	(0.010)	(0.008)	(0.009)	(0.008)	---	---	(0.005)	(0.012)
Observations	15259	15259	17539	17539	---	---	15259	15242
Number of Workers	4784	4784	4796	4796	---	---	4784	4778

Note: This table describes the results of estimating a rank-ordered logit model model fit to workers' preferences over outside firms specified in the researcher-provided firm module (Panel A) or the worker-provided firm module (Panel B). The model includes controls for the randomized raise of the outside firm, an indicator for whether the worker currently works at that firm, and the log driving distance between that firm and the worker's current place of work. Columns 2 to 4 use alternative distance metrics: quadratic driving distance (2), log point-to-point distance (3), and log distance to the closest firm establishment (4). Columns 5 and 6 restrict the sample to initial and follow-up survey waves, respectively. Columns 1 to 6 use sampling weights; Column 7 is unweighted; Column 8 re-weights to match the German full-time population. Levels of significance: * 10%, ** 5%, and *** 1%.

B Additional Results and Robustness Checks

B.1 Robustness to Alternative Weighting Schemes

Our main analysis uses survey weights to account for the fact that we oversampled workers from firms that participated in the firm survey used in Caldwell, Haegele and Heining (2024). However, our results are robust re-weighting the sample to match the overall population of regular full-time workers in Germany. We use the following variables to match our sample: female, labor market experience, federal state, German citizenship, occupational group (i.e., labor market entrant, experienced non-manager, manager), industry (i.e., manufacturing, retail, professional), and pay.

Results in Section 3. Appendix Table B1 shows how re-weighting changes sample composition.

Table B1: Characteristics of Surveyed Workers: Re-weighted

	Unweighted			Sampling Weights		Population Weights	
	Mean	Std. Dev.	N	Mean	Std. Dev.	Mean	Std. Dev.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Demographics							
Female	0.31	(0.46)	9756	0.40	(0.49)	0.48	(0.50)
Age	33.35	(6.23)	9756	31.13	(5.18)	32.13	(6.39)
German Citizen	0.92	(0.27)	9756	0.89	(0.32)	0.87	(0.34)
College Degree	0.60	(0.49)	9756	0.53	(0.50)	0.33	(0.47)
Apprenticeship	0.33	(0.47)	9756	0.37	(0.48)	0.51	(0.50)
Employment							
Daily Pay	180.05	(60.61)	9756	143.42	(50.40)	101.41	(54.29)
Censored Pay	0.22	(0.41)	9756	0.06	(0.24)	0.04	(0.20)
Hours (Survey)	40.27	(5.99)	9756	40.36	(6.47)	38.89	(7.64)
CBA Covered (Survey)	0.59	(0.49)	9547	0.48	(0.50)	0.50	(0.50)
Manufacturing Sector	0.47	(0.50)	9756	0.22	(0.41)	0.18	(0.38)
Retail Sector	0.09	(0.28)	9756	0.09	(0.29)	0.13	(0.34)
Professional Sector	0.13	(0.34)	9756	0.15	(0.36)	0.05	(0.22)
Observations				9756			

Note: This table describes the individuals in our sample under different weighting schemes, indicated in each column.

Results in Section 4. Appendix Table B2 shows that our finding that many workers were informed about pay at the time they applied to their current firm is robust to re-weighting In our main results, 45% of workers report firm-specific information, we obtain 41% using population weights. Appendix Table B3 shows that, after re-weighting, we still find that workers perceive heterogeneous (in pay) outside options.

Table B3: Variation in Expected Pay at Other Firms: Robustness to Re-weighting

	Fraction Identical			
	Baseline	Unweighted	Population Weights	N
	(1)	(2)	(3)	(4)
A. Researcher-Provided Firms				
Initial Survey	0.26	0.23	0.29	3518
Follow-Up Survey	0.30	0.26	0.40	3057
B. Worker-Provided Firms				
All Workers	0.25	0.26	0.25	4305
All in Same State	0.22	0.27	0.32	507
All in Same District	0.26	0.33	0.45	172
All in Same Municipality	0.22	0.34	0.40	158

Note: This table presents re-weighted results analogous to those in Table 3. Column 1 presents our baseline specification. Columns 2 and 3 present unweighted and population-weighted results, respectively.

Table B2: Stated Knowledge at Time of Application: Robustness to Re-weighting

	Knew Pay in Position at This Company		Rough Idea of Pay in Industry or Region	Little or No Idea	Obs
	Exactly	Rough Idea			
	(1)	(2)	(3)	(4)	(5)
Baseline	18%	27%	29%	26%	9756
Unweighted	14%	36%	29%	22%	9756
Population Weights	15%	26%	27%	32%	9756

Note: The first row presents baseline results with sample weights (Figure 2). Rows 2 and 3 present unweighted and population-weighted results, respectively.

Results in Section 5. Appendix Table A3 documents that our results on search in Section 5.1 of the main text are robust to re-weighting. Appendix Table B4 below confirms that the results from the two alternative designs (Section 5.2) are also robust.

Results in Section 6. Appendix Table A6 shows that our main estimates of the incumbent premium are sensitive to reweighting. Appendix Figure B1 and Appendix Table B5 show other core results, including the simulation, are robust to population weighting.

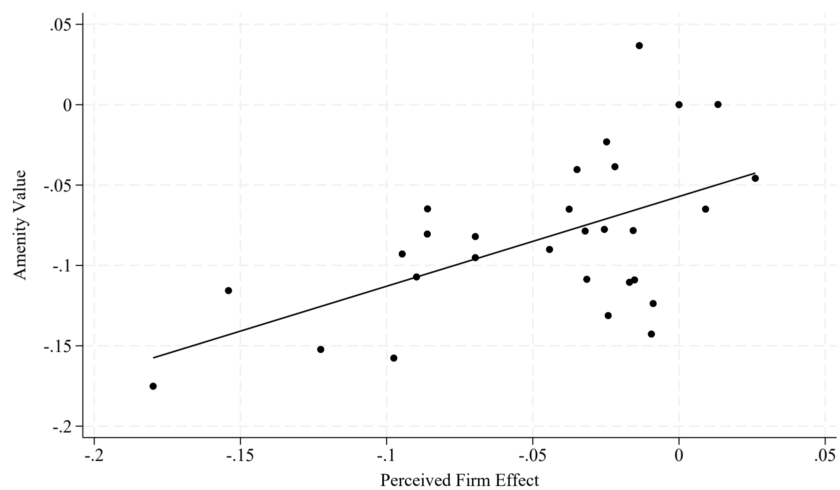
Results in Section 7. We continue to find evidence of firm-specific attachment under alternative weighting schemes, as shown in Appendix Table B6.

Table B4: Alternative Designs Linking Pay Expectations and Consideration: Robustness

	Stated Consideration			Free-Text Responses		
	Sample Weights	Unweighted	Population Weights	Sample Weights	Unweighted	Population Weights
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Perceived Firm Effect (Split-Sample)					
Firm Premium (Split-Sample)	1.014***	1.285***	0.605***	0.085***	0.157***	0.059***
	(0.127)	(0.127)	(0.127)	(0.013)	(0.013)	(0.013)
Observations	89742	89742	89600	224388	224388	224020
Number of Workers	9756	9756	9740	9756	9756	9740
	B. Observed Firm Effect					
Firm Premium (Observed)	0.173***	0.245***	0.091*	0.014***	0.012***	0.007
	(0.035)	(0.021)	(0.050)	(0.005)	(0.004)	(0.007)
Observations	89258	89258	89117	214632	214632	214280
Number of Workers	9756	9756	9740	9756	9756	9740
	C. Observed Log(Mean Daily Pay)					
Firm Mean Daily Pay	0.093***	0.149***	0.073***	0.008***	0.010***	0.004**
	(0.011)	(0.007)	(0.021)	(0.002)	(0.001)	(0.002)
Observations	89258	89258	89117	214632	214632	214280
Number of Workers	9756	9756	9740	9756	9756	9740

Note: This table probes the robustness of results in Table 6, which uses sampling weights, to no weighting (Columns 2 and 5) and population weights (Columns 3 and 6). Estimation details are in Section 5.2. Standard errors are clustered at the worker level. Levels of significance: * 10%, ** 5%, and *** 1%.

Figure B1: Perceived Pay Premia and Amenity Valuations: Robustness to Population Weighting



Note: This figure compares estimates of the perceived value of firm-specific amenities ($\frac{a_j}{\beta}$) and the perceived pay premia (ψ_j) for each of the researcher-provided firms. The methodology is the same as that described in Figure 6, but we replace sample weights with population weights. We plot a simple line of best fit.

Table B5: Relative Importance of Preferences and Beliefs

	Baseline		Unweighted		Population Weights	
	Search	Mobility	Search	Mobility	Search	Mobility
	(1)	(2)	(3)	(4)	(5)	(6)
A. Below Median-Wage Workers						
Full Transparency						
High Attachment	12%	7%	13%	7%	12%	7%
Low Attachment	12%	9%	13%	9%	12%	9%
Eliminate Attachment						
High Attachment	34%	24%	43%	22%	25%	18%
Low Attachment	14%	12%	20%	14%	11%	9%
B. All Workers						
Full Transparency						
High Attachment	8%	12%	4%	3%	8%	6%
Low Attachment	7%	10%	7%	6%	7%	7%
Eliminate Attachment						
High Attachment	19%	15%	12%	13%	14%	13%
Low Attachment	10%	8%	9%	9%	7%	6%

Note: This table probes the robustness of the baseline, sample-weighted results in Table 10 (Columns 1-2) under alternative weighting (Columns 3-6). All estimates assume the baseline level of compression ($\lambda = .5$, implying workers perceive 25% of actual between-firm variation in pay).

Table B6: Firm-Specific Preferences: Robustness to Population Weighting

	Preferences Over Outside Firms		Preferences Over All Firms		Preferences Over Considered or Incumbent Firm
	(1)	(2)	(3)	(4)	(5)
Log Raise (β)	7.676**	7.500**	8.843***	9.071***	15.139***
	(3.202)	(3.690)	(2.758)	(2.894)	(3.196)
Observations	4217	4217	5671	5671	3001
Number of Workers (Clusters)	1177	1177	1200	1200	1192
Distance Controls	Yes	Yes	Yes	Yes	Yes
Data Include Incumbent Ranking	No	No	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Firm-Consider Interactions	No	Yes	No	Yes	No
Firm-Incumbent Interactions	No	No	Yes	Yes	Yes
Amenities are Non-Zero $a_j = 0 \forall j$					
p-value	<.001	<.001	<.001	<.001	<.001
Degrees of Freedom	29	29	29	29	29
Consideration Doesn't Affect Valuations $\zeta_j = 0 \forall j$					
p-value	---	<.001	---	<.001	---
Degrees of Freedom	---	30	---	30	---
There is an Incumbent Effect $\iota_j = 0 \forall j$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	18	18	17
Insiders and Outsiders' Valuations Differ $\iota_j = \iota_k \forall j,k$					
p-value	---	---	<.001	<.001	<.001
Degrees of Freedom	---	---	17	17	16
Insiders and Outsiders Who Want to Move to the Firm Have Different Valuations $\iota_j = \zeta_j \forall j$					
p-value	---	---	---	<.001	---
Degrees of Freedom	---	---	---	18	---

Note: This table presents results analogous to Table 11, in which the regressions are weighted to match the overall population of full-time German workers. We provide more details on estimation in Section 6. Standard errors are clustered at the worker level.

B.2 Robustness to Alternative Winsorization Schemes

Our preferred specifications winsorize workers' pay expectations at the 90% level. For most of our analysis this represents a conservative assumption. Our decision to winsorize expectations, for instance, leads us to underestimate the amount of within-worker variation in pay; this biases us against our finding that workers believe in a heterogeneous outside option.

The main piece of analysis which could be influenced by this decision is our analysis of the agreement in the firm pay premia between different demographic groups (Table 4). Intuitively, by reducing the variation in workers' expectations, winsorization could lead us to over-state the amount of agreement. Appendix Table B7 shows that we continue to find a strong correlation in the perceived pay premia across demographic groups when using the raw elicitations (Column 1), winsorizing at the 98% level (Column 2), or trimming at the 98% level (Column 4).[48]

[48] In unreported results we have probed the robustness of other findings to these decisions.

Table B7: Correlation in Perceived Pay Premia: Robustness to Data Cleaning

	Raw	98% Winsor	90% Winsor	98% Trim	90% Trim
	(1)	(2)	(3)	(4)	(5)
Split-Sample	0.52	0.51	0.98	0.94	0.99
Sex	0.43	0.44	0.93	0.91	0.96
CBA	0.48	0.51	0.96	0.94	0.98
College Education	0.52	0.53	0.97	0.96	0.98
Current Firm AKM Effect (Split at Median)	0.65	0.63	0.95	0.91	0.98
Searched in Past 6 Mo.	0.49	0.49	0.96	0.95	0.99
Knew Wages at Application	0.61	0.64	0.97	0.96	0.99
Easy to Get a Better Job	0.60	0.59	0.98	0.95	0.98
Tenure (Split at 2 Years)	0.31	0.34	0.96	0.96	0.99

Note: This table looks at the correlation in the estimates of ψ_j as estimated by fitting equation 7 for different samples of workers (indicated in each row) under different data cleaning procedures (indicated in each column). Column 1 uses workers' raw pay expectations. Columns 2 and 3 (our baseline) use winsorized pay expectations. Columns 4 and 5 use trimmed pay expectations.

B.3 Provision of Worker-Provided Firms and Pay Responsiveness

Most of our analysis relies on the researcher-provided firm module. However, we also present several specifications which rely on data from the worker-provided firm module of the initial survey. Appendix Table B8 shows that, if anything, workers who do not provide firm names are less responsive to changes in outside pay, suggesting that this module may understate how inframarginal to their firm workers are.

Table B8: Stated Mobility Preferences and Provision of Firms

	Baseline	Same Commute	Same Growth	Initial Wave	Follow-Up
	(1)	(2)	(3)	(4)	(5)
Provided Specific Firm Names	1.612**	1.944***	2.391***	8.985*	2.816***
	(0.681)	(0.724)	(0.819)	(4.754)	(0.911)
	11898	5715	5676	6199	5699
Did Not Provide Specific Firm Names	0.833	1.694**	1.467*	3.919	2.206**
	(0.660)	(0.752)	(0.776)	(2.459)	(0.871)
	9974	4202	4211	5803	4171

Note: This table presents separation elasticities for workers who did (Row 1) and did not (Row 2) provide names of specific firms in the worker-provided firm module. We use data from the discrete choice experiments including researcher-provided firms. Standard errors computed using the delta method are presented in parentheses beneath the estimates. Sample sizes are presented below the standard errors. Levels of significance: * 10%, ** 5%, and *** 1%.

B.4 Workers Believe Firms Vary in Amenities: Robustness

Section 7.1 documents that workers believe firms vary in pay and non-pay characteristics. In the follow-up survey, we embedded additional discrete choice experiments, which allow us to confirm that this workers' believe this is true even when the training and learning opportunities would be the same at each of the outside firms.[49] We fit our baseline rank-ordered logit model to workers' preferences—as elicited in these experiments—using only the rankings on outside firms (i.e. excluding the incumbent). Appendix Table B9 shows that, across all models, we can reject the null that there are no unobserved firm-specific factors such as amenities which drive workers' preferences (beyond those we specified would remain the same).

Table B9: Workers Have Firm-Specific Preferences: Robustness

	Follow-Up Survey		
	Baseline	Same Growth	Same Training and Learning Opportunities
	(1)	(2)	(3)
Log Raise (β)	8.395***	8.941***	8.290***
	(0.800)	(0.815)	(0.763)
Observations	10012	9971	10001
Number of Workers (Clusters)	3375	3352	3363
Test: Ex Ante Firm Effects are Zero			
p-value	<.001	<.001	<.001
Chi-Squared Statistic	314.789	335.119	300.256
Degrees of Freedom	27	27	27

Note: This table presents logit coefficients (first row) and p-values and test statistics (remaining rows) from fitting a rank-ordered logit model to workers' stated preferences in the discrete choice experiments in the follow-up survey that include researcher-provided firms. The data include workers' choices over outside firms in the baseline scenario (Column 1), if the firms provided equal the training and learning opportunities (Column 2) or if their work would stay constant (Column 3). Regressions use sampling weights.

B.5 Robustness to Allowing for Preference Heterogeneity

To allow for preference heterogeneity, we estimated a random coefficient model allowing variation in workers' wage and commute preferences. In particular, we allow preferences to take the form:

$$u_{ij} = \beta_i \log w_{ij} + m_i \log d_{ij} + a_j + \epsilon_{ij}.$$

Because we consider workers' preferences over outside firms, there are no indicators for whether a firm is the current incumbent. We include both the log travel time between a worker's current workplace and the provided firm and an indicator for whether we cannot compute this distance.

Table B10 shows there is substantial variation in preferences. However, the valuations are similar to those of our baseline model ($\rho = 0.91$) and we continue to see a positive correlation between perceived premia and amenity valuations.

[49] We included similar discrete choice experiments in the initial survey, but only for the worker-provided firms.

Table B10: Random Coefficient Model Estimates

	Estimate (1)	Standard Error (2)
Model Estimates		
Mean		
Log(Wages)	15.08	0.80
Log(Distance)	-0.34	0.03
Variance		
Log(Wages)	25.10	1.89
Log(Distance)	0.20	0.04
Missing Distance	-0.41	0.35
Correlations		
With Baseline Amenity Values	0.92	
With Baseline Perceived Premia	0.73	
Comparison Values		
Correlation Between Baseline Amenities and Premia	0.58	

Note: The first panel of this table presents the estimates and standard errors from fitting a random coefficient rank-ordered model to workers' stated preferences over outside researcher-provided firms. The model includes the randomized raise, for the log distance between the worker and the firm, for the order in which the firm was presented, and for firm fixed effects. We allow preferences on distance and wages to be heterogeneous. The second panel presents the correlation between the amenity values from this specification and those in our baseline specification and with perceived wage premia. The third panel presents the baseline correlation between amenities and wage premia, for comparison.

C Relationship with Findings in Jäger et al. (2024)

A natural question arises as to why our results—demonstrating that workers possess detailed, firm-specific information about outside wages—differ from Jäger et al. (2024) (JRRS), who report that workers often have limited or inaccurate information about outside options. Our findings align with Guo (2025), who also documents accurate worker perceptions of external wage offers.

In this section, we explain the differences in the studies. We focus on two key elements. First, JRRS elicited a measure which combines beliefs about pay with beliefs about offer rates and preferences. By contrast, we elicited a measure of beliefs about pay. We provide a speculative reconciliation of the findings by noting that workers at low-wage firms receive offers at lower rates than similar workers at high-wage firms; this is true both among searchers and non-searchers. Our findings echo those from the literature (Faberman et al., 2022; Faberman, Mueller and Şahin, 2025) which has shown that heterogeneity in the intensity and productivity of search can resolve several puzzles in the literature. Second, we note it is difficult to separate survey anchoring (due to priming or question order effects) from true anchoring in beliefs.

C.1 Similarity of Empirical Settings

JRRS use two samples for their analysis: (1) a sample of 1,068 German job-seekers surveyed in the German Socio-Economic Panel (GSEOP) and (2) a sample of German workers recruited via internet panels. JRRS use the first dataset to describe workers' beliefs about outside options; they use the second to present experimental evidence on the relationship between biased beliefs and search and to re-examine their findings on anchoring. Our sample (13,680 workers surveyed by the Institute for Employment Research) is similar to the first dataset. Both surveys were fielded by well-known and well-respected entities in Germany to representative samples. Jäger et al. (2024) use data from questions embedded in a wave of the German Socio-Economic Panel. We use data from a novel survey conducted by the IAB.

C.2 Differences in What and How Beliefs are Elicited

A key result in JRRS is that "workers wrongly anchor their beliefs about outside options on their current wage" (abstract). Specifically, they report that a large share of workers (41%) believe they would earn the "same pay" at their outside option. In the GSOEP module, they first asked workers whether workers who switch jobs (to or from their employer) receive the "same pay" or "higher pay" or "lower pay" (with follow-up questions if they did not indicate the worker would receive the same pay); they then asked a similar question about whether the worker themself would earn the "same pay" at another employer if forced to switch firms. In the internet panel, they first asked what a worker currently made and then (immediately afterwards) asked what they thought they would make if forced to switch firms (Appendix Section C.5).

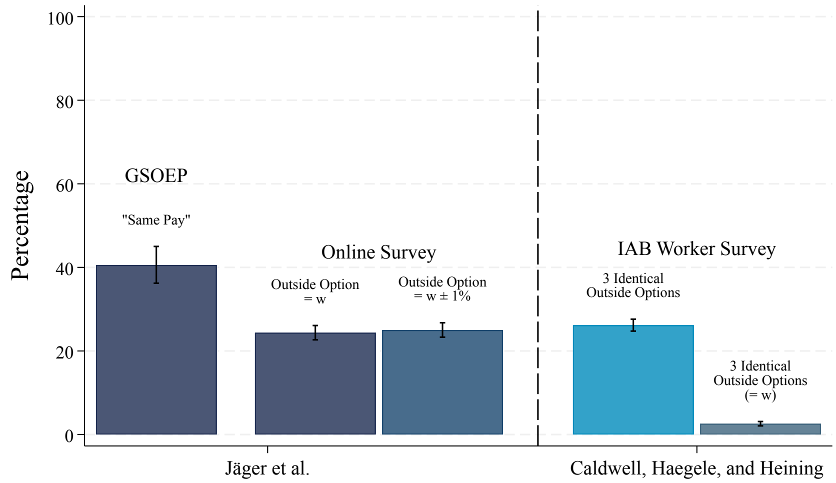
Differences in Approach. We differ both in what we elicit and in methodology. First, because we were interested in beliefs about pay (and how these varied across outside firms), we directly asked workers how much they believed they would earn if employed at each of three specific outside firms. These questions did not require assumptions about offer probabilities or other workers' preferences. By contrast, to form a belief about the expected wage following a forced job-to-job transition, workers must consider both the wages at outside firms and the probability they receive an offer from those firms. To form a belief about the wage changes of movers, workers must consider both these factors, as well as what factors lead workers to switch jobs and what factors workers consider when weighing outside offers.[50]

Second, because we could link all responses to administrative records, we did not ask respondents for their current salary. Research demonstrates that survey respondents anchor responses to previously provided or elicited numbers, even if those numbers are arbitrary or irrelevant (Tversky and Kahneman, 1974). Standard textbooks on survey methodology caution that priming effects can significantly affect responses (Schwarz, 1999; Dillman, Smyth and Christian, 2014). This suggests that eliciting a worker's current wage before eliciting their expectation about outside wages—as was done in the Internet panel used to provide a causal link between beliefs and search—would lead respondents to anchor to this number.

Comparison of Results. Appendix Figure C1 compares the extent of anchoring across the two studies. The first bar shows that 41% of workers in the GSOEP panel selected the "same pay" option. Their internet panel exhibits much lower anchoring, with fewer respondents exactly matching their outside wage expectations to their current pay (second and third bars). Differences observed

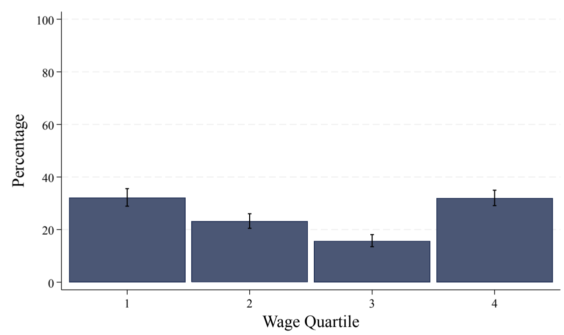
[50]We did collect a summary measure—the perceived ease with which workers could find a more preferred job elsewhere—which aligns somewhat more closely with that in JRRS, which we discuss below.

Figure C1: Comparison of Anchoring Behavior



Note: This figure compares our estimates (right of the dashed line) of the prevalence of anchoring (measured as indicated above each bar) to those in Jäger et al. (2024) (left of the dashed line).

Figure C2: Share of Workers Reporting Identical Salaries: Heterogeneity by Wage

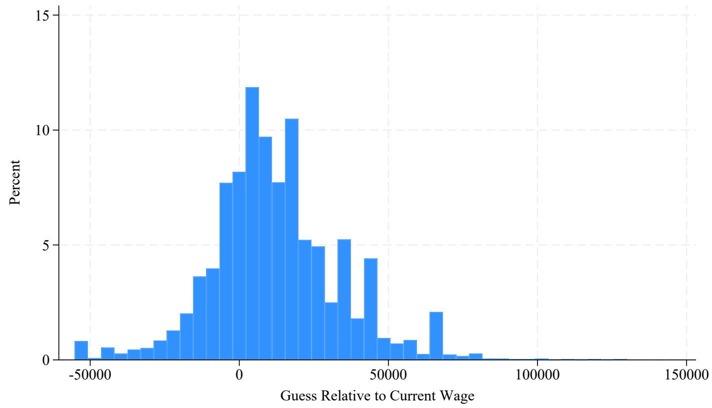


Note: This figure presents our estimates of the prevalence of anchoring (measured as the share of respondents indicating the same expected wage at all three outside options) by wage quartile.

between the GSOEP and internet panel samples may arise from several factors, including population differences or the precise interpretation respondents gave to the phrase "same pay"—whether respondents understood this literally or interpreted it more loosely, as within a monetary range. In our data, the share of respondents reporting identical wage expectations across different outside firms (fourth bar) is similar to the internet panel from JRRS.

Survey-induced anchoring occurs when respondents provide answers which are more similar (or, the same) to responses they have previously provided. Because JRRS asked internet respondents (the sample used for their survey experiment) for their current wage immediately before asking for their outside option pay, it is natural to think that any survey-induced anchoring will lead respondents to provide beliefs which are anchored to this number. Because we asked respondents for the wage they expect at each of three outside firms, it is natural to expect respondents

Figure C3: Anchoring in Beliefs at Workers' Current Salary



Note: This figure plots the difference between workers' expected pay at outside firms, relative to their pay at their current firm.

may anchor beliefs to the first number they provide.[51] A comparison of the third and fourth bar indicates that workers who provide the same three numbers do not typically provide their wage (not elicited in this survey, taken from administrative Social Security records). Appendix Figure C2 reveals minimal heterogeneity in the proportion of respondents reporting identical salaries across different wage groups. Appendix Figure C3 illustrates the distribution of the difference between workers' firm-specific wage expectations and their current salaries; this figure shows we do not see a spike in pay expectations precisely at respondents' current wage. Although the distribution is centered around zero, it is smoothly bell-shaped, displaying no substantial clustering at zero.

C.3 Economic Explanation for Differences in Findings

Setting aside potential issues with survey anchoring, there is a coherent economic story—the combination of Bayesian updating about wages with heterogeneity in offer probabilities—which can plausibly reconcile the descriptive results in both studies.

JRRS (Section IV.C) suggest Bayesian updating as a potential explanation: workers rationally incorporate their own wage experience and skill into their beliefs about outside pay. This would explain why workers provide more accurate estimates about their own potential wages (as elicited directly in our study) compared to the realized pay changes of other workers or their own pay given a forced transition (as elicited in their study). As discussed in JRRS, aside from the finding that workers in low-wage jobs disproportionately underestimate outside pay, Bayesian updating accounts for the empirical patterns.

Rational pessimism about offer probabilities—heterogeneous by firm wage premium—could reconcile the results of the two studies. Appendix Table C1 shows that, conditional on education, occupation, and experience, workers at low-wage firms receive significantly less job-related information than similar workers at higher-wage firms (Column 2). This pattern is robust to controlling

[51] Any survey-induced anchoring in our survey would reduce the variance in perceived wage premia and thus bias our analysis against detecting that workers perceive systematic firm wage premia and actively direct their search based on expected pay differences—central results we robustly document.

Table C1: Search Productivity by Firm Wage Premium

	All			Recent Search	
				No	Yes
	(1)	(2)	(3)	(4)	(5)
Current Firm Wage Effect	0.377***	0.182***	0.179**	0.212*	0.231***
	(0.055)	(0.057)	(0.069)	(0.118)	(0.081)
Person Wage Effect			0.048*	0.155**	0.030
			(0.028)	(0.064)	(0.030)
Observations	9319	9319	7506	2065	5441

Note: This table examines the relationship between the wage premium of a worker's firm and whether they received information on outside jobs in the previous six months. Column 1 presents results from a bivariate regression; Columns 2-5 include controls for experience and for education and occupation dummies. Columns 3-5 control for the person effect from an AKM wage regression. Columns 4–5 split by whether a worker has searched for employment in the previous six months. Regressions use sampling weights. Levels of significance: * 10%, ** 5%, and *** 1%.

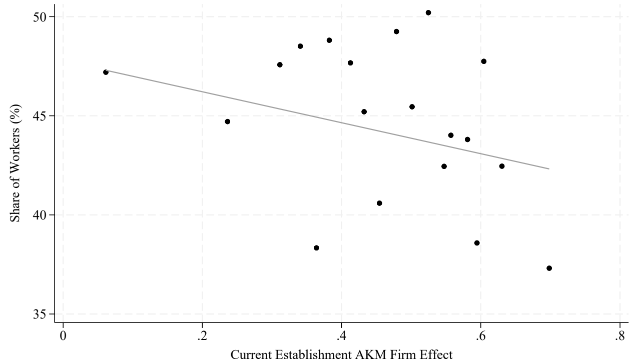
for unobserved differences in worker skill by controlling for the AKM worker effects (Column 3) and to focusing on workers who are not actively searching (Column 4).[52] This explanation is also consistent with the finding by Miano (2023) that information treatments on search difficulty affect behavior, while information treatments on wages do not.

C.4 Additional Result: Summary Measure of Outside Options

We also elicited a summary measure: how easy a worker believed it would be to get a better job. Caldwell, Haegele and Heining (2024) show this correlates with objective measures of outside options (Online Appendix Table A8) and predicts pay renegotiation (Table 5). Appendix Figure C4 shows that workers at lower-wage firms are generically more optimistic about the ease with which they could find a preferred job than workers at higher-wage firms.

[52]This pattern echoes findings in a large literature which has documented differences in the productivity of search (typically higher among employed than unemployed workers) and showed that these differences explain several other puzzles in the literature (Faberman et al., 2022; Faberman, Mueller and Şahin, 2025; Cederlöf et al., 2025).

Figure C4: Share of Workers Who Believe it Would be “Easy” To Get a Better Job



Note: This is a binned scatterplot of whether a worker said it would be “easy” to find a job they would prefer to their current position against their current AKM firm effect.

C.5 Jäger et al. (2024) Belief Elicitations

In the GSOEP JRRS elicited beliefs as follows:

3. **Beliefs About Firm Pay** Think of the typical employee with work experience that switches from another employer to your employer. Would this employee receive a lower, higher or the same pay compared to his previous employer?

   - Higher pay
   - Same pay
   - Lower pay

   [Asked only if previous answer is not “Same pay”] How much lower/higher would the monthly pay before taxes of this employee be (in percent) after the switch compared to his/her prior employer?

   - Between 0% and 2%
   - Between 2% and 5%

   ⋮

   - Between 50% and 75%
   - More than 75% (in data normalized to 87.5%)

   Think of the typical employee with work experience that switches from your current employer to another employer. Would this employee receive a lower, higher or the same pay compared to his previous employer?

   - Higher pay
   - Same pay
   - Lower pay

   [Asked only if previous answer is not “Same pay”] How much lower/higher would the monthly pay before taxes of this employee be (in percent) after the switch compared to his/her prior employer?

   - Between 0% and 2%
   - Between 2% and 5%

   ⋮

   - Between 50% and 75%
   - More than 75% (in data normalized to 87.5%)

11. **Posterior About Outside Option: Point Belief** Imagine that you were forced to leave your current job and that you had 3 months to find a job at another employer in the same occupation. Do you think that you would find a job that would offer you a higher overall pay, the same pay or a lower pay?

    - Higher pay
    - Same pay
    - Lower pay

    [Asked only if previous answer is not "Same pay"] What do you think: how much more/less would you earn in that new job?

    - Between 0 and 50 EUR
    - Between 50 and 100 EUR

      ⋮
    - Between 2000 and 3000 EUR
    - More than 3000 EUR (in data normalized to 3500 EUR)

Their main results report the share of workers who indicate "same pay" in response to these questions. In more quantitative exercises they build expected outside options by taking the midpoint of the selected range[53] and signing it depending on whether the respondent chose "higher pay" or "lower pay"; outside option beliefs for "same pay" respondents are coded as 0%.In the online survey, they elicited beliefs in a single step, immediately after eliciting current wages:

4. **Wage Income** How high is your current monthly gross income from work in EUR before taxes?

    __EUR
5. **Beliefs About Personal Outside Option** Imagine that you were forced to leave your current job and that you had 3 months to find a job at another employer. In the job with another employer, how much would you receive per month as gross employment income in EUR?

    __EUR

D Theoretical Appendix

Time. Discrete, $t = 0, 1, 2, \ldots$.

Workers and firms. Unit mass of workers; J firms in the set $\mathcal{J} = \{1, \ldots, J\}$.

Flow utility.

$$u_{ik} = \beta w_{ik} + a_k + \varepsilon_{ik}, \qquad \text{(employment)} \tag{11}$$

$$u_{i0} = a_0 + \varepsilon_{i0} \qquad \text{(unemployment).} \tag{12}$$

where $\varepsilon_{ik} \overset{i.i.d.}{\sim}$ Type I EV$(0, 1)$.

[53]In cases in which a respondent expected their wage to change by "[m]ore than 75%", the authors record this response as 87.5%.

Beliefs.

$$\widetilde{w}_{ik} = w_{ik} + b_i + \eta_{ik}, \quad \mathbb{E}[\eta_{ik} \mid w_{ik}, b_i] = 0. \tag{13}$$

Search technology. Worker i employed at k chooses an application set $A \subseteq \mathcal{J} \setminus \{k\}$, paying $\kappa|A|$. Each $k \in A$ yields a job offer with probability p_k (independent). The current match is destroyed with probability λ.

Switching costs. Accepting any outside offer incurs a one-time cost of $s_i \geq 0$.

Discounting. Common discount rate $\delta \in (0, 1)$.

Bellman equation. Define

$$\widetilde{V}_{ik} = \widetilde{u}_{ik} \tag{14}$$

$$+ \max_{A\subseteq\mathcal{J}\setminus\{k\}} \Big\{ -\kappa|A| + \delta\,\mathbb{E}_{B(A)}\big[(1-\lambda)\,\max\{\widetilde{V}_{ik}, \max_{m\in B(A)}(\widetilde{V}_{im} - s_i)\} \tag{15}$$

$$+ \lambda\,\max\{V_{i0,t+1}, \max_{m\in B(A)}(\widetilde{V}_{im} - s_i)\}\big]\Big\}. \tag{16}$$

Lemma. *Contraction Mapping*

The operator $\mathcal{T}$ that maps a bounded function V to the right-hand side of the above Bellman equation is a contraction on $(\mathcal{X}, \|\cdot\|_\infty)$ with modulus δ. Hence a unique bounded fixed point $\widetilde{V}$ exists.

Proof. Let $V, V' \in \mathcal{X}$. For any state s, the difference $|(\mathcal{T}V)(s) - (\mathcal{T}V')(s)| \leq \delta\,\mathbb{E}\big[\|(V - V')(s')\|_\infty\big] \leq \delta\|V - V'\|_\infty$. Taking sup over s yields $\|\mathcal{T}V - \mathcal{T}V'\|_\infty \leq \delta\|V - V'\|_\infty$. Because $\delta < 1$, Banach's fixed-point theorem applies. $\square$

Definition (Directed Search). Worker i engages in *directed* search if there exists at least one worker–firm pair (i, k, ℓ) with $\widetilde{w}_{ik} \neq \widetilde{w}_{i\ell}$ such that a marginal perturbation of $(\widetilde{w}_{ik}, \widetilde{w}_{i\ell})$, holding their empirical distribution fixed, changes the worker's optimal A.

Key Decision: The worker employed at firm j chooses the set A that maximizes

$$\max_{A\subseteq\mathcal{J}\setminus\{j\}} \Bigg\{ -\kappa\,|A| \;+\; \delta\,\mathbb{E}_{B(A)}\Big[(1-\lambda)\,\max\Big\{\overbrace{V_{ij}}^{\text{stay}},\ \overbrace{\max_{m\in B(A)}\big(\tilde{V}_{im} - s_i\big)}^{\text{switch employers}}\Big\}$$

$$+ \lambda\,\max\Big\{\underbrace{V_{i0}}_{\text{unemployed}},\ \underbrace{\max_{m\in B(A)}\big(\tilde{V}_{im} - s_i\big)}_{\text{switch employers}}\Big\}\Big]\Bigg\}.$$

where $\mathbb{E}_{B(A)}[\cdot]$ is the expectation over the random event of which firms B (subsets of A) actually extend offers, each with probability p_k. As before we use the notation $\tilde{V}_{ij}$ in cases where the worker's decision is based on beliefs about wages at firm j and V_{ij} when the worker's decision is based on the true wages at firm j.

Prediction 1: Probability of Searching Increases with Increased Perceived Outside Options and Decreased Switching Costs *If any $\tilde{V}_{ik}$ increases, the probability that worker i chooses a non-empty application set $A \neq \emptyset$ weakly increases. Equivalently, a worker's incentive to search is (weakly) increasing in her perceived outside options.*

Proof Sketch.

1. *Define search vs. no-search.* Let $\Phi(\{\tilde{V}_{ij}\})$ be the *maximum* expected payoff over all $A \subseteq \mathcal{J} \setminus \{j\}$, net of application costs. That is,

$$\Phi(\{V_{ij}\}) = \max_{A \subseteq \mathcal{J} \setminus \{j\}} \Bigg\{ -\kappa\,|A| \;+\; \delta\,\mathbb{E}_{B(A)}\Big[(1-\lambda)\,\max\Big\{ \overbrace{V_{ij}}^{\text{stay}},\; \overbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i\big)}^{\text{switch employers}} \Big\} + \lambda\,\max\Big\{ \underbrace{V_{i0}}_{\text{unemployed}},\; \underbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i\big)}_{\text{switch employers}} \Big\}\Big]\Bigg\}.$$

If the worker ends up choosing $A = \emptyset$, it means:

$$\emptyset \in \operatorname*{argmax}_{A \subseteq \mathcal{J} \setminus \{j\}} \Bigg\{ -\kappa\,|A| \;+\; \delta\,\mathbb{E}_{B(A)}\Big[(1-\lambda)\,\max\Big\{ \overbrace{V_{ij}}^{\text{stay}},\; \overbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i\big)}^{\text{switch employers}} \Big\} + \lambda\,\max\Big\{ \underbrace{V_{i0}}_{\text{unemployed}},\; \underbrace{\max_{m \in B(A)} \big(\tilde{V}_{im} - s_i\big)}_{\text{switch employers}} \Big\}\Big]\Bigg\}.$$

2. *Increase some $\tilde{V}_{ik}$.* Suppose $\tilde{V}_{ik}$ rises. The payoff from any set A that includes k increases, since with probability $p_k > 0$ the worker may receive the offer. The payoff from any set that does not include k is unchanged. Thus the *optimal* objective, Φ weakly increases.r
3. *Hence, search is at least as likely.* If the worker did not find any profitable $A \neq \emptyset$ before, a rise in $\tilde{V}_{ik}$ may make some A' profitable.
4. Notice that the same argument works when s_i is reduced.

Prediction 2: Fewer Mistaken Non-Search Outcomes Under Directed Search *When workers have firm-specific beliefs $\{\tilde{V}_{ik}\}$, they are less likely to fail to search when at least one firm is potentially profitable. By contrast, if they cannot match beliefs to firm identities, they may not search even when a single high-value firm would justify searching.*

Proof Sketch.

1. Consider two otherwise identical workers who differ in whether or not they can match beliefs to firm identities. The worker who cannot assign beliefs to firms searches when the expected payoff exceeds the cost, κ:

$$\delta \frac{1}{J} \sum_k p_k[(1-\lambda)\text{max}(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\text{max}(\tilde{V}_{ik} - s_i - V_{i0}, 0)] > \kappa. \tag{17}$$

The worker who can assign beliefs to firm identities searches when

$$\max_k \Big\{ \delta p_k[(1-\lambda)\text{max}(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\text{max}(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \Big\} > \kappa \tag{18}$$

2. Define mistaken non-search as a case where the worker fails to search (the corresponding inequality above fails), but the search is worthwhile under full information. Suppose the directed searcher (the second worker) fails to search. Then

$$l \in \underset{k}{\text{argmax}} \Big\{ \delta p_k[(1-\lambda)\text{max}(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\text{max}(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \Big\} \tag{19}$$

$$\delta p_l[(1-\lambda)\max(\tilde{V}_{il} - s_i - V_{ij}, 0) + \lambda\max(\tilde{V}_{il} - s_i - V_{i0}, 0)] \le \kappa \tag{20}$$

This implies the non-directed searcher also fails to search because:

$$\delta \frac{1}{J} \sum_k p_k[(1-\lambda)\max(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\max(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \tag{21}$$

$$\le \delta \frac{1}{J} \sum_k p_l[(1-\lambda)\max(\tilde{V}_{il} - s_i - V_{ij}, 0) + \lambda\max(\tilde{V}_{il} - s_i - V_{i0}, 0)]$$

$$= \delta p_l[(1-\lambda)\max(\tilde{V}_{il} - s_i - V_{ij}, 0) + \lambda\max(\tilde{V}_{il} - s_i - V_{i0}, 0)]$$

3. Now consider the case where the non-directed searcher fails to search:

$$\delta \frac{1}{J} \sum_k p_k[(1-\lambda)\max(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\max(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \le \kappa. \tag{22}$$

Suppose

$$l \in \underset{k}{\text{argmax}} \Big\{ \delta p_k[(1-\lambda)\text{max}(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\text{max}(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \Big\} \tag{23}$$

and that l is not a member of the set containing the corresponding $\arg\min$ (i.e. that expected gains are not identical across firms). Then, the expected value of applying to firm l is strictly greater than a random application. There is some level of κ such that the directed searcher searches:

$$\delta \frac{1}{J} \sum_k p_k[(1-\lambda)\max(\tilde{V}_{ik} - s_i - V_{ij}, 0) + \lambda\max(\tilde{V}_{ik} - s_i - V_{i0}, 0)] \tag{24}$$

$$\leq \kappa < \delta p_l[(1-\lambda)\max(\tilde{V}_{il} - s_i - V_{ij}, 0) + \lambda \max(\tilde{V}_{il} - s_i - V_{i0}, 0)].$$

Prediction 3: Directed Search Implies Higher Application Rates for Higher-Wage Firms, All Else Equal *If the worker engages in directed search*, then—holding amenities a_k and offer probabilities p_k constant—the probability of applying to firm k increases with $\tilde{w}_{ik}$. Conversely, if the worker believes all firms pay the same wage, then the probability of applying to a specific firm is uncorrelated with w_{ik}.

Proof Sketch.

1. *Worker's objective over sets A.* For each set $A \subseteq \mathcal{J} \setminus \{j\}$, define

$$U_j(A) \;=\; -\,\kappa\,|A| + \delta \mathbb{E}_{B(A)}\big[(1-\lambda)\max\{\max_{m\in B(A)}(\tilde{V}_{im}-s_i), V_{ij}) + \lambda \max\{\max_{m\in B(A)}\{(\tilde{V}_{im}-s_i)\}, V_{i0}\}\}\big].$$

   The worker picks A^* to maximize $U_j(A)$ given the probability distribution of offer sets B from each application set A.
2. *Incremental value of adding k.* Let $\Delta_{k;j,A} = U_j(A \cup \{k\}) - U_j(A)$. As $\tilde{w}_{ik}$ (and thus $\tilde{V}_{ik}$) increases, any state in which firm k extends an offer becomes more valuable. Hence $\Delta_{k;j,A}$ *weakly* increases.
3. *Probability of including k in A^*.* The event "$\text{Apply}_{ik} = 1$" means $k \in A^*$. Since A^* is chosen to maximize $U_j(\cdot)$, if $\Delta_{k;j,A^*}$ becomes positive after an increase in $\tilde{w}_{ik}$, the worker is more likely to include k. Hence Apply_{ikt} is weakly increasing in $\tilde{w}_{ik}$.
4. *No perceived variation.* If $\tilde{w}_{ik} = \bar{w}_i$ for all k, then workers do not base application decisions on the wages of individual firms at all, so the choice of A is uncorrelated with w_{ik}.

Conclusion. These results show that the predictions that—(1) higher perceived outside options boost overall search, (2) application choices increase with perceived wages, and (3) firm-specific beliefs reduce mistaken non-search—all extend to the *multi-firm application* environment. The core logic is that an increase in $\tilde{V}_{ik}$ raises the marginal expected value of including firm k in the application set, thereby increasing the likelihood of search and the correlation between wages and application decisions.

To see that the responsiveness of a firm's applications to changes in its own wage is increasing in the share of directed searchers, decompose the number of applicants to firm j, n_j, into the contributions from two groups of workers:

$$n_j = N \times \left[\underbrace{\alpha \times \mathcal{D}_j(a_j, \{\tilde{\mathbf{w}}_i\}, \{\kappa_i\})}_{\text{directed portion}} + \underbrace{(1-\alpha) \times \mathcal{R}_j(\{\tilde{\mathbf{w}}_i\}, \{\kappa_i\})}_{\text{non-directed portion}}\right],$$

where $\alpha \in [0,1]$ is the share of workers who direct their search, and $\mathcal{D}_j$ ($\mathcal{R}_j$) is the share of workers who direct (do not direct) their search who apply to j. The wage elasticity of applications η^j is

$$\eta^j = \frac{\partial n_j}{\partial w_j} \times \frac{w_j}{n_j} = \frac{w_j}{n_j} \times N \times \left[\alpha \frac{\partial \mathcal{D}_j}{\partial w_j} + (1-\alpha)\underbrace{\frac{\partial \mathcal{R}_j}{\partial w_j}}_{\approx 0}\right] \approx N \times \alpha \times \frac{w_j}{n_j}\left[\underbrace{\frac{\partial \mathcal{D}_j}{\partial w_j}}_{+}\right], \tag{25}$$

where $\partial \mathcal{D}_j / \partial w_j > 0$ as long as $\partial \tilde{w}_{ij} / \partial w_j > 0$ because the probability each worker who directs their search applies is increasing in their own beliefs (prediction 3).[54]

E Survey and Administrative Data

This appendix describes the survey modules designed for this paper, as well as the additional data sources. An additional survey appendix, available here, describes the logistics of inviting workers to the survey and response rates. Much of that material is included in the main text and appendix of Caldwell, Haegele and Heining (2024). The full text of the initial and follow-up questionnaires are available on the authors' websites here.

E.1 Researcher-Provided and Worker-Provided Firm Modules

Two separate survey modules elicit firm-specific information from respondents. Appendix Figure E1 provides an overview of the survey flow. We randomized the provision of many of the follow-up questions to keep the length of the survey manageable. After workers provided the names of firms they would consider, we randomized them into three groups (with probability 50%, 25%, and 25%) and presented them with additional questions accordingly.

In the researcher-provided firm module we asked a random 50% of workers to provide their expected pay at and rankings over three researcher-provided firms. We selected these firms at random from the seven researcher-provided firms we provided in the question on consideration; whether a firm was selected was unrelated to whether a worker said they would consider applying.

The follow-up survey proceeded similarly. However, we only posed the worker-provided firm module to workers who declined to complete it in the initial wave. We posed researcher-provided firm questions on pay expectations and preferences to all workers in this wave.

E.1.1 Selecting Researcher-Provided Firms

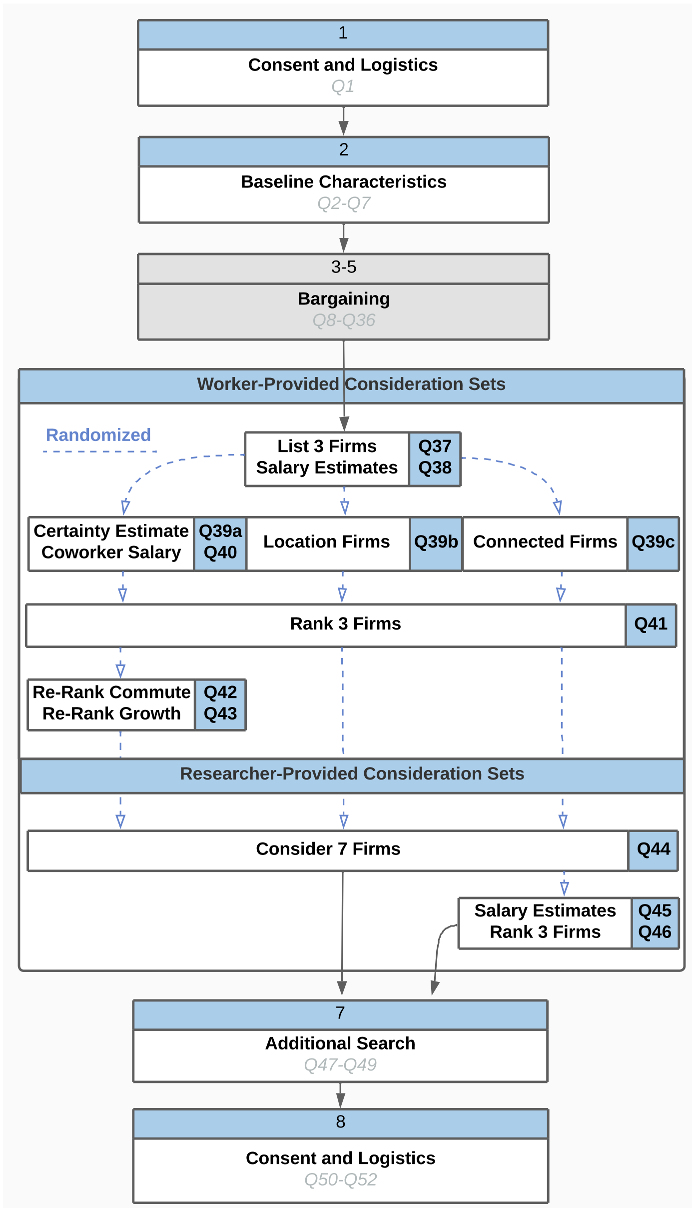
We had to select a subset of German firms to ask workers about in the researcher-provided firm module. Due to IAB's strong confidentiality requirements, we could not select firms on the basis of firm-to-firm flows or networks.[55] Further, these requirements do not allow us to export the list of firms included in our final list from the IAB. We instead describe the systematic process through which we selected firms for inclusion in the survey.

Selection Process. To balance the criteria described in Section 3.3, we focused on 30 well-known German firms. We included 20 of these in the initial survey, and 30 in the follow-up survey. For the initial survey, we used two publicly-available lists of German firms to select our 20 firms: the top 40 publicly listed firms (Appendix Table E1) and the top 100 family-owned firms in Germany.

[54] If firm j is small, $\partial \mathcal{R}_j / \partial w_j \approx 0$ because a wage change at firm j will influence workers' overall search behavior only through its influence on their beliefs about the overall wage distribution; any increase in applications will be spread across all firms.

[55] While we link survey respondents' answers about specific firms to the administrative data, we were not allowed to *export* firm names from the administrative data for inclusion in the survey.

Figure E1: Flow of the Initial Survey



Note: This figure provides an overview over the modules in the worker survey. The main questions used for our analysis are posed in the worker-provided and researcher-provided modules.

While the first list contains large, well-known firms, the second list also includes somewhat smaller firms.

To select firms from these lists, we first grouped them by their industry. We prioritized firms that were in the six major industries covered by our worker sample.[56] Because manufacturing is by far the most important industry in our sample, we included multiple manufacturing firms in our list. Second, we excluded firms with poor name recognition (e.g., those not known under their official name) or with ambiguous names. Third, within each industry, we prioritized firms that participated in our firm-survey on wage-setting (Caldwell, Haegele and Heining, 2024).[57] All of the 20 firms included in the initial wave of the survey can be linked to the Social Security records.

Initial Set of Firms. We verified that the final set of 20 firms included in the initial survey met the criteria listed above in two ways. First, we compared the firms to those provided by the 518 respondents to an IAB pilot conducted in February 2022. In that pilot, we asked respondents to name five firms that they would consider applying to if they were interested in switching firms. Ten of the 20 firms we selected were listed at least once. This test allows us to verify both that the firms we identified as well-known (from the publicly listed set) were likely to be self-reported by respondents and that the firms we believed to be lesser-known (from the family-owned set) were less likely to be listed. Second, to ensure familiarity with the chosen firms, we surveyed 50 German Prolific users in June 2022. This pilot confirmed that the firms on our final list are generally well-known to respondents (though there was substantial variation), and that the firms that are less well-known are those drawn from the second list.

Expanded Set of Firms. For the follow-up survey, we selected 10 additional firms. We again focused on firms that are reasonably well-known in Germany. In selecting these firms, we made sure that we have firms in the same industry across different regions. We also made sure to have variation in how large and popular the new firms were to avoid only including the most desirable German employers. All 10 additional firms were named by respondents in the initial survey.

E.1.2 Researcher-Provided Firm Module: Randomization Assessment

In the researcher-provided firm module, we performed several randomizations within the survey code. We posed the initial question in researcher-provided firm module—which asked respondents whether they would consider applying to each of these firms—to all survey respondents. In the initial survey we randomized each respondent into one of three lists of seven firms, each of which contained one of two randomly chosen focal firms: there were six randomization groups. We asked respondents whether they considered each of seven (rather than 20) firms to reduce the burden of the survey. We performed a similar randomization for the follow-up Appendix Table E3 shows that the randomization of workers to firms in the initial survey (Columns 1 to 3) and follow-up survey (Columns 4 to 6) was successful. Appendix Table E4 shows analogous results for the randomization of pay raises; as before, there is no evidence that the randomization failed. We find only two p values less than .05 (for college education in the follow-up survey) across the 72 test

[56]The gap in frequency between the sixth and seventh most frequent industry was large.

[57]For privacy reasons, we are not allowed to release the list of firms that participated in that survey; 9 of the 20 firms included in the initial wave of the survey participated in this survey.

Table E1: DAX 40 Firms

DAX 40
adidas AG
Airbus SE
Allianz SE
BASF SE
Bayer
Beiersdorf AG
BMW AG
Brenntag SE
Continental AG
Covestro AG
Daimler Truck Holding AG
Delivery Hero SE
Deutsche Bank AG
Deutsche Boerse AG
Deutsche Post AG
Deutsche Telekom AG
E.ON SE
Fresenius Medical Care AG & Co. KGaA
Fresenius SE
Hannover Rueck SE
HeidelbergCement AG
Henkel AG & Co. KGaA
Infineon AG
Linde plc
Mercedez-Benz Group AG
Merck KGaA
MTU Aero Engines AG
Muencher Rueckversicherungs-Gesellschaft AG
Porsche Automobil Holding SE
PUMA SE
QIAGEN NV
RWE AG
SAP SE
Sartorius AG VZ
Siemens AG
Siemens Healthineers AG
Symrise AG
Volkswagen AG VZ
Vonovia SE
Zalando SE

Note: This reproduces the list of the top 40 publicly listed firms as of July 2022.

Table E2: Top 100 Family-Owned Firms

Top 100 German Family-Owned Firms	
Volkswagen AG	Oldendorff Carriers GmbH & Co. KG
Schwarz Gruppe	Asklepios Gruppe
BMW Group	DER KREIS Einkaufsgesellshaft fur Kuche & Wohnen mbH & Co. KG
Aldi Diskounter (Nord+Süd)	DRÄXLMAIER Group
Robert Bosch GmbH	Norma Lebensmittelfilialbetrieb Stiftung & Co. KG
Fresenius Gruppe	Voith GmbH & Co. KGaA
Continental AG	FTI Touristik GmbH
Heraeus Holding GmbH	Helm AG
Merckle Gruppe	Müller Holding Ltd. & Co. KG
Metro AG	Andreas Stihl AG & Co. KG
Ceconomy AG	CLAAS KGaA mbH
Henkel AG & Co. KGaA	Jungheinrich AG
Merck KGaA	Mann+Hummel GmbH
Boehringer Ingelheim Pharma GmbH & Co. KG	Wilh. Werhahn KG
Bertelsmann SE & Co. KGaA	Trumpf GmbH + Co. KG
Adolf Wuerth GmbH & Co. KG	GOLDBECK GmbH
Otto GmbH & Co. KG	Sartorius AG
Schaeffler AG	Diersch & Schröder GmbH & Co. KG
Droege Group AG	Drägerwerk AG & Co. KGaA
Knauf Gruppe	Enercon GmbH
dm-drogerie markt GmbH+Co. KG	Dürr AG
Dirk Rossmann GmbH	Krones AG
Maxingvest AG	Webasto SE
Marquard & Bahls AG	Rehau AG + Co.
Mahle GmbH	SEW-Eurodrive GmbH & Co. KG
Freudenberg Gruppe	AURELIUS Equity Opportunities SE & Co. KGaA
Tengelmann Twenty-One KG	Vorwerk Gruppe
REMONDIS-Gruppe	Cremer Gruppe
Globus Gruppe	Franz Haniel & Cie. GmbH
B. Braun Melsungen AG	AUNDE Group SE
DKV MOBILITY SERVICES HOLDING GmbH + Co. KG	Messer Group GmbH
MHK Group	Alfred Kärcher SE & Co. KG
Carl Zeiss AG	AUGUST STORCK KG
BAUHAUS AG	EW Group GmbH
Dr. August Oetker KG	Diehl Gruppe
Tönnies Holding GmbH & Co. KG	Haribo GmbH & Co. KG
Knorr-Bremse AG	Festo AG & Co. KG
E/D/E Group	Viessmann Werke GmbH & Co. KG
Bechtle AG	Hubert Burda Medien
Dachser SE	PHW Gruppe (Wie- senhof etc.)
Hornbach Holding AG	Vaillant Deutsch- land GmbH & Co. KG
United Internet AG	SMS Group
Bartels-Langness Handelsgesellschaft mbH & Co. KG	InterSnack Gruppe
Brose Fahrzeugteile GmbH & Co.	MEYER NEPTUN Gruppe
CORDES & GRAEFE KG	Fressnapf Tiernahrungs GmbH
Eberspächer Grup- pe GmbH & Co. KG	Friedhelm Loh Group
Deichmann SE	Rohde & Schwarz GmbH & Co. KG
Wacker Chemie	OTTO FUCHS Gruppe
Miele & Cie. KG	Gauselmann AG
L. Possehl & Co. mbH	Hellmann Worldwide Logistics

Note: This reproduces the list of the Top 100 German family-owned firms, as compiled by Die Deutsche Wirtschaft on April 28th 2022 (Deutsche Wirtschaft, 2022).

Table E3: Randomization Assessment: Assignment of Workers to Provided Firms

	Initial Survey			Follow-Up		
	Firm Group	Firm	Firm Quality	Firm Group	Firm	Firm Quality
	(1)	(2)	(3)	(4)	(5)	(6)
Demographics						
Female	0.375	0.576	0.570	0.094	0.064	0.570
Age	0.094	0.332	0.817	0.886	0.783	0.817
German Citizen	0.334	0.374	0.195	0.591	0.697	0.195
Education						
College	0.526	0.643	0.947	0.005	0.014	0.947
Apprenticeship	0.496	0.714	0.607	0.069	0.146	0.607
Employment and Earnings						
Daily Earnings	0.228	0.137	0.552	0.941	0.971	0.552
Earnings are Censored	0.391	0.764	0.893	0.682	0.822	0.893
Weekly Hours (Survey)	0.085	0.128	0.106	0.451	0.639	0.106
Covered by a CBA (Survey)	0.351	0.785	0.736	0.882	0.965	0.736
Sector						
Manufacturing	0.998	0.999	0.941	0.481	0.704	0.941
Retail	0.628	0.945	0.297	0.813	0.818	0.297
Professional	0.730	0.980	0.785	0.399	0.360	0.785

Note: Firm group is an indicator for the set of firms a worker is assigned to, firm is an indicator for the firm a worker is assigned to, and firm quality is the objective pay premium (Bellmann et al., 2020). We perform separate regressions of each covariate (indicated in the row) on the characteristics indicated in the column. Each entry provides the p-value from an F-test that all of the included regressor(s) (other than the constant) are equal to zero. P-values are calculated using standard errors clustered at the worker level.

performed. This is not unexpected given the large number of tests; further, because almost all of our analysis is within worker, potential imbalance would not affect our results.

E.1.3 Cleaning Worker-Provided Firms

To use data from the worker-provided firm module, we had to clean the firm names that workers provided. In the initial survey, respondents (overall, including those dropped from our main analysis sample) provided 5,828 unique strings. Of these, we could assign 5,016 (86%) to uniquely identifiable firms. We performed this assignment manually through online searches to ensure that the strings matched a unique firm.

Of the 812 strings that could not be uniquely identified, the majority are unspecific firm types (e.g., "City government", "police"); a small share (10%) represent non-response (e.g., "I do not want to answer") or unidentifiable content (e.g., "XXX"). Our manual assignment procedure, which corrects for differences in spelling and pools subdivisions of the same firm, assigns the 5,016 identifiable strings to 3,018 unique firms. Of these 3,018 firms, 2,013 can be matched to establishments in the IAB records. Among the firms that we cannot link to the IAB, 29% are foreign entities and 37% are in the public sector; these firms may not have any workers in Social Security-covered employment in Germany. We link 79% of the firms that were at least named twice—for which we also collect additional firm characteristics (Appendix Section E.1.4)—to the IAB records. We obtained the municipality of firms' headquarters for 99% of linked firms.

Table E4: Randomization Assessment: Assignment of Pay Offers to Firms

	Initial Survey	Follow-Up
	(1)	(1)
Number of Employees	0.39	0.31
Sector		
Manufacturing	0.64	0.64
Retail	0.44	0.09
Professional Services	0.71	0.15
Information Services	0.33	0.92
Transportation	0.21	0.34
Finance	0.74	0.27
Other Firm Characteristics		
HQ in Eastern Germany	0.78	0.59
Year of Incorporation	0.79	0.20
Financial Characteristics		
Total Assets per Employee	0.94	0.36
Fixed Assets per Employee	0.85	0.30

Note: We regress each covariate (indicated in the row) on the randomly assigned pay offer, controlling for the position of the firm (whether listed first, second, or third), and clustering standard errors at the worker level. Each entry is the p-value from a test that the coefficient on the pay offer is zero.

E.1.4 Measuring Firm Characteristics Using the IAB Social Security Records

We follow standard linkage approaches at the IAB to match the firms to the IAB data. We assign each firm the employment-weighted average for all matched establishments. For each establishment, we collect the following information: number of full-time employees, average pay, share of women, share of college educated workers, and the AKM establishment effect. We use the AKM establishment effects as estimated by Bellmann et al. (2020).

To examine the geographic coverage of our firms, we use information on the location of all firm establishments. Based on the headquarter addresses of the researcher-provided and worker-provided firms, we assigned each firm a 7-digit municipality code ("Gemeindeschlüssel") that represents the finest geographical distinction in the IAB records. We have this code for over 99% of linked firms. Based on the municipality code, we can aggregate to the district ("Kreis") and state ("Bundesland").

We measure distance as the driving distance between a worker's workplace and the headquarters of the indicated firm. In some specifications, we use an alternative measure of distance, based on the driving distance between a worker's current place of work and the closest establishment of the provided firm (rather than the largest). We have verified our results are robust to using this alternative measure; many of these specifications are in the appendix.

E.1.5 Additional Characteristics of Researcher-Provided and Worker-Provided Firms

Appendix Table 2 shows that both the researcher-provided firms and the worker-provided firms span a broad range of the German labor market.[58] While the most common industry is manufactur-

[58]Our current dataset captures worker-provided firms that were named at least twice as well as a randomly drawn set of 200 worker-provided firms that were named only once.

ing, our sample also captures other major sectors, such as retail, professional services, information services, and finance. Appendix Table E5 shows that, when including non-headquarters establishments, the firms operate in all German states. The researcher-provided firms capture a large span of the major occupational groups in the labor market, with the exception of agriculture (Appendix Figure A1).

These firms represent important employers in the German labor market. The researcher-provided firms employ over 1.8 million German workers (4% of the workforce). 63% are on lists of the largest German employers (Deutsche Wirtschaft, 2022) and 53% are rated among the most popular employers in Germany (Statista 2023). In total, the 30 researcher-provided firms received over 70,000 reviews on Kununu, almost 3% of total reviews. Nine are in the top 20 by number of reviews. The firms also received more than 39.1 million page views, almost 3% of total page views. Seven are in the top 20 by page views. In the past 10 years, 23% of respondents have worked at least at one of the 30 researcher-provided firms, highlighting the relevance of these firms for the German workforce.

The worker-provided firms listed at least twice collectively employ over 6.2 million German workers. They have received over 295,000 reviews and over 190 million page views.

E.1.6 Cleaning Workers' Expected Pay

In both firm-specific modules, we elicited what workers believe they would earn at specific outside firms. The questionnaire included free-form text entries. The survey tool required responses to be numeric or missing to proceed to the subsequent question. Workers responses are high quality. Less than 3% of workers who provide at least one pay estimate, do not provide all three requested estimates. The vast majority of pay estimates are within a reasonable range. Only 5% of workers report numbers that are below the annual pay that a full-time worker would receive if they earned the minimum wage. Less than 0.1% of provided pay estimates are above 1,000,000 Euros. There are no numbers reported that can be classified as gibberish (e.g., "999999").

We clean these free-form text responses in several ways. First, we scale up pay which was input in thousands. Second, to deal with outlier observations we winsorize workers' expected pay at the 90% level. As we document in Appendix B.2, we have implemented alternative winsorization schemes (e.g., at the 98% level) or trimming schemes and obtained similar results. To generate Figure A4, we scale the researcher predictions (from 2022) by the CPI when comparing our predictions to the workers' predictions in the follow-up survey (conducted in 2024).

E.2 Additional Data Sources

IEB Social Security Records We used the German Social Security records, which are assembled by the Institute for Employment Research (IAB) into the Integrated Employment Biographies (IEB) database, to select the sample of workers to whom we fielded the survey. These data capture all private-sector and public-sector employees with Social Security contributions. They contain information on employee demographics (e.g., gender, age, and education), employer information (e.g., sector and location), and job-spell-based information (e.g., full-time status, daily pay, and occupation). We impute daily pay for individuals whose pay is top-coded.

Table E5: Coverage of the Worker- and Researcher-Provided Firms

	German Labor Market	Researcher-Provided Firms	Worker-Provided Firms	
			Unweighted	Weighted
	(1)	(2)	(3)	(4)
Region				
Baden-Württemberg	0.14	0.13	0.13	0.35
Bavaria	0.19	0.33	0.25	0.45
Berlin	0.05	0.07	0.05	0.24
Brandenburg	0.03	0.00	0.00	0.00
Bremen	0.01	0.00	0.00	0.00
Hamburg	0.03	0.07	0.08	0.32
Hesse	0.08	0.10	0.11	0.28
Lower Saxony	0.09	0.07	0.05	0.35
Mecklenburg Western Pomerenia	0.02	0.00	0.00	0.00
Northrhine-Westphalia	0.20	0.17	0.05	0.29
Rhineland Palatinate	0.05	0.07	0.02	0.07
Saarland	0.01	0.00	0.00	0.00
Saxony	0.04	0.00	0.00	0.00
Saxony-Anhalt	0.02	0.00	0.00	0.00
Schleswig Holstein	0.04	0.00	0.00	0.00
Thuringia	0.02	0.00	0.00	0.00
Sector				
Information and Communication	0.05	0.10	0.07	0.27
Manufacturing	0.08	0.57	0.31	0.49
Professional Services	0.19	0.10	0.13	0.35
Retail	0.22	0.07	0.12	0.30
Number of Employees				
1-49	0.97	0.00	0.01	0.08
50-249	0.03	0.00	0.04	0.07
250+	0.01	1.00	0.95	0.11

Note: This table compares firm characteristics for the universe of all German firms (Column 1) to our sample of researcher-provided (Column 2) and worker-provided firms (Columns 3 and 4). Column 4 re-weights the worker-provided firms by the number of times they were listed by survey respondents. Information on all German firms stems from Hiersemenzel, Sauer and Wohlrabe (2022). The sample of worker-provided firms contains only those that were named at least twice and for which we collected this data. We present the breakdown into major sectors and exclude the 'other' category.

Orbis We collected information on firms' assets, year of incorporation, sector based on the 4-digit NACE industry code, the headquarters zip-code, and the number of employees from Orbis.[59] For each variable from Orbis, we select the last year that the data is available; we CPI-adjust monetary values. We match firms based on firm name and address. We match 100% of the researcher-provided and 79% of worker-provided firms named at least twice in the initial survey.

Kununu We collected firm-level information from Kununu, the leading employer review platform in Germany. Kununu is similar to the review platform Glassdoor but has a stronger presence in Germany. We linked firms in the Kununu data to our data based on firm name. Among the researcher-provided firms, 100% of firms appear on Kununu. Among the worker-provided firms that were named at least twice in the initial survey, this is true for 88% of firms. As of February 2023 (when we collected the Kununu data), Kununu contained reviews for almost 200,000 German firms. In total, these firms received over 2.5 million ratings and had around 1.5 billion page views.

Employer Ratings We hand-collected information on firms' reputations from three established employer rankings; we use data from 2022. First, we identify whether a firm is listed as one of the largest German employers (Deutsche Wirtschaft, 2022). Second, we identify whether a firm is listed as one of the most popular employers in Germany based on a rating of worker reviews (Statista, 2023). Third, we obtain information about each firm's brand recognition, which may affect workers' familiarity with the firm (Kantar, 2023).

[59]Since Orbis draws on firms' balance sheet information, the number of employees may include employees outside of Germany. We hand-collected the number of employees firms have in Germany from the firms' websites.

F Survey Validations

This appendix describes a number of validation exercises which confirm responses in our worker survey accurately describe workers' beliefs and preferences. Additional validation exercises were reported in the online appendix to Caldwell, Haegele and Heining (2024).

F.1 Comparison with External Surveys

As noted in Section 3, the survey contained a bargaining module which was designed for the analysis in Caldwell, Haegele and Heining (2024). The table below (Table F1) reproduces the analysis from that paper. Each entry shows the coefficient on the "firm policy" from a regression of workers' responses (elicited in the worker survey fielded by the IAB) on firm responses (elicited in the firm survey, conducted a year prior by the ifo Institute). For instance, the first entry shows that at firms that say they have some workers covered by a CBA, workers are more likely to say that they themselves are covered by a collective bargaining agreement. Column 3 shows that whether a firm elicits salary expectations predicts whether a worker says they provided these expectations before receiving an initial salary offer (workers may choose to provide this, even if the firm does not elicit this at the application stage).

Overall, the correlation between worker and firm responses indicates that parties in both surveys (conducted two years apart, by separate partners) were attentive and provided accurate information.

Table F1: Comparing Firm and Worker Responses (from Caldwell, Haegele and Heining, 2024)

	CBA Coverage		New Hire Bargaining				
			Binary				Continuous
	All	Recent Entrants	Provided Salary Expectations	Asked for More	Asked for ≥6% More	Asked and Got More	Asked and Got More
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Firm Policy	0.604***	0.622***	0.175***	0.099***	0.112***	0.122***	0.956***
	(0.054)	(0.058)	(0.059)	(0.030)	(0.031)	(0.030)	(0.245)
Constant	0.121***	0.145***	0.490***	0.288***	0.159***	0.170***	0.817***
	(0.021)	(0.037)	(0.052)	(0.024)	(0.023)	(0.023)	(0.167)
Observations	5466	637	652	627	623	624	621
Firms (Clusters)	321	120	152	144	143	144	143

Note: This table validates firms' survey responses by comparing firms' responses with those of workers at the same firm. Each column presents results from a different regression of worker response indicators (indicated in the column) on indicators for the relevant firm strategy. Columns 1 and 2 include individuals who have not changed their firm in the previous two years (and who are still at the firm indicated in the firm survey). Because we only elicited bargaining histories for workers who had joined their firm within the previous three years, the remaining columns include individuals who have been at their firm for between two and three years. Standard errors are clustered at the firm level. Levels of significance: * 10%, ** 5%, and *** 1%.

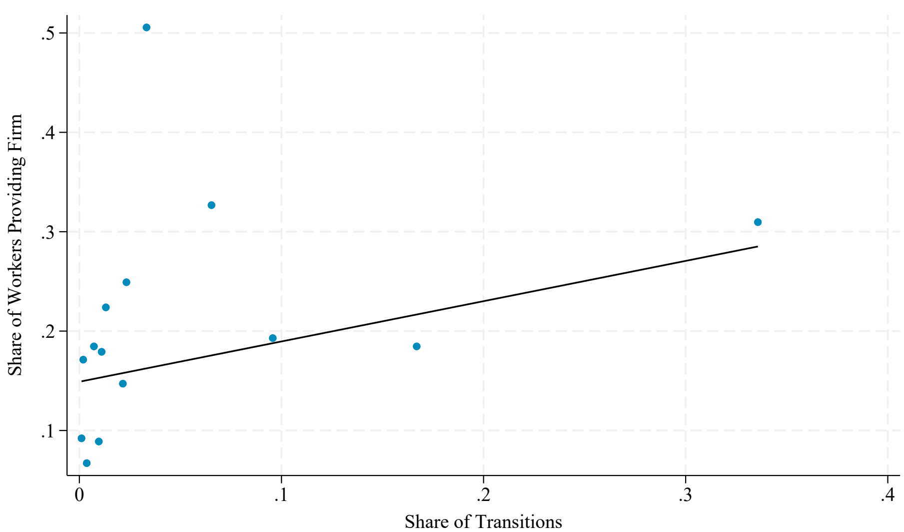
F.2 Comparison with Administrative Records

Another way we validated the survey was by comparing workers' responses to administrative records. Here we focus on a check which align workers' perceptions of their outside options to objective measures.

For each firm in our dataset we identified the set of firms provided by workers of that firm. We calculate the share of workers providing each of the worker provided firms. We then identified each of the firms our surveyed workers work at and used the administrative records to identify the historical destinations of workers moving from these firms. For each of these potential destinations (including those never mentioned by surveyed workers), we calculated the share of workers moving to these firms.

Figure F1 shows workers' provided firms are firms which workers from their firm have historically moved to. The strong positive relationship confirms that workers are not naming aspirational "dream" firms, but rather the specific employers that constitute the actual relevant labor market for their colleagues.

Figure F1: Relationship Between Stated and Revealed Destinations



Note: For each origin-destination pair we calculate the share of workers at the origin firm who move to a given destination firm between 2010-2019. We then calculate the share of workers at this firm who name each destination firm as a place they would apply if they wanted to switch firms. This is a binned scatterplot of the former share (x-axis) against the latter (y-axis).

F.3 Internal Consistency

Finally, we performed a long series of internal consistency checks to ensure workers' responses to distinct questions were aligned with their observed behavior and with other questions in the survey. We report the most important checks here. In previous work on bargaining, we reported an additional set of internal consistency checks, focusing on questions related to bargaining (Caldwell, Haegele and Heining, 2024). These checks confirm the finding in this appendix: workers'

Table F2: The Correlation Between Workers' Stated Outside Options and Objective Measures

	Easy to Find a Better Job (0/1)			Ease of Finding a Better Job (0-3)		
		Search in Past 6 Months?			Search in Past 6 Months?	
	Full Sample	Yes	No	Full Sample	Yes	No
	(1)	(2)	(3)	(4)	(5)	(6)
	A. Received an Offer in Previous 6 Months					
Any Offers	0.146***	0.130***	0.136***	0.234***	0.186***	0.265***
	(0.019)	(0.022)	(0.041)	(0.032)	(0.035)	(0.074)
Constant	0.436***	0.474***	0.351***	1.413***	1.495***	1.230***
	(0.012)	(0.014)	(0.020)	(0.020)	(0.023)	(0.037)
Observations	9641	7008	2633	9641	7008	2633
	B. Number of Job Offers in Previous Six Months					
Number of Offers	0.062***	0.054***	0.063***	0.100***	0.078***	0.126***
	(0.008)	(0.008)	(0.017)	(0.012)	(0.014)	(0.031)
Constant	0.437***	0.475***	0.349***	1.415***	1.497***	1.226***
	(0.011)	(0.014)	(0.020)	(0.019)	(0.022)	(0.036)
Observations	9619	6996	2623	9619	6996	2623
	C. Was Contacted with Job Information in Previous Six Months					
1{Contacted}	0.118***	0.100***	0.085**	0.200***	0.153***	0.175***
	(0.019)	(0.022)	(0.037)	(0.031)	(0.036)	(0.066)
Constant	0.424***	0.466***	0.350***	1.389***	1.479***	1.227***
	(0.014)	(0.018)	(0.022)	(0.024)	(0.029)	(0.041)
Observations	9649	7015	2634	9649	7015	2634

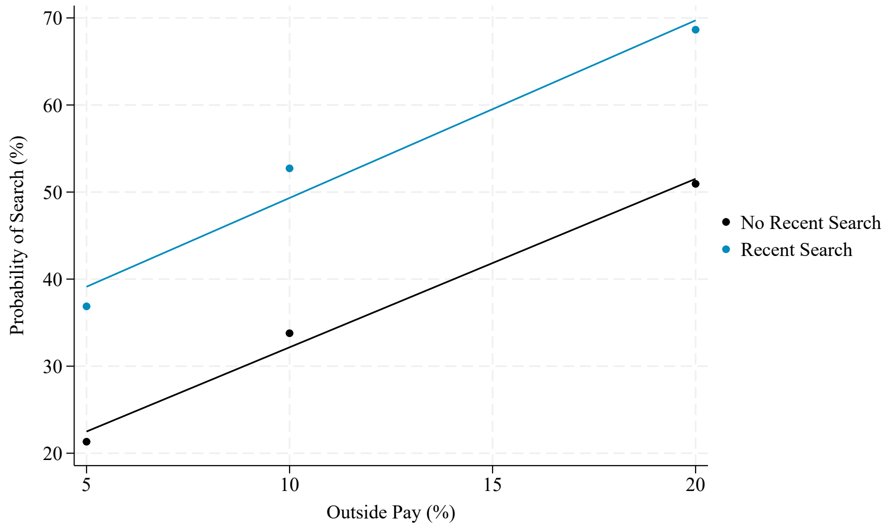
Note: This table examines the correlation between workers' stated outside options and other characteristics. The dependent variable in Columns 1 to 3 is an indicator for whether the worker said it would be "easy" or "very easy" to get an offer from a firm they preferred. The dependent variable in Columns 4 to 6 is a continuous measure, which ranges from 0-3 where 0 is "very difficult" and 3 is "very easy". Each panel presents results from a bivariate regression with robust standard errors. The sample in Columns 1 and 4 includes all workers who participated in the worker survey. The remaining columns look at the subset of these workers who report they did (Columns 2 and 5) or did not (Columns 3 and 6) search for a job in the previous six months. Levels of significance: * 10%, ** 5%, and *** 1%.

responses to distinct questions are aligned in the way we would expect. This suggests that workers were attentive in the survey and that they accurately reported their preferences.

First, we examined whether workers' subjective perceptions of their outside options were aligned with more objective measures elicited in the same survey. Table F2 shows that they are: workers who say they have better outside options are more likely to have received job offers in the previous six months or to have been contacted with job information. This is true both for workers who have engaged in search recently and for those who have not. The cross-question consistency confirms workers were attentive within the survey. Note that this table replicates analysis in Caldwell, Haegele and Heining (2024) but includes a broader sample of workers (i.e. not simply those at surveyed firms).

Second, we examined whether workers' stated probability of search was related to their observed search behavior. Appendix Figure F2 plots the stated search probability against randomized outside pay, split by whether the worker reported actual search activity in the prior six months.

Figure F2: Search Probabilities and Outside Pay by Recent Search Activity



Note: This figure plots a binned scatterplot of the probability a worker says they would search for outside employment against the stated (randomized) market pay. The black dots and line present results for workers who did not search in the 6 months before the survey; the blue dots and line present results for workers who searched for outside employment in the 6 months before the survey.

Consistent with revealed preference, stated search probabilities are uniformly higher for workers who report recent actual search. However, we cannot reject equality of the slopes (the p-value on the interaction is .5).

Finally, we examined whether workers' responses to questions about expected pay and mobility are consistent across modules and question formats. We estimate a "worker optimism" fixed effect by regressing the deviation between subjective and objective pay predictions on worker dummies. We perform this estimation separately by module and normalize the fixed effects to have a standard deviation of 1. Panel A of Figure F3 shows a strong positive correlation between a worker's optimism in the researcher-provided module and their optimism in the worker-provided module.

We estimate a "worker attachment" fixed effect by fitting an OLS model to the discrete choice experiments. For each worker-firm pair we construct an indicator for whether the worker prefers the incumbent firm to the outside firm. We regress this indicator on the randomized difference in pay and on worker fixed effects. Panel B shows that workers who require a high premium to leave their firm in the researcher-provided module also require a high premium in the worker-provided module. Note that, by construction, the mean within each module is zero.

Panels A and B of Figure F3 confirm that the survey captures persistent worker-specific traits (beliefs and preferences) rather than transient noise.

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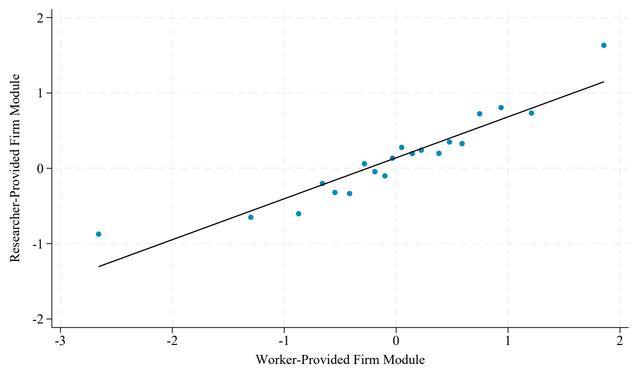
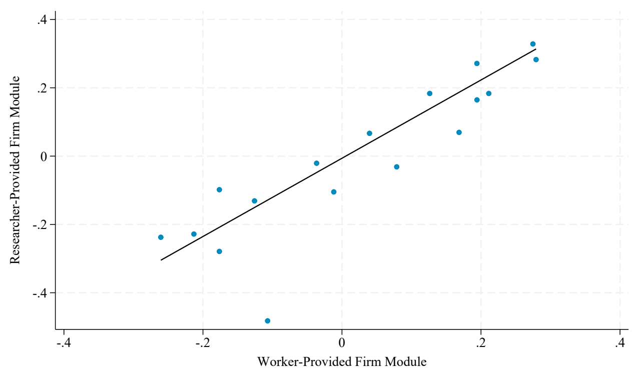
Figure F3: Bias and Attachment By Module

A. Comparison of Deviations From Objective Predictions (Standardized)



B. Comparison of Worker Attachment



Note: We compute standardized measures of bias by regressing the worker-firm deviations from researcher predictions on worker fixed effects. We take the fixed effects to be our measure of bias and normalize them so that they have a standard deviation of 1. We do this separately for the researcher-provided and worker-provided firm modules; Panel A presents a binned scatterplot of these measures against each other. We compute measures of worker attachment by regressing indicators for whether a worker would prefer to stay at their incumbent firm on the randomized wage offer and on worker fixed effects. We normalize the worker estimates to be mean zero in each module. Panel B presents a binned scatterplot of the worker-specific estimates from the researcher-provided firm module against analogous estimates from the worker-provided firm module.

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G Survey Logistics

This appendix provides additional details on how we selected and invited survey participants. We also describe how we randomly assigned incentives to participate and discuss survey response patterns. This section replicates some of the results and discussion presented in the appendix to Caldwell, Haegele and Heining (2024), which analyzes the implementation and response rates to the initial survey. This document also provides the questionnaires.

G.1 Implementation of the Initial Survey

G.1.1 Sample Construction

We used German Social Security records to identify participants for the survey. Our eligible pool consisted of workers who were—as of December 30, 2020—between the ages of 25 and 50, employed at a full-time job, and who had been at their current establishment for fewer than eight years. To manage the large number of letters, we mailed the survey in batches. For the first batch, we selected 75% of the sample (N=82,500) by randomly sampling from the set of eligible workers at firms that participated in a firm survey we conducted and then linked to IAB records in Caldwell, Haegele and Heining (2024). We oversampled these workers so that we would have appropriate power for the main analysis in Caldwell, Haegele and Heining (2024). We selected the remaining 25% (N=27,500) at random from (a random 5% sample of) workers at non-surveyed firms. We selected all of the workers (N=25,000) for the second batch from the random 5% sample of eligible workers at non-surveyed firms. We mailed reminders to a random 25% subset of individuals in the first batch who had not responded to the initial invitation when we prepared the second mailing. In spring 2024, we invited all respondents from the initial survey who provided panel consent to participate in a follow-up survey.

G.1.2 Invitations

After we identified workers for inclusion in the survey, a specialized department at the IAB pulled their addresses. This approach followed the standard protocol for surveys through the IAB. The IAB fielded the survey and the director of the IAB signed the invitation to participate. We mailed invitations to the initial survey between June 2022 and December 2022. We described the survey to potential respondents as a scientific study on pay progression in Germany. To manage the large number of letters, we staggered the mailings.

We invited respondents via mail instead of e-mail or phone because the Federal Employment Agency in Germany (*Bundesagentur für Arbeit*) only has e-mail and phone numbers of individuals who have recently been unemployed or have participated in re-employment measures. Postal addresses are available for all workers. In the invitation, we informed respondents that the survey would take approximately 10 minutes to complete. Appendix Figure G1 shows (a translation of) the wording of the invitation; the sample we include is for a worker randomized into one of the gift card treatments.

In one batch of the invitations, a printing issue led to some cases where endorsement and cover letters were mixed up. This meant that two different addresses (endorsement person a, cover person b) ended up in one envelope. Letters were sent to the addresses provided on the endorsement letter,

Figure G1: Original Invitation (English Translation)

INSTITUT FÜR ARBEITSMARKT- UND BERUFSFORSCHUNG
Die Forschungseinrichtung der Bundesagentur für Arbeit

Bei Rückfragen wenden Sie sich bitte an:
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«First name» «Last name»
«Street address»
«Zip code» «City»

Anschreiben-ID: «anschreiben_id»
Nürnberg, «date»

Scientific Study on Wage Flexibility in Germany

Hello «First name» «Last name»,

The Institute for Employment Research (IAB) is conducting a scientific study to understand how the salary progression of employees in Germany is changing. We are therefore interested in your experience in the labor market and would like to invite you to a survey. By participating, you support the IAB in advising political decision-makers and thus help to improve economic and social policy in Germany.

In a nutshell - we will interview you via the Internet
The survey will not take more than **10 minutes** of your time. Your participation is of course voluntary and anonymous. To get to the survey, you can use the display **QR code** or click on the following link:
https://umfragen.iab.de/goto/LF-befragung

Your personal password for participation is: **«survey_password»**

Participating pays off
As a thank you for your participation, we are raffling off 1,500 vouchers, each with a value of 10 euros.

Your information is confidential
The safety of your personal data is important to us. We assure you that your information will be treated with strict confidentiality in accordance with the statutory data protection regulations and will only be used for scientific purposes. Your answers cannot be linked to your person. You will find additional data protection explanations attached.

Thank you for your cooperation and for your trust!

Kind regards

Prof. Bernd Fitzenberger Ph.D.
Direktor des Instituts für Arbeitsmarkt- und Berufsforschung (IAB)

Note: This is a screenshot of the translated cover letter sent to workers randomized to the high-gift-card but the cover letter information included the name and password of a different individual. If an individual who received an incorrect mailing participated in the survey they therefore would have been linked to the wrong survey. Based on inspection of the frequency of this error in letters that were returned due to incorrect addresses (which is an independent issue, but allows us to analyze the frequency of the error), we calculate that at most 431 people among the set of 17,772 recipients in this batch were likely affected (2.4%), assuming the share among returns is the same as in the overall sample. People may likely have realized the mix-up and may not have participated. Since this only affected people with endorsement letters (which were randomly assigned), this may have reduced the response rate for this group.

G.1.3 Exogenous Variation in Incentives to Participate

So that we could analyze patterns of non-response, we introduced random variation in individuals' incentives to participate. We did this through (1) randomized financial incentives, (2) randomized endorsement letters, and (3) randomized follow-up.

Gift Card Lottery Information on the gift card lottery was randomly assigned to random subsets of the first 110,000 workers selected for batch 1 and was included with the initial invitation. 10,000 workers were selected to be in a raffle for a 5 euro gift card and 20,000 workers were selected to be in a raffle for a 10 euro gift card. Workers were informed about the lottery in the cover letter, which provided the gift card value, as well as the number of gift cards that we would raffle off. After the survey closed in January 2023, the IAB conducted the gift card lottery by randomizing among' participants who started the survey. Winners were informed either via e-mail or mail, according to the preferences indicated in the survey.

Endorsement Letters The endorsement letter was randomly assigned to a random subset of 82,500 of the first 110,000 workers selected for batch 1 and was included with the initial invitation to participation. The letter was signed by one of the 2021 Nobel Prize winners in economics, identified both as one of the 2021 Laureates and as a previous collaborator of the IAB. The letter highlighted the importance of scientific labor market research and urged recipients to complete the survey.

Reminder Mailings There were 99,698 initial non-responders. We randomly selected 25,000 to receive a follow-up letter. The reminder letters had nearly identical wording to the initial invitations, but reminded individuals that they had previously been invited to participate in the survey. The letters included the same information on data protection as before. Individuals who had been included in the gift card raffle in the initial invitation received reminders of their offer to participate. Individuals who had been randomized to receive endorsement letters did not receive a second endorsement letter.

G.1.4 Balance Check

Appendix Table G1 describes the workers we invited for the survey. As Columns 2 and 3 indicate, conditional on the strata used for selection (whether an individual is at a surveyed firm), there is no difference in the characteristics of eligible workers and those invited to participate in our survey. Columns 4 and 5 compare those selected for the treatment and control groups for the endorsement letters and gift card treatments. Conditional on the strata used to assign these treatments, we find no difference in the characteristics of those selected and those not selected.[60] Column 6 shows that among those eligible to receive a reminder (initial non-responders in batch 1), there is no difference between those selected to receive a reminder and those not selected.

G.1.5 Response Rate and Consent

The survey was open until January 15, 2023. We received 13,680 total responses. Subtracting the number of letters that could not be delivered, the survey had an effective response rate of 11.4%. This response rate is much higher than those of other surveys at the IAB that invite respondents for the first time (Haas et al., 2021). Among the 13,680 individuals who started the survey, 11,868 completed it; this represents a completion rate of 74%.[61] The median response time among individuals who completed the survey was 9 minutes.

We asked participants for their consent to link their answers to the employer-employee data at the IAB. We have 10,134 complete responses with linkage consent, which we link to the IAB records. While this direct consent is necessary under German privacy laws to link the survey data to other data sources, we are able to analyze the raw and unlinked data for both consenters and non-consenters. We also asked participants for their consent to participate in follow-up surveys. 8,416 respondents who provided consent for this linkage also provided consent to be contacted for future survey waves. Among the 11,868 complete responses, this represents a panel consent rate of 83%.

G.1.6 Impact of Randomized Incentives on Response Rates

Appendix Table G2 shows that neither the gift cards nor the endorsement letter had a statistically significant (or economically meaningful) impact on response rates in the initial survey. By contrast, the reminder message increased response rates by 4 percentage points among workers who did not initially respond to the survey. Because both the endorsement letter and gift card information were only visible to individuals who opened the initial mailer, one plausible interpretation is that much of the initial non-response was driven by individuals simply ignoring our initial invitation.

G.1.7 Selection into Non-Response

We follow Dutz et al. (2021) in analyzing the characteristics of compliers. We focus on only individuals in batch 1 of the initial survey, because this is the batch in which we implemented the randomization of incentives. We do not characterize endorsement letter or gift card compliers because these incentives did not have a significant impact on response rates. Appendix Table G3 describes three populations of workers. Column 1 describes "early always takers": those who responded to the survey before we mailed the reminder. Column 2 describes the "late always takers":

[60] We grouped individuals into two groups based on their federal state of residence and randomly assigned gift cards within these strata. We randomly assigned endorsement letters without regard to state.

[61] We define a response as complete if a respondent clicked through to the (second to last) question eliciting consent for participating in a follow-up survey. We do not require respondents to have answered every question to be counted as complete responses. The survey did not require individuals to respond to particular questions.

Table G1: Randomization Assessment

	Eligible Mean	Selection: Wave 1 - Eligible	Selection: Wave 2 - Eligible	Wave 1 Randomization: Lottery - No Lottery	Wave 1 Randomization: Letter - No Letter	Wave 1 Randomization: Reminder - No Reminder
	(1)	(2)	(3)	(4)	(5)	(6)
Demographics						
Female	0.32	0.63	0.33	0.48	0.41	0.32
Age	33.02	0.29	0.94	0.75	0.73	0.82
German Citizen	0.77	0.22	0.65	0.75	0.67	0.22
College Education	0.34	0.59	0.47	0.06	0.35	0.39
Apprenticeship	0.45	0.90	0.67	0.64	0.51	0.69
Daily Earnings	129.12	0.22	0.39	0.23	0.47	0.09
Occupation Group						
Manager	0.04	0.96	0.45	0.95	0.71	0.57
Recent Entrant	0.34	0.86	0.84	0.95	0.06	0.01
Sector						
Manufacturing	0.32	0.86	0.72	0.01	0.74	0.44
Retail	0.10	0.47	0.40	0.53	0.47	0.41
Professional	0.10	0.95	0.42	0.08	0.44	0.70
Establishments		24928	21248	7253	19600	7204
Workers		110000	25000	30000	82500	25000
F-statistic		0.693	0.444	1.558	0.982	1.039
p-value		0.747	0.937	0.104	0.460	0.408

Note: This table documents that the random assignment of survey invitations and incentives was successful. Column 1 describes workers eligible for inclusion in our worker survey. Each entry in Columns 2 to 6 presents the p-value for a two-sided test of the coefficient from a regression of the characteristic in each row on the treatment in each column, controlling for the strata used for random assignment. P-values are calculated using robust standard errors. At the bottom of the table we present the F-statistic and associated p-value from a test of whether the coefficients on all of the coefficients are jointly zero. Columns 2 and 3 show that, conditional on the strata used for selection (whether an individual worked at a surveyed firm in 2020), selected individuals are not statistically distinguishable for non-selected individuals. Columns 4 to 6 show that, conditional on the strata used for random assignment, individuals selected to receive each of the three types of incentives, are not distinguishable from those who were not selected. We only randomized these incentives to workers in batch 1.

Table G2: Impact of Randomized Incentives on Response Rates in the Initial Survey

	Endorsement	Gift Card		
	Letter	Level	Binary	Reminder
	(1)	(2)	(3)	(4)
Treatment	0.000	-0.000	-0.002	0.040***
	(0.002)	(0.000)	(0.002)	(0.001)
Observations	109995	109995	109995	99698

Note: This table analyzes the effect of the randomized incentives on the likelihood that invited individuals completed the survey and provided linkage consent. Each coefficient stems from a separate regression of survey completion on an indicator for the respective incentive, conditional on the strata used for random assignment. Column 1 focuses on endorsement letters. Columns 2 and 3 focus on gift cards. Column 4 focuses on the survey reminder. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

those who we randomized into not receiving a reminder, but who nonetheless responded after we mailed the reminders. Column 3 describes the reminder compliers. As Column 2 indicates, virtually all of the always takers responded before we mailed the reminde r. Column 3 indicates that the reminder compliers are, relative to the early always takers, somewhat more likely to be male or covered by a collective bargaining agreement.

In addition to examining selection into response, we also examine selection into providing consent to link survey responses to administrative data and to be invited to participate in future surveys. Appendix Table G4 describes the characteristics of invited individuals (Column 1) to those who completed the survey and provided linkage consent (Column 2) and to those who additionally provided consent to participate in future surveys (Column 4). Columns 3 and 5 compare the samples in Columns 2 and 4 to the samples in Columns 1 and 2, respectively. We find modest differences in respondent characteristics with respect to gender, age, occupation type, and sector. For instance, while the female share is 30% among invited individuals, it is 32% among those who responded and provided consent. German citizens on the other hand were much more likely to respond than non-citizens. This is not surprising: these workers are likely more comfortable with the German language (we fielded our survey in German) and may feel more of an obligation to contribute to research on the German labor market. We also find meaningful differences with respect to education and earnings: more educated workers were more likely to respond to the survey. This may reflect the fact that they are more likely to be familiar with the IAB. Columns 6 and 7 show workers who responded to the follow-up survey were again positively selected on citizenship, education, and earnings.

While there is some positive selection into the survey, our final sample includes workers at a broad set of firms, including low-pay firms (Figure G2).

Table G3: Characteristics of Reminder Compliers and Always Takers in the Initial Survey

	Always Takers		Reminder
	Early	Late	Compliers
	(1)	(2)	(3)
Shares	0.05	0.000	0.03
Female	0.27	0.50	0.24
Age	33.85	34.36	34.02
German	0.94	0.93	0.91
College Degree	0.63	0.64	0.60
Apprenticeship	0.31	0.21	0.34
Daily Pay	186.04	202.40	184.97
Censored Pay	0.26	0.29	0.29
Hours Worked	40.63	41.39	40.37
CBA	0.61	0.71	0.65
Sector			
Manufacturing	0.56	0.64	0.56
Retail	0.07	0.00	0.05
Professional	0.17	0.14	0.20
AKM Fixed Effects			
Person Effect	4.47	4.33	4.44
Firm Effect	0.52	0.58	0.54
Risk Preferences	6.22	5.57	6.19
Outside Options	1.41	1.14	1.41
Did Not Provide Names	0.45	0.50	0.50

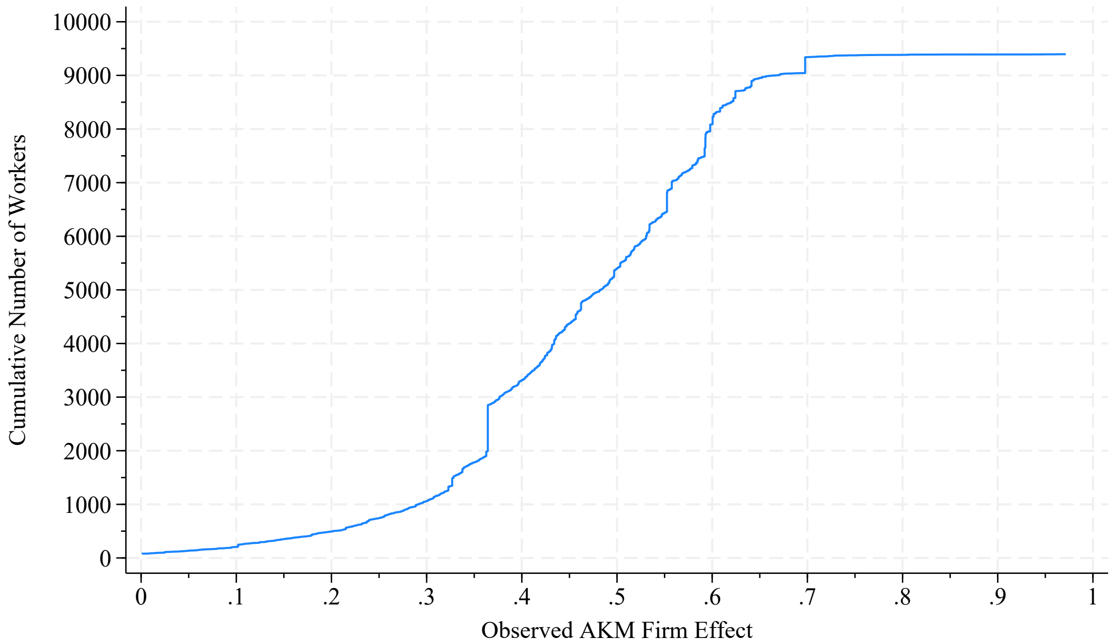
Note: This table compares the characteristics of early and late always takers to reminder compliers. Early always takers are workers in the control group who responded before we mailed the (randomized) reminders. Late always takers are workers in the control group who responded after we mailed the reminders. Reminder compliers are workers who responded after being mailed a reminder. Following Dutz et al. (2021), we estimate the reminder compliers' average characteristics via an instrumental variables regression with Y_i(1-R_{i1})R_{i2} as the outcome variable, (1-R_{i1})R_{i2} as the endogenous variable, and Z_i as the instrument. Y_i is the characteristic of interest, R_{i1} is an indicator for responding before we mailed the reminders, R_{i2} is an indicator for responding after we mailed the reminders, and Z_i is an indicator for having received a reminder. Shares reported in Row 1 are out of all workers invited to respond to the initial survey (Column 1), those who neither responded earlier nor received a reminder (Column 2), and those who did not respond earlier but did receive a reminder (Column 3).

Table G4: Non-Response and Consent

	Invited	Linkage Consent		Panel and Linkage		Responded to Follow-Up	
	Mean	Mean	Difference Rel. Invited	Mean	Difference Rel. Linked	Mean	Difference Rel. Invited
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Demographics							
Female	0.30	0.32	0.02 ***	0.32	0.01	0.31	-0.01
	(0.46)	(0.46)	(0.00)	(0.47)	(0.01)	(0.46)	(0.01)
Age	33.63	33.33	-0.32 ***	33.33	-0.02	33.41	0.14
	(6.59)	(6.23)	(0.06)	(6.14)	(0.17)	(6.16)	(0.14)
German Citizen	0.81	0.92	0.12 ***	0.92	0.03 ***	0.94	0.02 ***
	(0.39)	(0.27)	(0.00)	(0.26)	(0.01)	(0.24)	(0.01)
College Education	0.39	0.59	0.22 ***	0.60	0.07 ***	0.65	0.07 ***
	(0.49)	(0.49)	(0.01)	(0.49)	(0.01)	(0.48)	(0.01)
Apprenticeship	0.45	0.33	-0.12	0.32	-0.05	0.29	-0.06
	(0.50)	(0.47)	(0.00)	(0.47)	(0.01)	(0.45)	(0.01)
Daily Earnings	146.03	169.79	25.69 ***	170.92	6.61 ***	175.02	7.27 ***
	(60.77)	(56.71)	(0.59)	(56.67)	(1.50)	(55.27)	(1.24)
Occupation Group							
Manager	0.05	0.06	0.02 ***	0.06	0.01 *	0.07	0.01 **
	(0.21)	(0.24)	(0.00)	(0.24)	(0.01)	(0.25)	(0.01)
Recent Entrant	0.26	0.20	-0.06 ***	0.20	-0.02 *	0.19	-0.02 **
	(0.44)	(0.40)	(0.00)	(0.40)	(0.01)	(0.39)	(0.01)
Sector							
Manufacturing	0.44	0.46	0.03 ***	0.46	0.00	0.47	0.01
	(0.50)	(0.50)	(0.01)	(0.50)	(0.01)	(0.50)	(0.01)
Retail	0.10	0.09	-0.01 ***	0.09	0.01	0.09	0.00
	(0.30)	(0.28)	(0.00)	(0.28)	(0.01)	(0.28)	(0.01)
Professional	0.10	0.13	0.04 ***	0.13	0.01	0.13	0.00
	(0.29)	(0.34)	(0.00)	(0.34)	(0.01)	(0.34)	(0.01)
Establishments	42705	3556		2983		1457	
Observations	134995	10134		8416		3664	

Note: This table describes the characteristics of workers we invited to complete the survey (Column 1) to workers who completed the survey and provided consent to link their responses to the administrative data (Column 2), to the subset of these workers who additionally provided consent to participate in follow-up surveys (Column 4) and those that participated in the follow-up survey (Column 6) Columns 3, 5 and 7 present the strata-adjusted differences between the samples indicated in the header and sub-header. For instance, Column 3 reports the difference between workers who completed the survey and provided linkage consent to the set of invited workers. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

Figure G2: Sample Size by AKM Firm Effect



Note: This figure shows the cumulative number of workers in our sample (y-axis) who, at the time of sampling, worked at an establishment whose AKM firm effect was at or below the value indicated on the x-axis. Workers whose firms do not have associated estimates are not included in this graph.

G.2 Implementation of the Follow-Up Survey

In spring 2024, we invited all respondents from the initial survey who provided panel consent to participate in a follow-up survey. Participants who did not provide an e-mail address in the initial survey were contacted via mail by the IAB. The invitation letter was very similar to that of the initial survey, but reminded respondents of their participation and introduced the follow-up survey. Among the participants who did provide an e-mail address in the initial survey, we randomized the initial mode of contact: 25% received an invitation sent by the IAB via e-mail and 75% first received an invitation letter via mail (identical to those participants who did not provide an e-mail) and then received another invitation via e-mail. Within each group of respondents, we randomly selected 75% to receive a further reminder (either by mail or e-mail).

The survey was open for 14 weeks and received a 50% response rate. The completion rate was 92%. The median response time among individuals who completed the survey was 9 minutes. Among those who completed the follow-up survey, 90% provided linkage consent.

G.2.1 Impact of Randomized Incentives on Response Rates

We asked individuals in the initial wave of the survey whether we could contact them for a follow-up survey. We also asked these individuals whether we could contact them via e-mail. Most workers indicated they would prefer to be contacted by e-mail in the follow-up survey. In the

Table G5: Impact of Randomized Incentives on Response Rates

	Provided an E-Mail			
	No	Yes		
	(1)	(2)	(3)	(4)
Reminder Letter	0.079***			
	(0.019)			
Initial Letter		0.230***		0.232***
		(0.015)		(0.015)
Reminder E-mail			0.070***	0.077***
			(0.016)	(0.016)
Constant	0.360***	0.270***	0.393***	0.210***
	(0.017)	(0.013)	(0.014)	(0.018)
Observations	3405	5011	5011	5011

Note: This table analyzes the effect of the randomized incentives on the likelihood that invited individuals completed the follow-up survey and provided linkage consent. In the initial survey, we asked individuals whether we could contact them for a follow-up survey and whether we could contact them via e-mail (which they then provided). Column 1 focuses on workers who did not provide an e-mail address with which to contact them for the follow-up survey. Columns 2 to 4 focus on workers who did provide an e-mail address for this purpose. Each column presents the coefficients from a regression among workers in the indicated sample; the dependent variable is an indicator for whether the individual responded to the follow-up survey and provided consent to link this new response to the administrative records; the independent variables are an indicator for the survey incentive(s) indicated in the rows and a constant. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

follow-up survey, we groups individuals by whether they provided an e-mail address and targeted incentives accordingly.

Overall, about 50% of workers we invited to complete the follow-up survey successfully did so. Appendix Table G5 shows that each of the reminders in this follow-up had a significant impact on response rates. Column 1 examines the response rates of individuals in the letter (non-email) sample. For these workers, we randomized a single incentive: the provision of a reminder letter (as in the initial survey wave). As Column 1 indicates, those that did not receive the reminder (25% of the sample) had a response rate of just under 40%; the reminder boosted response rates by 8 percentage points.

Among workers who provided an e-mail address, we cross-randomized two incentives. First, we mailed a random subset (75%) of these individuals a letter before we e-mailed them regarding the survey. Second, we sent an e-mail reminder to a random subset (75%, cross-randomized) of these individuals. Both incentives were highly effective. The initial letter boosted response rates by 23 percentage points (Column 2). The e-mail reminder boosted response rates by 7 percentage points (Column 3).

G.2.2 Selection Into Non-Response

Analogous to the initial survey, we analyze the characteristics of compliers for the follow-up survey following Dutz et al. (2021). Appendix Table G6 describes three groups of individuals: early always takers (Column 1), late always takers (Column 2), and reminder compliers (Column 3). As in the initial survey, almost all of the always takers responded before the reminder was sent. Column 3 indicates that the reminder compliers are very similar to the early always takers.

H Questionnaire

Note that while we included questions for workers who indicated they were, at the time of the survey, self-employed or non-employed, our survey was targeted at employed workers. We exclude self-employed and non-employed individuals from our analysis (in all papers) as the number of workers in these categories was low.

H.1 English Translation of Questionnaire

This section provides the questionnaires of both the initial and follow-up survey. From both surveys, we omit questions about negotiation that were designed for and described in detail in Caldwell, Haegele and Heining (2024). The appendix of that paper provides more information on those questions. We first provide the English translation of both questionnaires, followed by the original wording in German.

H.1.1 Main Survey

1. In order to keep the survey short, we would like to include data in the evaluation that are available at the Federal Employment Agency in Nuremberg. This is, for example, information about your previous employment.

   We strictly comply with all data protection regulations or these are strictly controlled by the IAB. All information and answers are evaluated and presented anonymously, i.e. without name, address and contact information. Nobody can therefore recognize which answers you have given. There is also no transfer of data that would identify you. Further information on data protection can be found https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx.

   The evaluation is carried out for purely scientific purposes, i.e., not for commercial purposes such as advertising or marketing. Your consent is of course voluntary.

   Do you agree to your data being linked?

   a. Yes, I agree to the link

   b. No, I do not agree to the link

2. What best describes your current main employment?
   (a) Employed with social security contributions
   (b) Self-employed
   (c) Not employed
   (d) Other

Table G6: Characteristics of Reminder Compliers and Always Takers in the Follow-Up Survey

	Always Takers		Reminder
	Early	Late	Compliers
	(1)	(2)	(3)
Shares	0.35	0.01	0.13
Female	0.32	0.50	0.31
Age	33.37	32.55	33.36
German	0.94	0.91	0.93
College Degree	0.65	0.77	0.61
Apprenticeship	0.29	0.14	0.33
Daily Pay	175.12	157.38	171.34
Censored Pay	0.22	0.18	0.19
Hours Worked	40.45	40.48	40.52
CBA	0.56	0.33	0.61
Sector			
Manufacturing	0.47	0.36	0.48
Retail	0.09	0.09	0.09
Professional	0.13	0.18	0.13
AKM Fixed Effects			
Person Effect	4.45	4.41	4.42
Firm Effect	0.46	0.44	0.45
Risk Preferences	6.09	5.68	6.22
Outside Options	1.40	1.45	1.41
Did Not Provide Firm Names	0.44	0.41	0.43

Note: This table compares the characteristics of early and late always takers to reminder compliers for the follow-up survey. Early always takers are workers in the control group who responded before we mailed the (randomized) reminders. Late always takers are workers in the control group who responded after we mailed the reminders. Reminder compliers are workers who responded after being mailed a reminder. Following Dutz et al. (2021), we estimate the reminder compliers' average characteristics via an instrumental variables regression with $Y_i(1\text{-}R_{i1})R_{i2}$ as the outcome variable, $(1\text{-}R_{i1})R_{i2}$ as the endogenous variable, and Z_i as the instrument. Y_i is the characteristic of interest, R_{i1} is an indicator for responding before we mailed the reminders, R_{i2} is an indicator for responding after we mailed the reminders, and Z_i is an indicator for having received a reminder. The shares reported in Row 1 are out of all workers invited to respond to the follow-up survey (Column 1), those who neither responded earlier nor received a reminder (Column 2), and those who did not respond earlier but did receive a reminder (Column 3).

Module: Employment and Bargaining

Submodule: Employed or missing

3. When did you first join your current company?
   (a) In the past 6 months
   (b) 6-12 months ago
   (c) 1-2 years ago
   (d) 2-3 years ago
   (e) >3 years ago
4. Is your current position covered by a CBA agreement (i.e., are you paid according to a CBA)?
   (a) Yes
   (b) No
   (c) I don't know
5. How many hours do you work in a typical week? {fill in}
6. At the time that you first applied to your current company, did you know anyone who worked at the company?
   (a) Yes
   (b) No, but I knew someone who used to work at the company
   (c) No, I did not know anyone
7. At the time that you applied, did you know what salary you would earn?
   (a) I had no or very little idea
   (b) I only had a rough idea what is paid in my region or sector
   (c) I had at least a rough idea what this company pays for the position
   (d) I knew exactly what this company pays for the position
8. During the past six months, have you done any of the following? Please select all that apply.
   (a) Looked at job postings
   (b) Updated public resume or employment information (e.g., Xing, LinkedIn)
   (c) Reached out to people in my network for information about potential job opportunities
   (d) Applied to a job at another company
9. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g., sent you a job opening or offered to provide a referral)?
   (a) Yes
   (b) No
10. During the past six months, did you receive any job offers from other companies?
   (a) Yes
   (b) No
11. During the past six months, did your company offer you a salary increase without you asking?
   (a) Yes
   (b) No
12. During the past six months, did you actively ask for an increase in salary?
   (a) Yes

(b) No

Submodule: Self-Employed or Non-Employed

13. How many hours do you work in a typical week?{fill in}
14. During the past six months, have you done any of the following? Please select all that apply.

    o Looked at job postings

    o Updated public resume or employment information (e.g., Xing, LinkedIn)

    o Reached out to people in my network for information about potential job opportunities

    o Applied for jobs
15. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g. sent you a job opening or offered to provide a referral)?

    (a) Yes
    (b) No
16. During the past six months, did you receive any job offers?

    (a) Yes
    (b) No

Module: Worker-Provided Firms

17. Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to? Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. "PlaceHolder Inc"). These can be companies without current job vacancies.

    (a) {fill in Company 1}
    (b) {fill in Company 2}
    (c) {fill in Company 3}
    (d) I do not want to answer this question [exclusive option]
18. **[Employed]** What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

    (a) {fill in Company 1}
    (b) {fill in Company 2}
    (c) {fill in Company 3}
19. What best describes how certain you feel about your pay estimates?

Company Name	Very Uncertain	Uncertain	Somewhat Certain	Certain	Very Certain
{fill in Company 1}					
{fill in Company 2}					
{fill in Company 3}					

20. Some of these companies may have subsidiaries in multiple locations. Which location did you have in mind?

Company Name	Location
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

21. Do you know any employees at these companies (e.g. former coworkers, friends and family)?

Company Name	Yes, I know current employees	Yes, I know former employees	No, I do not know any who ever worked at this co
22. {fill in Company 1}			
{fill in Company 2}			
{fill in Company 3}			

23. Suppose one of your colleagues got offers from the following companies for a position like yours. Which company do you think offers the highest gross annual pay? Please rank the offers from 1 (highest pay) to 3 (lowest pay).

Company Name	Ranking
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

24. Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 3 (least likely to take).

Company Name	Ranking
{fill in Company 1}	
fill in Company 2}	
{fill in Company 3}	

25. **[Employed]** Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {15%/15%} raise	
Remain at Current Firm at Current Pay	

26. **[Employed]** Now we would like to ask you to re-rank the firms under 2 scenarios. First, suppose that you have the same three offers, but would be able to have the same **commute/route** to work. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain at Current Firm at Current Pay	

27. **[Employed]** Now suppose that you have the same set of offers, but instead of the commute staying the same your **career growth** (future raises/promotions) are identical at each option. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain at Current Firm at Current Pay	

Module: Researcher-Provided Firms

28. Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.
    (a) Company 1
    (b) Company 2
    (c) Company 3
    (d) Company 4
    (e) Company 5
    (f) Company 6
    (g) Company 7
29. **[Employed/Self-Employed]** We are now interested in your perception of job opportunities available at different companies. What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company Name	Pay in Euros
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

30. **[Non-Employed]** We are now interested in your perception of job opportunities available at different companies. What do you think your gross annual pay would be if you worked at these companies in a position similar to your most recent one?

Company Name	Pay in Euros
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

31. **[Employed]** Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain at Current Firm at Current Pay	

32. **[Self-employed]** Now suppose you can remain self-employed or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Option	Ranking
{fill in Company 1} with a {10%/15%} raise	
{fill in Company 2} with a {10%/15%} raise	
{fill in Company 3} with a {5%/20%} raise	
Remain Self-employed at Current Pay	

33. **[Non-employed]** Now suppose you can remain non-employed or join any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Option	Ranking
{fill in Company 1} with a {10%/15%} higher income	
{fill in Company 2} with a {10%/15%} higher income	
{fill in Company 3} with a {5%/20%} higher income	
Remain Non-employed	

34. In your opinion, how easy would it be for you to obtain a job offer from a different company that you would prefer to your current position?
    (a) Very Easy
    (b) Easy
    (c) Difficult
    (d) Very difficult
35. If you had to move to a new company within the next {one/three/six} months, what best describes where you would search for a job? This only applies to positions **without** a home office option.
    (a) I would search for jobs which are not further away from my place of residence than {fill-in} Kilometers
36. Finally, we would like to ask you to assess yourself. Are you generally a person who is willing to take risks or do you try to avoid taking risks? Please choose a value on the scale below, where the value 0 means "not at all willing to take risks" and the value 10 means "very willing to take risks".
    (a) 0 (Not at all willing to take risks) 1 2 3 4 5 6 7 8 9 10 (Very willing to take risks)

Module: Consent

37. To thank participants for their time, we are raffling off {1500}/{1000} gift cards with a value of {10}/{5} Euros. If you would like to participate in the raffle, please provide your email address below. _______________ {open-field that accepts e-mail addresses only}
38. Your information supports IAB's research on employees' salary progression in Germany. Since current labor market dynamics are rapidly changing, we would like to survey you again in a few months. Would you like to continue to contribute to providing well-founded information on employees' circumstances in Germany? Then we ask you to provide us with your email address so that we can question you again. Of course, we will only use the e-mail address to write to you for our survey. You can of course unsubscribe from the survey at any time. To which email address can we send our invitation?
    (a) {open field that accepts e-mail addresses only}

(b) I do not want to be contacted by email.

39. May we then invited you again by mail?
    (a) Yes
    (b) No

H.1.2 Follow-up Survey

1. In order to keep the survey short, we would like to include data in the evaluation that are available at the Federal Employment Agency in Nuremberg. This is, for example, information about your previous employment. We strictly comply with all data protection regulations or these are strictly controlled by the IAB. All information and answers are evaluated and presented anonymously, i.e., without name, address and contact information. Nobody can therefore recognize which answers you have given. There is also no transfer of data that would identify you. Further information on data protection can be found https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx. The evaluation is carried out for purely scientific purposes, i.e. not for commercial purposes such as advertising or marketing. Your consent is of course voluntary. Do you agree to your data being linked?
    (a) Yes, I agree to the linkage
    (b) No, I do not agree the linkage
2. Have you started a new job at a different company since {month_survey1 2022/2023}?
    (a) Yes
    (b) No
3. **[Q2=Yes]** What best describes your current main employment?
    (a) Employed with social-security contributions
    (b) Self-employed
    (c) Not employed
    (d) Other
4. During the past six months, have you done any of the following? Please select all that apply.
    (a) Looked at job postings
    (b) Updated public resume or employment information (e.g., Xing, LinkedIn)
    (c) Reached out to people in my network for information about potential job opportunities
    (d) Applied to a job at another company
5. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g., sent you a job opening or offered to provide a referral)?
    (a) Yes
    (b) No
6. During the past six months, how many job offers from other companies did you receive?
    (a) 0
    (b) 1
    (c) 2
    (d) 3 or more

Module: Worker-Provided Firms

7. Next, we are interested in how you perceive employment opportunities in Germany. Even if you don't know something exactly, we would like to ask you to share your best guess with us. Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to? Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g., "Place-Holder Inc"). These can be companies without current job vacancies.
   (a) {fill in Company 1}
   (b) {fill in Company 2}
   (c) {fill in Company 3}
   (d) I do not want to answer this question [exclusive option]
8. **[Q7=d]** What best describes why you did not provide any company names in the previous question?
   (a) I am not comfortable sharing this information
   (b) There are companies I would consider, but the survey interface was too difficult
   (c) I could not think of specific companies
   (d) I have no interest in switching companies and have not thought about it at all
9. **[Q7 has only 1 or 2 firms provided]** What best describes why you did not provide three company names in the previous question?
   (a) I am not comfortable sharing this information
   (b) There are more companies I would consider, but the survey interface was too difficult
   (c) I could not think of any other specific companies
   (d) I have no interest in switching companies and have not thought much about it
10. **[Q7 is not missing]** What do you think your **gross annual pay** would be if you worked at these companies in a position similar to your current position?

Company	**Gross Annual Pay in Euros**
{Company 1}	
{Company 2}	
{Company 3}	

11. **[Q7 is not missing]** Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the pay specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{Company 1} with a pay of{}	
{Company 2} with a pay of{}	
{Company 3} with a pay of{}	
Remain at current company with a pay of{}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Researcher-Provided Firms

12. Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.
    (a) Company 1
    (b) Company 2
    (c) Company 3
    (d) Company 4
    (e) Company 5
    (f) Company 6
    (g) Company 7
13. What do you think your **gross annual pay** would be if you worked at these companies in a position similar to your current position?

Company	Gross Annual Pay in Euros
{Company 1}	
{Company 2}	
{Company 3}	

14. **[Q13 is missing]** Even if you are unsure, we are interested in your guess of how these companies compare with respect to what they pay. Please rank the companies from 1 (highest pay) to 3 (lowest pay).

Company	Ranking
{Company 1}	
{Company 2}	
{Company 3}	

15. Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the pay specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{Company 1} with a pay of {}	
{Company 2} with a pay of {}	
{Company 3} with a pay of {}	
Remain at current company with a pay of {}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

16. Now we would like to ask you to rerank the companies under two scenarios. First suppose that you have the same three offers but would be able to have the same **commute** to work as before. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{Company 1} with a pay of {}	
{Company 2} with a pay of {}	
{Company 3} with a pay of {}	
Remain at current company with a pay of {}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

17. Now suppose that you have the same set of offers, but instead of the commute staying the same you knew that the **training and learning opportunities** at each of the three firms was the same as at your current company. Please rank the offers from 1 (most likely to take) to 4 (least likely to take)

Job Offer	Ranking
{Company 1} with a pay of{}	
{Company 2} with a pay of {}	
{Company 3} with a pay of{}	
Remain at current company with a pay of {}	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Applications

18. Suppose you planned to move to a new company in the next {one/three/six} months, how many different companies do you think would you apply to?
    (a) The number of companies I would apply to is: _____
19. **[Randomize who sees Q19 (50%) and who sees Q20/Q21 (50%)]** In your opinion, what are the main reasons that make employees hesitant to switch jobs? Please choose the two most important reasons.
    (a) Personal ties to coworkers
    (b) Location
    (c) Pay
    (d) Benefits/ company culture
    (e) Reluctance to undergo changes
    (f) Lack of opportunities at other companies
20. Suppose you are comparing job opportunities at two different companies: Company 1 pays 10% above the market average and Company 2 pays 30% above the market average.Which company do you think attracts more qualified applicants per opening?
    (a) Company 1
    (b) Company 2
    (c) Both attract the same number of applicants
21. Which company do you think provides better non-wage amenities (e.g., home office, child-

care subsidy)?

(a) Company 1
(b) Company 2
(c) Both provide the same non-wage amenities

22. Imagine you were to discover that other companies in your area pay {5%/10%/20%} more than your current employer. How likely is it that you would start applying for jobs at other companies?

(a) Please type in a number between 0% (I would definitely not apply) and 100% (I would start applying immediately): []%

23. How sure would you have to be that you would get a job at one of the higher-paying companies for you to start applying for a new job?

(a) At least 90% sure
(b) At least 75% sure
(c) At least 50% sure
(d) At least 25% sure
(e) I would start applying in any case

24. When considering a potential job opportunity, how important are each of the following features to you? [Dropdown scale with 1 - not important at all, 2 - slightly important, 3 - moderately important, 4 - very important, 5 - extremely important]

(a) Opportunity for reduced hours
(b) Home office option
(c) Predictable schedule (including overtime)

25. To what extent do you agree with the following statements? [Dropdown scale 1 (totally disagree) 2 3 4 5 6 7 (totally agree)]

(a) "I have confidence in my capabilities."
(b) "If I were comparing companies to apply to, I would factor in the amount of competition from other applicants."
(c) "I would be reluctant to apply for a job if the probability I would get an offer is low."

Module: Person-Specific Information

26. At the end of the survey, we are interested in your personal background. Are you married?

(a) Yes
(b) No

27. Do you have children?

(a) Yes
(b) No

H.2 Original German Questionnaire

H.2.1 Initial Survey

Module: Consent

1. Um die Befragung kurz zu halten, würden wir gerne bei der Auswertung Daten einbeziehen,

die bei der Bundesagentur für Arbeit in Nürnberg vorliegen. Dabei handelt es sich zum Beispiel um Informationen zu Ihrer bisherigen Berufstätigkeit. Wir halten alle datenschutzrechtlichen Bestimmungen streng ein bzw. diese werden vom IAB streng kontrolliert. Alle Angaben und Antworten werden in anonymisierter Form, also ohne Namen, Anschrift und Kontaktinformationen, ausgewertet und dargestellt. Niemand kann daher erkennen, welche Antworten Sie gegeben haben. Es erfolgt auch keine Weitergabe von Daten, die Ihre Person erkennen lassen. Weitere Informationen zum Datenschutz finden Sie unter https://www.iab.de/de/befragungen/lohnfle in-deutschland.aspx). Die Auswertung erfolgt zu rein wissenschaftlichen Zwecken, das heißt nicht für kommerzielle Zwecke wie Werbung oder Marketing. Ihre Zustimmung ist selbstverständlich freiwillig. Sind Sie mit der Zuspielung Ihrer Daten einverstanden?

   (a) Ja, ich bin mit einer Zuspielung einverstanden.
   (b) Nein, ich bin mit einer Zuspielung nicht einverstanden.

2. Was trifft am besten auf Ihre aktuelle Haupterwerbstätigkeit zu?
   (a) Sozialversicherungspflichtig beschäftigt
   (b) Selbstständig
   (c) Nicht erwerbstätig
   (d) Sonstiges

Module: Employment and Bargaining

Submodule: Employed or missing

3. Bitte beantworten Sie folgende Fragen in Bezug auf Ihren Hauptarbeitgeber. Seit wann sind Sie in Ihrem jetzigen Unternehmen beschäftigt?
   (a) Seit weniger als 6 Monaten
   (b) Seit 6-12 Monaten
   (c) Seit 1-2 Jahren
   (d) Seit 2-3 Jahren
   (e) Seit mehr als 3 Jahren
4. Ist Ihre Stelle tarifgebunden (d.h. werden Sie nach Tarifvertrag bezahlt)?
   (a) Ja
   (b) Nein
   (c) Ich weiß nicht
5. Wie viele Stunden arbeiten Sie in einer typischen Woche?{fill in}
6. Als Sie sich zum ersten Mal bei Ihrem jetzigen Unternehmen beworben haben, kannten Sie jemanden, der bei diesem Unternehmen beschäftigt war?
   (a) Ja.
   (b) Nein, aber ich kannte jemanden, der in der Vergangenheit bei diesem Unternehmen beschäftigt war.
   (c) Nein, ich kannte niemanden.
7. Wussten Sie zum Zeitpunkt Ihrer Bewerbung welches Gehalt Sie verdienen würden?
   (a) Ich hatte keine oder nur sehr wenig Ahnung.
   (b) Ich hatte nur eine ungefähre Vorstellung davon, was in meiner Region oder Branche bezahlt wird.
   (c) Ich hatte zumindest eine ungefähre Vorstellung, was dieses Unternehmen für die Stelle

bezahlt.

   (d) Ich wusste genau, was dieses Unternehmen für die Stelle bezahlt.

8. In den vergangenen sechs Monaten …… haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.

   (a) Stellenausschreibungen angesehen
   (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
   (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
   (d) Sich auf eine Stelle in einem anderen Unternehmen beworben

9. In den vergangenen sechs Monaten …… hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, eine Empfehlung für Sie auszusprechen)?

   (a) Ja
   (b) Nein

10. In den vergangenen sechs Monaten …… haben Sie Stellenangebote von anderen Unternehmen erhalten?

    (a) Ja
    (b) Nein

11. In den vergangenen sechs Monaten…… hat Ihr Unternehmen Ihnen eine Gehaltserhöhung angeboten, ohne dass Sie danach gefragt haben?

    (a) Ja
    (b) Nein

12. In den vergangenen sechs Monaten……. haben Sie proaktiv nach einer Gehaltserhöhung gefragt?

    (a) Ja
    (b) Nein

13. Wie viele Stunden arbeiten Sie in einer typischen Woche{fill in}
14. In den vergangenen sechs Monaten …… haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.

    (a) Stellenausschreibungen angesehen
    (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
    (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
    (d) Auf eine Stelle beworben

15. In den vergangenen sechs Monaten …… hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, Empfehlung für Sie auszusprechen)?

    (a) Ja
    (b) Nein

16. In den vergangenen sechs Monaten ...... haben Sie Stellenangebote erhalten?
    (a) Ja
    (b) Nein

Module: Worker-Provided Firms

17. Als Nächstes interessiert uns, wie Sie Beschäftigungsmöglichkeiten in Deutschland wahrnehmen. Auch wenn Sie etwas nicht genau wissen, möchten wir Sie bitten, Ihre Einschätzung mit uns zu teilen. Angenommen, Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln. Bei welchen Unternehmen würden Sie erwägen, sich zu bewerben? Bitte geben Sie drei Unternehmen an, bei denen Sie sich bewerben würden und die Beschäftigte in Stellen wie Ihrer einstellen (z.B. „Musterunternehmen GmbH“). Das können auch Unternehmen ohne aktuelle Stellenangebote sein.
    (a) {Unternehmen 1 eintragen}
    (b) {Unternehmen 2 eintragen}
    (c) {Unternehmen 3 eintragen}
    (d) Ich möchte diese Frage nicht beantworten [exklusive Option]
18. **[Employed]** Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

19. Was beschreibt am besten, wie sicher Sie sich bei Ihren Gehaltsschätzungen sind?

Unternehmen	Sehr unsicher	Unsicher	Etwas sicher	Sicher
{Unternehmen 1}				
{Unternehmen 2}				
{Unternehmen 3}				

20. Einige dieser Unternehmen haben möglicherweise Niederlassungen in mehreren Orten. An welchen Ort hatten Sie jeweils gedacht?

Unternehmen	Ort
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

21. Kennen Sie Mitarbeiter bei diesen Unternehmen (z.B. ehemalige Kollegen, Freunde oder Familienmitglieder)?

Unternehmen	Ja, ich kenne derzeitige Mitarbeiter	Ja, ich kenne ehemalige Mitarbe
{Unternehmen 1}		
{Unternehmen 2}		
{Unternehmen 3}		

22. Nehmen Sie nun an, einer Ihrer Kollegen bekommt Jobangebote von folgenden Unternehmen für eine Stelle wie Ihre. Was glauben Sie, welches Unternehmen bietet das höchste Bruttojahresgehalt? Bitte ordnen Sie die Angebote von 1 (höchstes Gehalt) bis 3 (niedrigstes

Gehalt).

Unternehmen	Rangfolge
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

23. **[Employed]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

24. **[Self-Employed]** Angenommen, Sie könnten selbstständig bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Mit aktuellem Gehalt selbstständig bleiben	

25. **[Non-employed]** Angenommen, Sie könnten weiterhin nicht erwerbstätig sein oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Einkommenserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Einkommen	
{Fill-in 2} mit {10%/15%} höherem Einkommen	
{Fill-in 3} mit {15%/15%} höherem Einkommen	
Weiterhin keine Erwerbstätigkeit	

26. **[Employed]** Nun möchten wir Sie bitten, die Rangfolge der Unternehmen unter zwei Szenarien neu einzustufen. Nehmen Sie zuerst an, Sie hätten wieder die drei Jobangebote zur Auswahl, aber Sie hätten den gleichen Arbeitsweg wie bisher. Bitte ordnen Sie die Angebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

27. **[Employed]** Statt dem Arbeitsweg, nehmen Sie nun an, dass Ihre Karriereentwicklung (zukünftige Gehaltserhöhungen/Beförderungen) für alle Optionen identisch ist. Bitte ordnen Sie die

Angebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {10%/15%} höherem Gehalt	
{Fill-in 2} mit {10%/15%} höherem Gehalt	
{Fill-in 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

Module: Researcher-Provided Firms

28. Falls Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, würden Sie erwägen, sich bei folgenden Unternehmen zu bewerben? Bitte kreuzen Sie alle zutreffenden Antworten an.
    (a) Unternehmen 1
    (b) Unternehmen 2
    (c) Unternehmen 3
    (d) Unternehmen 4
    (e) Unternehmen 5
    (f) Unternehmen 6
    (g) Unternehmen 7
29. **[Employed or Self-Employed]** Uns interessiert nun, wie Sie Stellenangebote in verschiedenen Unternehmen wahrnehmen. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

30. **[Non-Employed]** Uns interessiert nun, wie Sie Stellenangebote in verschiedenen Unternehmen wahrnehmen. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie Ihrer letzten Stelle arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

31. **[Employed]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {15%/10%} höherem Gehalt	
{Unternehmen 12} mit {15%/10%} höherem Gehalt	
{Unternehmen 3} mit {15%/15%} höherem Gehalt	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

32. **[Self-Employed]** Angenommen, Sie könnten selbstständig bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung.

Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {15%/10%} höherem Gehalt	
{Unternehmen 2} mit {15%/10%} höherem Gehalt	
{Unternehmen 13} mit {15%/15%} höherem Gehalt	
Mit aktuellem Gehalt selbstständig bleiben	

33. **[Non-Employed]** Angenommen, Sie könnten weiterhin nicht erwerbstätig sein oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Einkommenserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

34.

Jobangebot	Rangfolge
{Unternehmen 1} mit {15%/10%} höherem Einkommen	
{Unternehmen 2} mit {15%/10%} höherem Einkommen	
{Unternehmen 3} mit {15%/10%} höherem Einkommen	
Weiterhin keine Erwerbstätigkeit.	

35. Was glauben Sie, wie einfach wäre es für Sie, ein Stellenangebot von einem anderen Unternehmen zu erhalten, das Sie Ihrer jetzigen Stelle vorziehen würden?
    (a) Sehr einfach
    (b) Einfach
    (c) Schwierig
    (d) Sehr schwierig
36. Wenn Sie {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} das Unternehmen wechseln müssten, was beschreibt am besten, wo Sie nach einer Stelle suchen würden? Es geht dabei ausschließlich um Stellen ohne Home-Office Option.
    (a) Ich würde nach Stellen suchen, die nicht weiter entfernt von meinem Wohnort sind als. {eintragen} Kilometer
37. Abschließend interessiert uns Ihre Selbsteinschätzung. Sind Sie generell ein risikobereiter Mensch oder versuchen Sie Risiken zu vermeiden? Verwenden Sie dazu bitte eine Skala von 0 bis 10. Der Wert 0 bedeutet „gar nicht risikobereit" und der Wert 10 „sehr risikobereit". Mit den Werten dazwischen können Sie Ihre Einschätzung abstufen.
    (a) 0 (gar nicht risikobereit) 1 2 3 4 5 6 7 8 9 10 (sehr risikobereit)

Module: Consent

38. Als Dankeschön für Ihre Teilnahme verlosen wir {1500}/{1000} Geschenkkarten im Wert von {10}/{5} Euro. Wenn Sie an der Verlosung teilnehmen möchten, geben Sie bitte Ihre E-Mail-Adresse an. {open field that accepts e-mail addresses only}
39. Ihre Angaben unterstützen das IAB bei seiner Forschung zu Gehaltsentwicklung von Beschäftigten in Deutschland. Da sich die aktuellen Arbeitsmarktdynamiken stetig verändern, möchten wir Sie gerne in einigen Monaten erneut befragen. Wollen Sie auch weiterhin dazu beitragen, fundierte Informationen zur Gehaltsentwicklung in Deutschland bereitzustellen? Dann bitten wir Sie, uns Ihre E-Mail-Adresse zur Verfügung zu stellen, damit wir Sie erneut zu einer Befragung einladen können. Die Teilnahme daran ist freiwillig. Natürlich werden wir die E-Mail-Adresse ausschließlich nutzen, um Sie für unsere Befragung anzuschreiben. Sie können sich selbstverständlich jederzeit wieder aus der Befragung abmelden. An welche

E-Mail-Adresse dürfen wir unsere Einladung schicken?

(a) {open field that accepts e-mail addresses only}
(b) Ich möchte nicht per E-Mail kontaktiert werden.

40. Dürfen wir Sie dann postalisch erneut einladen?

(a) Ja
(b) Nein

H.2.2 Follow-up Survey

1. Um die Befragung kurz zu halten, würden wir gerne bei der Auswertung Daten einbeziehen, die bei der Bundesagentur für Arbeit in Nürnberg vorliegen. Dabei handelt es sich zum Beispiel um Informationen zu Ihrer bisherigen Berufstätigkeit. Wir halten alle datenschutzrechtlichen Bestimmungen streng ein bzw. diese werden vom IAB streng kontrolliert. Alle Angaben und Antworten werden in anonymisierter Form, also ohne Namen, Anschrift und Kontaktinformationen, ausgewertet und dargestellt. Niemand kann daher erkennen, welche Antworten Sie gegeben haben. Es erfolgt auch keine Weitergabe von Daten, die Ihre Person erkennen lassen. Weitere Informationen zum Datenschutz finden Sie unter https://www.iab.de/de/befragungen/lohnfle in-deutschland.aspx). Die Auswertung erfolgt zu rein wissenschaftlichen Zwecken, das heißt nicht für kommerzielle Zwecke wie Werbung oder Marketing. Ihre Zustimmung ist selbstverständlich freiwillig. Sind Sie mit der Zuspielung Ihrer Daten einverstanden?

   (a) Ja, ich bin mit einer Zuspielung einverstanden.
   (b) Nein, ich bin mit einer Zuspielung nicht einverstanden.

2. Haben Sie seit {month_survey1 2022/2023} eine neue Stelle bei einem anderen Unternehmen angetreten?

   (a) Ja
   (b) Nein

3. [Q2=1]Was trifft am besten auf Ihre aktuelle Haupterwerbstätigkeit zu?

   (a) Sozialversicherungspflichtig beschäftigt
   (b) Selbstständig
   (c) Nicht erwerbstätig
   (d) Sonstiges

4. In den vergangenen sechs Monaten... ...haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.

   (a) Stellenausschreibungen angesehen
   (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
   (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
   (d) Sich auf eine Stelle in einem anderen Unternehmen beworben

5. In den vergangenen sechs Monaten......hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, eine Empfehlung für Sie auszusprechen)?

(a) Ja
(b) Nein

6. In den vergangenen sechs Monaten......wie viele Stellenangebote von anderen Unternehmen haben Sie erhalten?
   (a) 0
   (b) 1
   (c) 2
   (d) 3 oder mehr

Module: Worker-Provided Firms

7. Als Nächstes interessiert uns, wie Sie Beschäftigungsmöglichkeiten in Deutschland wahrnehmen. Auch wenn Sie etwas nicht genau wissen, möchten wir Sie bitten, Ihre Einschätzung mit uns zu teilen. Angenommen, Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln. Bei welchen Unternehmen würden Sie erwägen, sich zu bewerben? Bitte geben Sie drei Unternehmen an, bei denen Sie sich bewerben würden und die Beschäftigte in Stellen wie Ihrer einstellen (z.B. „Musterunternehmen GmbH"). Das können auch Unternehmen ohne aktuelle Stellenangebote sein.
   (a) {Unternehmen 1 eintragen}
   (b) {Unternehmen 2 eintragen}
   (c) {Unternehmen 3 eintragen}
   (d) Ich möchte diese Frage nicht beantworten [exklusive option]
8. **[Q7=d]** Was beschreibt am besten, warum Sie in der vorherigen Frage keine Unternehmen angegeben haben?
   (a) Ich möchte diese Informationen nicht teilen
   (b) Es gibt Unternehmen, die ich in Betracht ziehen würde, aber die Eingabe in der Umfrage war zu kompliziert
   (c) Mir fallen keine konkreten Unternehmen ein
   (d) Ich habe kein Interesse daran, das Unternehmen zu wechseln und habe darüber bis jetzt nicht nachgedacht
9. **[Q7 has only 1 or 2 provided firms]** Was beschreibt am besten, warum Sie in der vorherigen Frage nicht drei Unternehmen angegeben haben?
   (a) Ich möchte diese Informationen nicht teilen
   (b) Es gibt Unternehmen, die ich in Betracht ziehen würde, aber die Eingabe in der Umfrage war zu kompliziert
   (c) Mir fallen keine weiteren, konkreten Unternehmen ein
   (d) Ich habe kein Interesse daran, das Unternehmen zu wechseln und habe darüber bis jetzt nicht nachgedacht
10. **[Q7 is not missing]** Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Fill-in 1}	
{Fill-in 2}	
{Fill-in 3}	

11. **Q7 is not missing]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort das unten angegebene Gehalt. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Researcher-Provided Firms

12. Falls Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, würden Sie erwägen, sich bei folgenden Unternehmen zu bewerben? Bitte kreuzen Sie alle zutreffenden Antworten an.

    (a) Firm 1
    (b) Firm 2
    (c) Firm 3
    (d) Firm 4
    (e) Firm 5
    (f) Firm 6
    (g) Firm 7

13. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

14. **[Q13 is missing]** Auch wenn Sie sich nicht sicher sind, interessieren wir uns sehr für Ihre Einschätzung, wie diese Unternehmen im Hinblick auf das Gehalt abschneiden. Bitte ordnen Sie die Unternehmen von 1 (höchstes Gehalt) bis 3 (niedrigstes Gehalt) ein.

Unternehmen	Rangfolge
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

15. Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der

genannten Unternehmen wechseln und erhalten sofort das unten angegebene Gehalt. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

16. Nun möchten wir Sie bitten, die Rangfolge der Unternehmen unter zwei Szenarien neu einzustufen. Nehmen Sie zuerst an, Sie hätten wieder die drei Jobangebote zur Auswahl, aber Sie hätten den gleichen Arbeitsweg wie bisher. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

17. Statt dem Arbeitsweg, nehmen Sie nun an, dass die Training- und Weiterbildungsmöglichkeiten der drei Unternehmen die gleichen sind wie in Ihrem aktuellen Unternehmen. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {} Gehalt	
{Unternehmen 2} mit {} Gehalt	
{Unternehmen 3} mit {} Gehalt	
Im jetzigen Unternehmen bleiben mit {} Gehalt	

[Note: The raises were randomized as follows: Workers group 1 saw the following raises for the four options: +15%/+10%/+10%/current pay. Workers in group 2 saw: +5%/+5%/+10%/-5%. Workers in group 3 saw: +5%/+15%/+15%/current. Workers in group 4 saw: +10%/+10%/-5%/current. Workers in group 5 saw: +10%/+5%/+10%/-5%.]

Module: Applications

18. Falls Sie planen würden {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, bei wie vielen verschiedenen Unternehmen würden Sie sich bewerben?

    (a) Die Anzahl der Unternehmen, bei denen ich mich bewerben würde, ist: _____

19. **[Randomize who sees Q19 (50%) and who sees Q20/Q21 (50%)]** Was sind Ihrer Meinung

nach die Hauptgründe, die Mitarbeitende zögern lassen, den Arbeitsplatz zu wechseln? Bitte wählen Sie die zwei wichtigsten Gründe aus.

(a) Persönliche Bindung zu den Arbeitskollegen
(b) Standort
(c) Gehalt
(d) Nebenleistungen/ Unternehmenskultur
(e) Abneigung gegen Veränderungen
(f) Fehlende Möglichkeiten bei anderen Unternehmen

20. Angenommen, Sie vergleichen die Stellenangebote bei zwei verschiedenen Unternehmen: Unternehmen 1 zahlt 10% über dem Durchschnitt im Arbeitsmarkt und Unternehmen 2 zahlt 30% über dem Durchschnitt im Arbeitsmarkt. Welches Unternehmen zieht Ihrer Meinung nach mehr qualifizierte Bewerbende pro Stelle an?

    (a) Unternehmen 1
    (b) Unternehmen 2
    (c) Beide ziehen die gleiche Anzahl von Bewerbenden an

21. Welches Unternehmen bietet Ihrer Meinung nach die besseren Nebenleistungen (z. B. Home Office, Kita-Zuschuss)?

    (a) Unternehmen 1
    (b) Unternehmen 2
    (c) Beide bieten die gleichen Nebenleistungen

22. Nehmen sie an, Sie stellen fest, dass andere Unternehmen in Ihrer Gegend {5%/10%/20%} mehr bezahlen als Ihr aktueller Arbeitgeber. Wie wahrscheinlich ist es, dass Sie sich auf eine neue Stelle bei einem anderen Unternehmen bewerben würden?

    (a) Bitte geben Sie eine Zahl zwischen 0% (ich würde mich auf keinen Fall bewerben) und 100% (ich würde mich sofort bewerben) an: ___%.

23. Wie sicher müssten Sie sein, dass Sie eine Stelle bei einem der besser bezahlenden Unternehmen bekommen würden, um sich für einen neuen Job zu bewerben?

    (a) Mindestens 90% sicher
    (b) Mindestens 75% sicher
    (c) Mindestens 50% sicher
    (d) Mindestens 25% sicher
    (e) Ich würde mich in jedem Fall bewerben

24. Falls Sie einen neuen Job in Betracht ziehen, wie wichtig ist jede der folgenden Eigenschaften für Sie? [Dropdown scale with 1 - Überhaupt nicht wichtig, 2 - Etwas wichtig, 3 - Mäßig wichtig, 4 - Sehr wichtig, 5 - Äußerst wichtig]

    (a) Möglichkeit für reduzierte Arbeitszeiten
    (b) Homeoffice-Option
    (c) Planbare Arbeitszeiten (einschließlich Überstunden)

Module: Person-Specific Information

25. Am Ende der Umfrage interessieren wir uns für Ihren persönlichen Hintergrund.Sind Sie verheiratet?

    (a) Ja

(b) Nein

26. Haben Sie Kinder?

    (a) Ja
    (b) Nein

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